



MP2787

7-Cell to 16-Cell, High-Accuracy Battery Monitor with Coulomb Counting

DESCRIPTION

The MP2787 is a robust battery management device, providing a complete analog front-end (AFE) monitoring. It is designed for multiple-cell series battery management systems (BMS). The device can support I²C or SPI communication. It supports connections for 7-cell to 16-cell series battery packs, with an absolute voltage exceeding 80V on particular pins.

The MP2787 integrates two separate analog-to-digital converters (ADCs). The first ADC measures each channel's differential cell voltages (up to 16 channels), die temperature, and 4-channel temperatures via external negative temperature coefficient (NTC) thermistors. The second ADC measures the charge/discharge current via an external current-sense resistor. The dual ADC architecture enables synchronous voltage and current measurements for cell and pack impedance monitoring.

When paired with an MPF4279x fuel gauge, the MP2787 can achieve a state-of-charge (SOC) error to within 2%.

The MP2787 provides robust hardware monitoring alarm functions, such as over-current (OC) interrupt, battery under-voltage (UV) interrupt, battery over-voltage (OV) interrupt, and high-/low-temperature interrupt. All of these alarms have configurable thresholds.

Internal passive balancing MOSFETs can be used to equalize mismatched cells, supporting up to 58mA. There is also the option to drive external balancing transistors (MOSFET or BJT).

The MP2787 is available in a TQFP-48 (7mmx7mm) package.

FEATURES

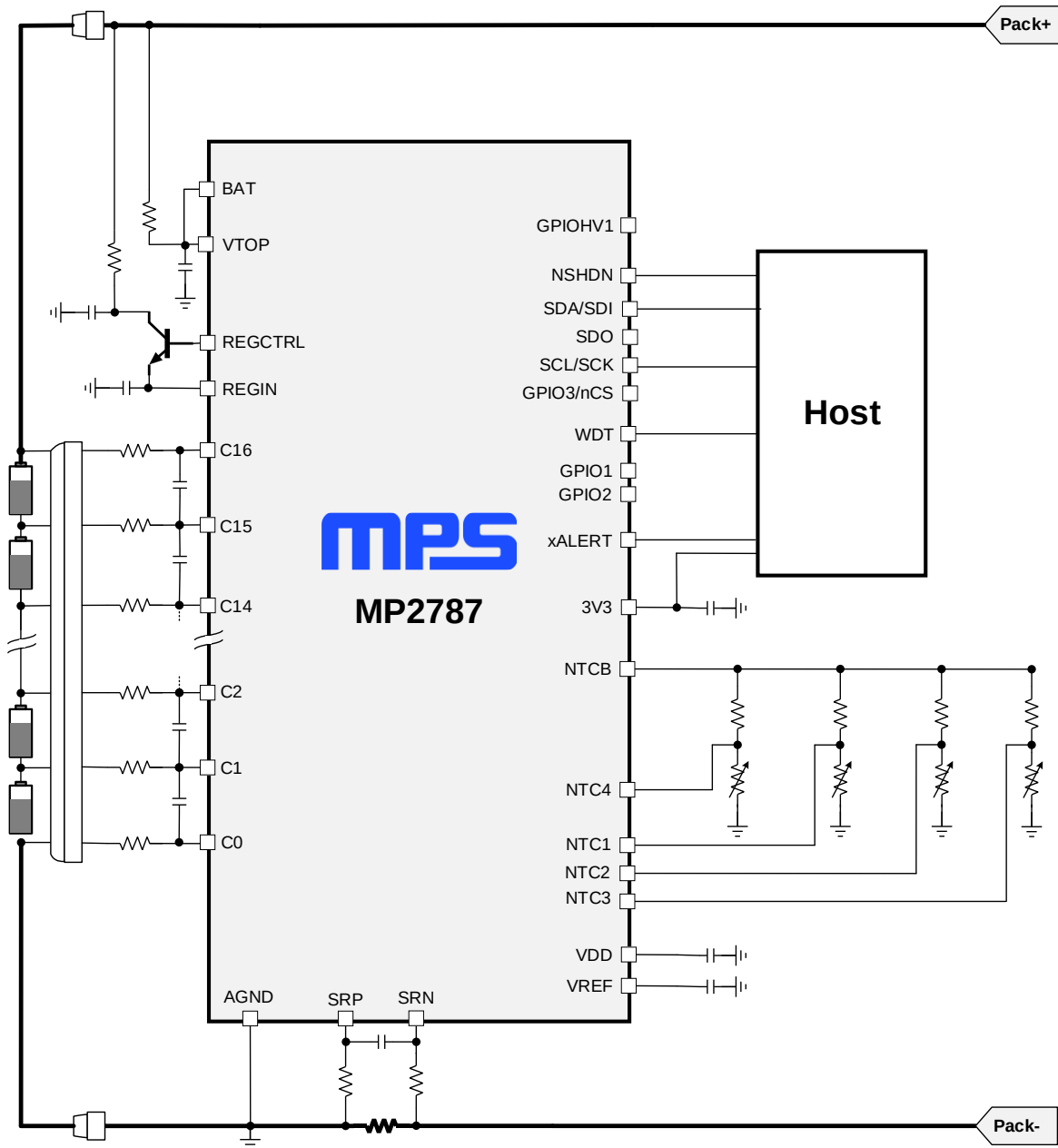
- Incorporates Dual Analog-to-Digital Converter (ADC) Architecture:
 - $<\pm 2\%$ State-of-Charge (SOC) Error with MPF4279x Fuel Gauge
 - Cell Voltage Measurement Error $<5\text{mV}$
 - Current/Coulomb Counter Error $<\pm 0.5\%$
 - Strictly Synchronized Current/Voltage Measurement for Impedance Sensing
- Hardware Configurable Alarms:
 - Charge/Discharge Over-Current (OC) Interrupt
 - Cell Under-Voltage (UV) and Over-Voltage (OV) Interrupt
 - Pack UV and OV Interrupt
 - Cell Low-/High-Temperature Interrupt
 - Die High-Temperature Interrupt
- Passive Cell Balancing up to 58mA per Cell:
 - Can Drive External Balancing Transistors
 - Automatic or Manual Control
- Additional Features:
 - Integrated 3.3V and 5V Low-Dropout (LDO) Regulators
 - High-Voltage and Low-Voltage GPIOs
 - Dedicated Thermistor Inputs
 - Open Wire Detection
 - Persistent Dead Battery Flag
 - Lockable Multiple-Time Programmable (MTP) Memory for Key Thresholds
- I²C or SPI Interface with 8-Bit Cyclic Redundancy Check (CRC)
- Random Cell Connection Tolerant
- Available in a TQFP-48 (7mmx7mm) Package

APPLICATIONS

- Power and Gardening Tools
- E-Bikes and E-Scooters
- Battery Backups and Uninterruptable Power Supplies (UPS)
- Energy Storage Systems (ESS)

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TYPICAL APPLICATION



ORDERING INFORMATION

Part Number*	Package	Top Marking	MSL Rating	Communication
MP2787DFP-xxxx**	TQFP-48 (7mmx7mm)	See Below	3	
MP2787DFP-0000	TQFP-48 (7mmx7mm)	See Below	3	I ² C, CRC disabled
MP2787DFP-0001	TQFP-48 (7mmx7mm)	See Below	3	I ² C, CRC enabled
MP2787DFP-0002	TQFP-48 (7mmx7mm)	See Below	3	SPI, CRC disabled
EVKT-MP2787-0000	Evaluation kit	N/A	N/A	N/A
EVKT-MP2787-0002	Evaluation kit	N/A	N/A	N/A

* For Tray, add suffix -T (e.g. MP2787DFP-xxxx-T).

** “xxxx” is the configuration code identifier for the register settings. Each “x” can be a hexadecimal value between 0 and F. The default codes are “0000”, “0001”, and “0002”. Contact an MPS FAE to create this unique number.

TOP MARKING

MPSYYWW
MP2787
LLLLLLLLLL

MPS: MPS prefix
 YY: Year code
 WW: Week code
 MP2787: Part number
 LLLLLLLLLL: Lot number

EVALUATION KIT EVKT-MP2787-0000

EVKT-MP2787-0000 kit contents (items below can be ordered separately):

#	Part Number	Item	Quantity
1	EV2787-0000-FP-00A	MP2787DFP-0000 I ² C evaluation board	1
2	EVKT-USBI2C-02	Includes one USB to I ² C communication interface, one USB cable, and one ribbon cable	1
3	Online resources	Include datasheet, user guide, product brief, and GUI	1

Order directly from MonolithicPower.com or our distributors.

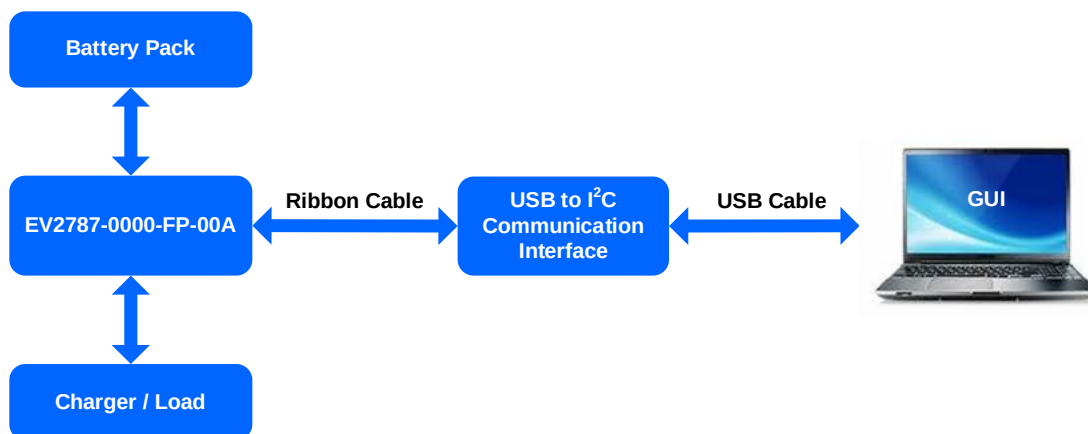


Figure 1: EV2787-0000-FP-00A Evaluation Kit Set-Up

EVALUATION KIT EVKT-MP2787-0002

EVKT-MP2787-0002 kit contents (items below can be ordered separately):

#	Part Number	Item	Quantity
1	EV2787-0002-FP-00A	MP2787DFP-0002 SPI evaluation board	1
2	EVKT-USBSPI-00	Includes one USB to SPI communication interface, one USB cable, and one ribbon cable	1
3	Online resources	Include datasheet, user guide, product brief, and GUI	1

Order directly from MonolithicPower.com or our distributors.

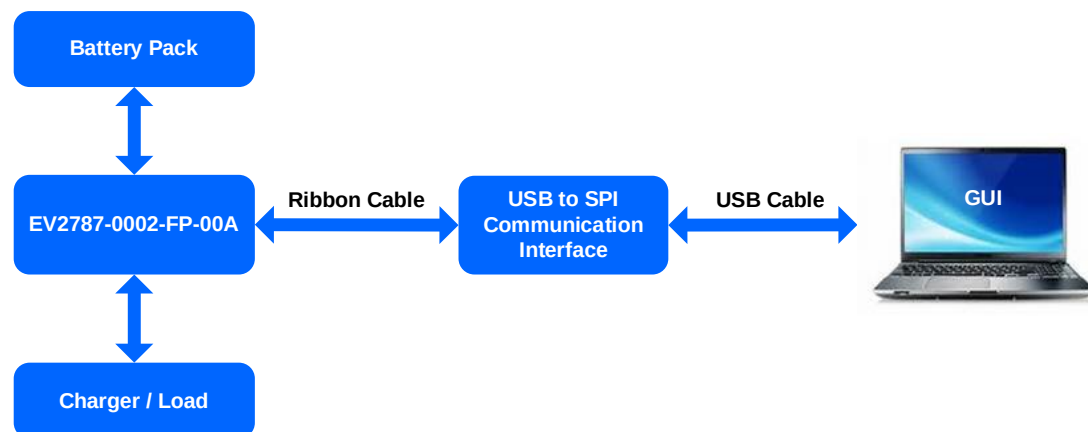
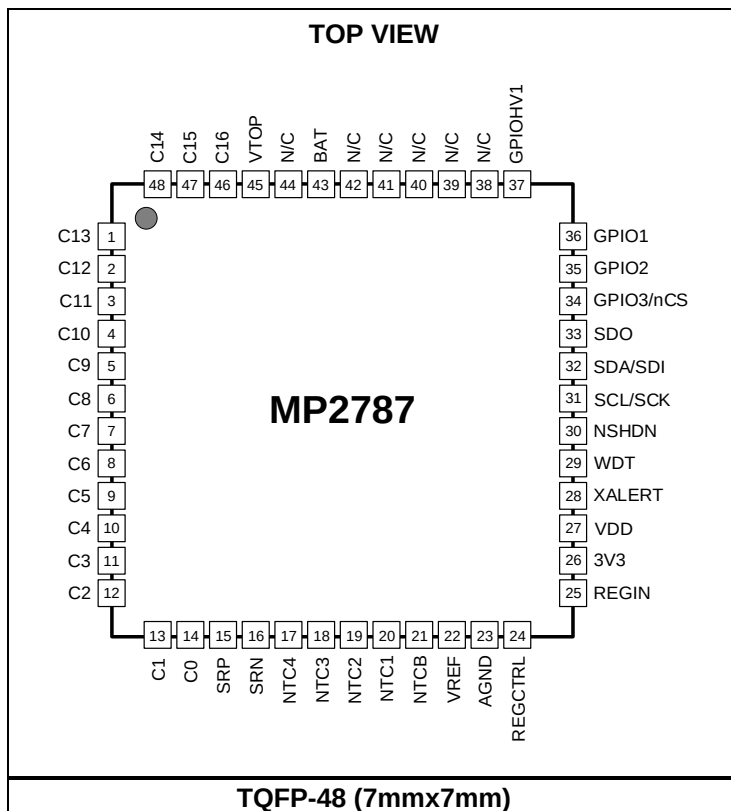


Figure 2: EV2787-0002-FP-00A Evaluation Kit Set-Up

PACKAGE REFERENCE



PIN FUNCTIONS

Pin #	Name	Type	Description
26	3V3	P	3.3V voltage output to drive external peripherals. Bypass the 3V3 pin with an external 1 μ F capacitor.
23	AGND	P	Ground. Connect the AGND pin near the positive connection (SRP) of the low-side sense resistor.
45	VTOP	P	Battery-side pack-sensing voltage and low-current positive supply pin. The VTOP pin must be connected to the top of the battery stack, which is the highest positive voltage in the battery pack.
46	C16	I	Connect to the positive pin of cell 16.
47	C15	I	Connect to the positive pin of cell 15.
48	C14	I	Connect to the positive pin of cell 14.
1	C13	I	Connect to the positive pin of cell 13.
2	C12	I	Connect to the positive pin of cell 12.
3	C11	I	Connect to the positive pin of cell 11.
4	C10	I	Connect to the positive pin of cell 10.
5	C9	I	Connect to the positive pin of cell 9.
6	C8	I	Connect to the positive pin of cell 8.
7	C7	I	Connect to the positive pin of cell 7.
8	C6	I	Connect to the positive pin of cell 6.
9	C5	I	Connect to the positive pin of cell 5.
10	C4	I	Connect to the positive pin of cell 4.
11	C3	I	Connect to the positive pin of cell 3.
12	C2	I	Connect to the positive pin of cell 2.
13	C1	I	Connect to the positive pin of cell 1.
14	C0	I	Connect to the negative pin of cell 1.
44	N/C		Not connected.
39	N/C		Not connected.
36	GPIO1	I/O	General-purpose pin 1.
35	GPIO2	I/O	General-purpose pin 2.
37	GPIOHV1	I/O	General-purpose, high-voltage pin 1.
30	NSHDN	I	Active-low shutdown input signal.
20	NTC1	I	Thermistor 1 terminal.
19	NTC2	I	Thermistor 2 terminal.
18	NTC3	I	Thermistor 3 terminal.
17	NTC4	I	Thermistor 4 terminal.
21	NTCB	O	NTC bias.
40	N/C		Not connected.
38	N/C		Not connected.
24	REGCTRL	P	Turn-on control for the external bipolar junction transistor (BJT) low-dropout (LDO) regulator.
25	REGIN	P	Internal regulator input. Connect an external 3.3 μ F bypass capacitor between the REGIN and AGND pins.
41	N/C		Not connected.
29	WDT	I/O	Watchdog timer.
28	xALERT	O	Interrupt alert output.
16	SRN	I	Negative sense pin.
15	SRP	I	Positive sense pin.
42	N/C		Not connected.
27	VDD	P	1.8V rail for internal use. Connect a 1 μ F bypass capacitor between the VDD and AGND pins.
43	BAT	P	Battery-side, low-current positive supply. Connect the BAT pin to the top of the battery stack, which is the highest positive voltage in the battery pack.

PIN FUNCTIONS (continued)

Pin #	Name	Type	Description
22	VREF	P	Analog-to-digital converter (ADC) reference voltage.
34	GPIO3/ nCS	I/O	Multi-function pin. This pin can be set as GPIO3, or it can be set for SPI cable selection.
31	SCL/SCK	I	Multi-function pin. This pin can be set as the I ² C interface clock or the SPI interface clock.
32	SDA/SDI	I/O	Multi-function pin. This pin can be set as the I ² C interface data or the SPI serial data input.
33	SDO	O	SPI serial data output.

ABSOLUTE MAXIMUM RATINGS ⁽¹⁾

GPIOHV1 to AGND.....-0.3V to +86V
 GPIOHV1 to VTOP.....0.3V
 VTOP to AGND.....-0.3V to +86V
 BAT to AGND.....-0.3V to +86V
 Cx - C (x - 1) (where x = 1 to 16)
-0.3V to +10V
 C16 to AGND-0.3V to +86V
 Cn to AGND (where n = 1 to 15).....
-0.3V to (n - 1) x 5.375 +7V
 C0 to AGND-0.5V to +5.7V
 SRP, SRN to AGND.....-0.5V to +6V
 REGCTRL to AGND-0.3V to +15V
 NSHDN to AGND.....-0.3V to +9V
 VDD to AGND-0.3V to +2V
 All other pins to AGND.....-0.3V to +6V
 Junction temperature (T_J)150°C
 Lead temperature.....260°C
 Storage temperature -65°C to +150°C

ESD Ratings

Human body model (HBM) ⁽²⁾..... 1.5kV
 Charged-device model (CDM) ⁽³⁾ 500V

Recommended Operating Conditions ⁽⁴⁾

VTOP voltage.....18V to 75.2V
 Cx - C (x - 1) (where x = 1 to 16) ⁽⁵⁾1V to 5V
 C0 to AGND-0.25V to +0.3V
 REGIN voltage4.5V to 5.5V
 Operating temperature (T_J) -40°C to +85°C
 SRP to SRN-100mV to +100mV

Thermal Resistance ⁽⁶⁾

Junction-to-ambient (R_{θJA}) 46.6°C/W
 Junction-to-case (top) (R_{θJC_TOP}) 14.5°C/W
 Junction-to-board (top) (R_{θJB_TOP}) 27.1°C/W

Notes:

- 1) Exceeding these ratings may damage the device.
- 2) Tested per ANSI/ESDA/JEDEC JS-001.
- 3) Tested per ANSI/ESDA/JEDEC JS-002.
- 4) The device is not guaranteed to function outside of its operating conditions.
- 5) The stack voltage should exceed 18V.
- 6) Metrics provided using set-up conditions compliant with EIA/JESD51-2, 7, and 8.

ELECTRICAL CHARACTERISTICS

Connected cells = 16, each cell voltage = 3.75V, $V_{TOP} = 60V$, $T_A = 25^{\circ}C$, unless otherwise noted.

Parameter	Symbol	Condition	Min	Typ	Max	Units
Supply Current and Leakage						
Total shutdown current	I _{VTOP_SHDN}	VTOP + REGIN current, NSHDN pin low, 3.3V off, T _A = 25°C		1		μA
		VTOP + REGIN current, NSHDN pin low, 3.3V off, T _A = -40°C to +85°C			2.5	μA
		VTOP + REGIN current, 3.3V on, NSHDN pin low, T _A = 25°C		13.5		μA
		VTOP + REGIN current, 3.3V on, NSHDN pin low, T _A = -40°C to +85°C			20	μA
Total active current	I _{VTOP_ACTIVE}	VTOP + REGIN current, communication interface enabled, hardware monitoring off, T _A = 25°C		23		μA
		VTOP + REGIN current, communication interface enabled, hardware monitoring off, T _A = -40°C to +85°C			30	μA
Cell leakage	I _{CX_LEAK}	T _A = 25°C	-200		+200	nA
		T _A = -40°C to +85°C	-600		+600	nA
Supported Series Cells						
Supported cell number ⁽⁷⁾	N _{CELL}		7		16	
VTOP supply voltage range	V _{VTOP_SUPPLY}	T _A = -40°C to +85°C	18		75.2	V
VTOP under-voltage lockout (UVLO) threshold	V _{VTOP_UVLO}	Falling edge, T _A = -40°C to +85°C	14.6	15.8	17	V
VTOP under-voltage (UV) hysteresis	V _{VTOP_UVLO_HYST}	T _A = 25°C		1.15		V
		T _A = -40°C to +85°C	0.78		1.5	V
Current Sense						
Current analog-to-digital converter (ADC) conversion time	t _{IADC}	16 bits (15 bits + sign)		2		ms
Current ADC measurement range	V _{SRPN}	T _A = -40°C to +85°C	-100		+100	mV
Current ADC measurement gain error	I _{ADC_GAIN_ERR}	16-bit conversion, full-scale range (FSR) = 100mV, SRP and SRN common mode: +300mV to -125mV from AGND, T _A = 25°C	-0.5		+0.5	%
		16-bit conversion, FSR = 100mV, SRP and SPN common mode: +300mV to -125mV from AGND, T _A = -40°C to 85°C	-1		+1	%
Current-sense offset ⁽⁸⁾	V _{IADC_ACC_OFFSET}	SRP - SRN = 0V, SRP and SRN common mode: +300mV to -125mV from AGND, T _A = -40°C to +85°C	-12		+12	μV

ELECTRICAL CHARACTERISTICS (continued)

Connected cells = 16, each cell voltage = 3.75V, $V_{TOP} = 60V$, $T_A = 25^{\circ}C$, unless otherwise noted.

Parameter	Symbol	Condition	Min	Typ	Max	Units
SRP and SRN leakage	I _{SRPN_LEAK}	IADC not running, T _A = -40°C to +85°C	-500		+500	nA
SRP and SRN differential input current	I _{SRPN_DIFF}	IADC running, SPR - SRN = 100mV, absolute value, T _A = -40°C to +85°C		0.5		μA
		IADC running, SPR - SRN < 5mV, T _A = -40°C to +85°C		50		nA
Analog-to-Digital Converter (ADC) Sigma Delta Voltage						
Voltage ADC conversion time	t _{VADC}	15 bits, T _A = 25°C		2		ms
Cell Voltage Measurement						
Cell ADC measurement range	V _{CELL}	T _A = -40°C to +85°C	1		5	V
Total cell measurement error ⁽⁹⁾	V _{CELL_ERR}	V _{CELL} = 2V to 4.5V, T _A = 25°C	-5		+5	mV
		V _{CELL} = 2V to 4.5V, T _A = -20°C to +65°C	-7.5		+7.5	mV
		V _{CELL} = 2V to 4.5V, T _A = -40°C to +85°C	-12.5		+12.5	mV
		V _{CELL} = 1V to 5V, T _A = -40°C to +85°C	-15		+15	mV
C0 to AGND voltage	V _{C0_TO_AGND}	C0 to AGND voltage drop resulting in less than 15mV error on cell 1, T _A = -40°C to +85°C	-0.25			V
Cell ADC input current	I _{IN_CELL_CONV}	Input current during ADC conversion when V _{CELL} = 5V, T _A = 25°C, cell 1 is measured by the C1 to C0 voltage		1.2		μA
Die Temperature						
Die temperature operating range	T _{DIE}		-40		+85	°C
Over-temperature (OT) analog shutdown threshold	T _{DIE_OTSD}			140	155	°C
Negative Temperature Coefficient (NTC) Temperature Measurement						
NTC voltage measurement range	V _{NTC}	Nominal NTC range within a percentage of NTCB, T _A = 25°C	0		100	%
NTC total measurement error	V _{NTC_ERR}	15-bit conversion, NTCB = 3.3V, T _A = -40°C to 85°C	-55		+55	LSB
NTCx input leakage	I _{NTC_IN_LEAK}	ADC off, on each NTC pin, T _A = -40°C to +85°C			250	nA
NTCx input leakage ADC on	I _{NTC_LEAK_CONV}	ADC converting NTCx, on each NTC pin, T _A = 25°C			250	nA

Notes:

- 7) Guaranteed by design.
- 8) The current measurement offset error is specified after completion of post-PCB assembly calibration. Contact an MPS FAE for the related application note.
- 9) No calibration required. The error can be further reduced by post-PCB assembly calibration. Contact an MPS FAE for the related application note.

ELECTRICAL CHARACTERISTICS (continued)

Connected cells = 16, each cell voltage = 3.75V, $V_{TOP} = 60V$, $T_A = 25^{\circ}C$, unless otherwise noted.

Parameter	Symbol	Condition	Min	Typ	Max	Units
VTOP Measurement						
VTOP measurement range	V _{VTOP}	T _A = -40°C to +85°C	18		75.2	V
VTOP total measurement error	V _{VTOP_ERR}	15-bit conversion (positive range), T _A = -40°C to +85°C	-250		+250	mV
GPIO Measurement						
GPIO measurement range	V _{GPIO}	T _A = 25°C	0		3.3	V
GPIO total measurement error	V _{GPIO_ERR}	15-bit ADC reading, T _A = 25°C	-15		+15	mV
		15-bit ADC reading, T _A = -40°C to +85°C	-25		+25	mV
GPIO input current during ADC reading	I _{GPIO_CONV_IN}	GPIO pins input current during ADC conversion, T _A = -40°C to +85°C		1		μA
Regulators Measurements						
REGIN measurement error	V _{REGIN_ERR}	15-bit ADC reading, T _A = 25°C	-15		+15	mV
		15-bit ADC reading, T _A = -40°C to +85°C	-25		+25	mV
3V3 total measurement error	V _{3V3_ERR}	15-bit ADC reading, T _A = 25°C	-15		+15	mV
		15-bit ADC reading, T _A = -40°C to +85°C	-25		+25	mV
VDD total measurement error	V _{VDD_ERR}	15-bit ADC reading, T _A = 25°C	-15		+15	mV
		15-bit ADC reading, T _A = -40°C to +85°C	-25		+25	mV
Hardware Protection						
Cell over-voltage (OV) and UV steps	V _{CELL_TH_STEP}			19.5		mV
Cell OV/UV step accuracy	V _{CELL_TH_ACC}	V _{CELL} = 2V to 4.5V, T _A = 0°C to 60°C	-19.5		+19.5	mV
		V _{CELL} = 1V to 5V, T _A = -40°C to +85°C	-40		+40	mV
Pack OV/UV steps	V _{PACK_TH_STEP}			19.5		mV
Deep discharge steps	V _{DEPTH_STEP}			19.5		mV
Deep discharge step accuracy	V _{DEPTH_STEP_ACC}	V _{CELL} = 2V to 4.5V, T _A = 0°C to 60°C	-19.5		+19.5	mV
		V _{CELL} = 1V to 5V, T _A = -40°C to +85°C	-40		+40	mV
OC fastest detection time	t _{OC_DGL_MIN}	Fastest deglitch setting and latency, T _A = -40°C to +85°C			120	μs
OC discharge 1x FSR value	V _{OC_DSG_FSR_1X}	Range and LSB selector = 0 (1x), FSR (max setting), T _A = 25°C		176		mV
OC discharge 1x full scale accuracy	V _{OC_DSG_FSR_1X_ACC}	Range and LSB selector = 0 (1x), FSR (max setting), T _A = -40°C to +85°C	-15		+15	%

ELECTRICAL CHARACTERISTICS (continued)

Connected cells = 16, each cell voltage = 3.75V, $V_{TOP} = 60V$, $T_A = 25^{\circ}C$, unless otherwise noted.

Parameter	Symbol	Condition	Min	Typ	Max	Units
OC discharge 1x offset	V _{OC_DSG_FSR_1X_OFF}	Range and LSB selector = 0 (1x), offset on smallest setting, T _A = -40°C to +85°C	-1.5		+1.5	mV
OC discharge 3x FSR value	V _{OC_DSG_FSR_3X}	Range and LSB selector = 1 (3x), FSR (max setting), T _A = 25°C		528		mV
OC discharge 3x full scale accuracy	V _{OC_DSG_FSR_3X_ACC}	Range and LSB selector = 1 (3x), FSR (max setting), T _A = -40°C to +85°C	-15		+15	%
OC discharge 3x offset	V _{OC_DSG_FSR_3X_OFF}	Range and LSB selector = 1 (3x), offset on smallest setting, T _A = -40°C to +85°C	-2		+2	mV
OC charge 1x FSR value	V _{OC_CHG_FSR_1X}	Range and LSB selector = 0 (1x), FSR (max setting), T _A = 25°C		80		mV
OC charge 1x full scale accuracy	V _{OC_CHG_FSR_1X_ACC}	Range and LSB selector = 0 (1x), FSR (max setting), T _A = -40°C to +85°C	-15		+15	%
OC charge 1x offset	V _{OC_CHG_FSR_1X_OFF}	Range and LSB selector = 0 (1x), offset on smallest setting, T _A = -40°C to +85°C	-1.5		+1.5	mV
OC charge 3x FSR value	V _{OC_CHG_FSR_3X}	Range and LSB selector = 1 (3x), FSR (max setting), T _A = 25°C		240		mV
OC charge 3x full scale accuracy	V _{OC_CHG_FSR_3X_ACC}	Range and LSB selector = 1 (3x), FSR (max setting), T _A = -40°C to +85°C	-15		+15	%
OC charge 3x offset	V _{OC_CHG_FSR_3X_OFF}	Range and LSB selector = 1 (3x), offset on smallest setting, T _A = -40°C to +85°C	-1.5		+1.5	mV
Cell Balancing						
On resistance for the balancing MOSFET	R _{DS(ON)_BAL_FET}	T _A = 25°C		28		Ω
		T _A = -40°C to +85°C	15		50	Ω
Open Wire						
Open-wire pull-up current	I _{OW_PUP}	T _A = 25°C		100		μA
Open-wire pull-down current	I _{OW_PD}	T _A = 25°C		100		μA
Low-Dropout Regulator (LDO) Supply and References						
REGCTRL output voltage	V _{REGCTRL}	V _{TOP} = 18V to 75.2V, with REGCTRL load current = 0mA and 1mA, T _A = -40°C to +85°C	5.45	5.6	5.75	V

ELECTRICAL CHARACTERISTICS (continued)

Connected cells = 16, each cell voltage = 3.75V, $V_{TOP} = 60V$, $T_A = 25^{\circ}C$, unless otherwise noted.

Parameter	Symbol	Condition	Min	Typ	Max	Units
REGIN voltage ⁽¹⁰⁾	V_{REGIN}	With external REGIN BJT (FZT853TA), $V_{TOP} = 18V$ to $75.2V$, load current = 1mA and 50mA, $T_A = -40^{\circ}C$ to $+85^{\circ}C$	4.65	5.1	5.4	V
REGIN analog UV	V_{REGIN_UV}	External power to REGIN, falling edge, $V_{TOP} = 18V$ to $75.2V$, $T_A = -40^{\circ}C$ to $+85^{\circ}C$	4.16	4.3	4.44	V
3V3 nominal voltage	V_{3V3}	$T_A = 25^{\circ}C$		3.3		V
3V3 output accuracy	V_{3V3_ACC}	$T_A = -40^{\circ}C$ to $+85^{\circ}C$	-5		+5	%
3V3 short-circuit current	I_{3V3_EFET}	External power to REGIN, $T_A = -40^{\circ}C$ to $+85^{\circ}C$	55	68	80	mA
VDD output voltage	V_{DD}	$T_A = 25^{\circ}C$		1.8		V
Reference voltage	V_{REF}	$T_A = 25^{\circ}C$		3.3		V
NTCB pull-up voltage	V_{NTCB}	NTCB enabled, no load, $T_A = 25^{\circ}C$		3.3		V
NTCB load	$I_{NTCB_MAXLOAD}$	NTCB enabled, max load, $T_A = 25^{\circ}C$		4		mA
		NTCB enabled, max load, $T_A = -40^{\circ}C$ to $+85^{\circ}C$	3.2		4.8	mA
3V3 analog UV	V_{3V3_UV}	Falling edge		2.85		V
GPIO						
GPIO input voltage (high)	V_{IH_GPIO}	Voltage high (V_H) = 3.3V or 5V, $T_A = -40^{\circ}C$ to $+85^{\circ}C$	$0.8 \times V_H$		V_H	V
GPIO input voltage (low)	V_{IL_GPIO}	$V_H = 3.3V$ or $5V$, $T_A = -40^{\circ}C$ to $+85^{\circ}C$	0		$0.2 \times V_H$	V
GPIO output voltage (high)	V_{OH_GPIO}	$V_H = 3.3V$ or $5V$, $I_{SOURCE} = 1.5mA$, $T_A = -40^{\circ}C$ to $+85^{\circ}C$	$V_H - 0.5$		V_H	V
GPIO output voltage (low)	V_{OL_GPIO}	$V_H = 3.3V$ or $5V$, $I_{SINK} = 1.5mA$, $T_A = -40^{\circ}C$ to $+85^{\circ}C$	0		0.4	V
GPIO push-pull resistor (up)	R_{GPIO_PUP}	GPIO high voltage = 3.3V, external power to REGIN		430		Ω
GPIO push-pull resistor (down)	R_{GPIO_PDOWN}	GPIO low voltage = 0V		110	200	Ω
GPIO pull-up mode	$R_{GPIO_WEAK_PUP}$			17		k Ω
GPIOHV1 Pin						
GPIOHV1 pin input voltage high level	V_{IH_GPIOHV}	$T_A = -40^{\circ}C$ to $+85^{\circ}C$	2.4			V
GPIOHV1 pin input voltage low level	V_{IL_GPIOHV}	$T_A = -40^{\circ}C$ to $+85^{\circ}C$			0.7	V
GPIOHV1 pin max pull-up current	I_{GPIOHV_PUP}			11		mA
GPIOHV1 pin max pull-down current	I_{GPIOHV_PDOWN}			8		mA

Note:

10) Guaranteed by design. The REGIN voltage is limited by $V_{REGCTRL}$ and is equal to $V_{REGCTRL} - V_{BE_ON}$.

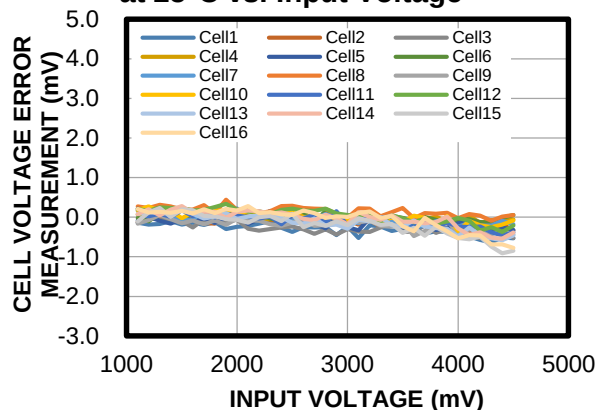
ELECTRICAL CHARACTERISTICS (continued)

Connected cells = 16, each cell voltage = 3.75V, $V_{TOP} = 60V$, $T_A = 25^{\circ}C$, unless otherwise noted.

Parameter	Symbol	Condition	Min	Typ	Max	Units
NSHDN Pin						
NSHDN internal pull-down	R _{SHDN_PD}	NSHDN = 0V to 5V		5		MΩ
NSHDN configuring voltage	V _{NSHDN_PROG}		7.5	7.6	7.7	V
NSHDN deglitch time to enter shutdown	t _{NSHDN_DGL_ENTER}			8		ms
NSHDN deglitch time to enter active	t _{NSHDN_DGL_ACTIVE}			4		ms
NSHDN falling threshold	V _{NSHDN_FALL}			1		V
NSHDN rising threshold	V _{NSHDN_RISE}	3V3 enabled at shutdown mode		2.65		V
		3V3 disabled at shutdown mode		2		V
WDT Pin						
Reset pulse length	t _{WDT_RSTPULSE_LEN}			10		ms
Internal Clock						
Internal clock frequency	f _{2M_CLOCK}	V _{TOP} = 18V to 75.2V, T _A = 25°C		2		MHz
Internal clock frequency	f _{2M_CLOCK}	V _{TOP} = 18V to 75.2V, T _A = -40°C to +85°C	1.85	2	2.15	MHz
Low-frequency internal clock	f _{32K_CLOCK}	V _{TOP} = 18V to 75.2V, T _A = 25°C	31.2	32	32.8	kHz
Low-frequency internal clock	f _{32K_CLOCK}	V _{TOP} = 18V to 75.2V, T _A = -40°C to +85°C	30.7		33.3	kHz
I ² C Communication Interface						
I ² C clock frequency	f _{I2C}	T _A = 25°C			400	kHz
SCL, SDA low input voltage	V _{ILOW}	T _A = -40°C to +85°C			0.25 x 3V3	V
SCL, SDA high input voltage	V _{IHIGH}	T _A = -40°C to +85°C	0.7 x 3V3			V
SCL, SDA input hysteresis	V _{IHYST}	T _A = -40°C to +85°C		0.2 x 3V3		V
SPI Communication Interface						
SPI clock frequency	f _{SPI}	T _A = 25°C			1.0	MHz
SPI input low	V _{SPI_IN_LOW}	SDI, CLK, T _A = -40°C to +85°C			0.25 x 3V3	V
SPI input high	V _{SPI_IN_HIGH}	SDI, CLK, T _A = -40°C to +85°C	0.7 x 3V3			V
SDO high-side (HS) on resistance	R _{SDO_DRV_HI}	I _{SOURCE} = 1.5mA, T _A = -40°C to +85°C	75	100	125	Ω
SDO low-side (LS) on resistance	R _{SDO_DRV_LOW}	I _{SINK} = 1.5mA, T _A = -40°C to +85°C	38	52	66	Ω

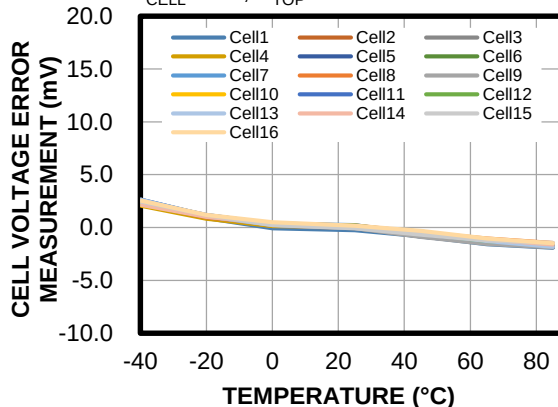
TYPICAL PERFORMANCE CHARACTERISTICS

Cell Voltage Error Measurement at 25°C vs. Input Voltage



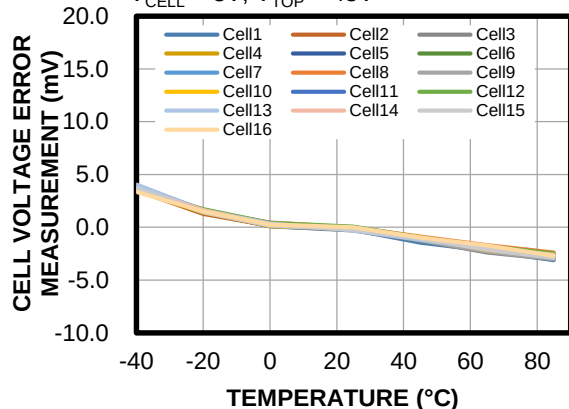
Cell Voltage Error Measurement vs. Temperature

$V_{CELL} = 2V, V_{TOP} = 32V$



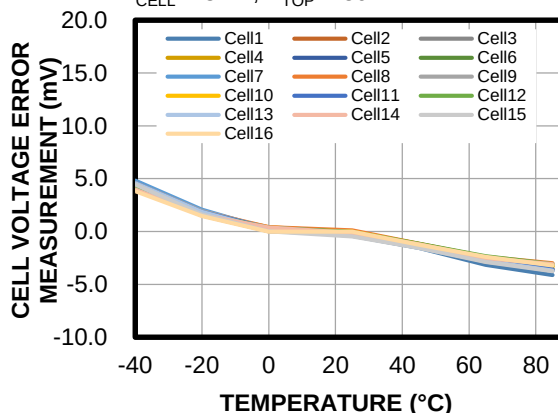
Cell Voltage Error Measurement vs. Temperature

$V_{CELL} = 3V, V_{TOP} = 48V$



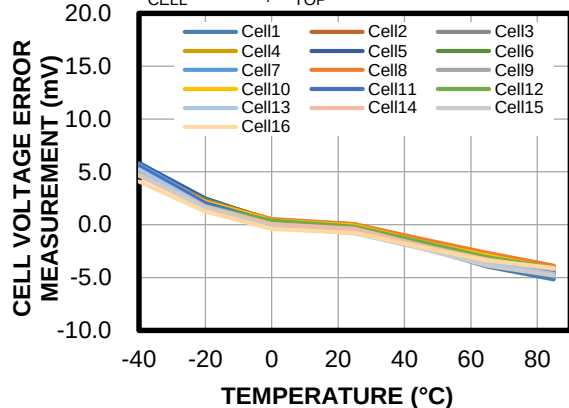
Cell Voltage Error Measurement vs. Temperature

$V_{CELL} = 3.7V, V_{TOP} = 59.2V$



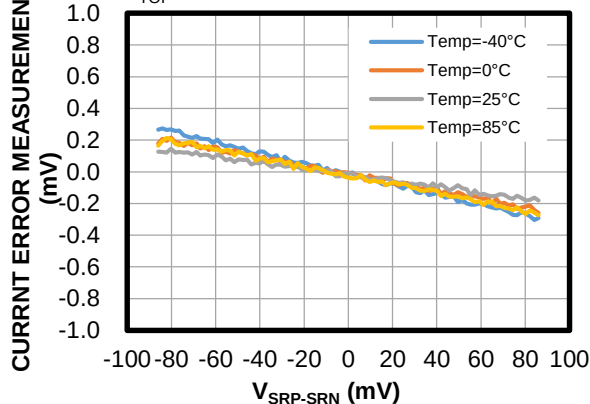
Cell Voltage Error Measurement vs. Temperature

$V_{CELL} = 4.5V, V_{TOP} = 72V$



Current Error Measurement vs.

$V_{SRP-SRN}$
 $V_{TOP} = 60V$



TYPICAL PERFORMANCE WHEN PAIRED WITH MPF4279X FUEL GAUGE

The MP2787 battery monitor includes strictly synchronized cell and pack voltage and current measurements, for the purpose of maximizing state-of-charge (SOC) determination. MPS' MPF4279x family of fuel gauges are designed to take advantage of this feature. This section illustrates the SOC accuracy of the MP2787 battery monitor when combined with MPS' MPF4279x fuel gauge family.

Constant-Current/Constant-Voltage (CC/CV) Charge and Dynamic Discharge Cycle

The next scenarios consist of charging a 10S1P ⁽¹¹⁾ battery using the typical CC/CV method, followed by a highly dynamic discharge at different ambient temperatures. The charge constant current rate is 1C, while the charge termination current in this example is 0.1C. The highly dynamic discharge corresponds to a typical e-bike's current profile, with an average current of 1C and maximum peak currents up to 2.8C. Figure 3 shows the current profile of the complete cycle at 25°C.

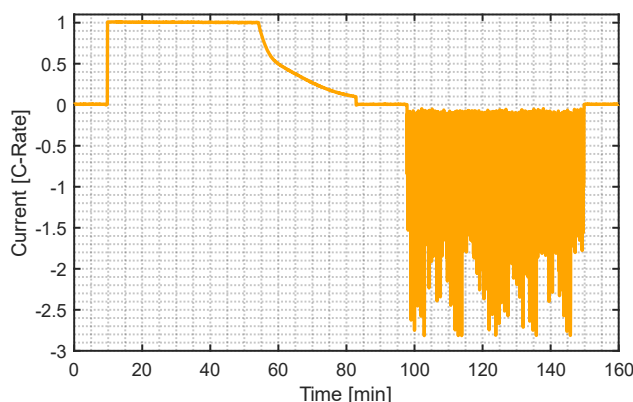


Figure 3: CC/CV Charge and Dynamic Discharge Current Profile

Figure 4 shows the performance of the combined MP2787 and MPF42791 for the CC/CV charge and dynamic discharge cycle at an ambient temperature of 25°C. During charge, the root-mean-squared (RMS) ⁽¹²⁾ and maximum pack SOC error are 0.61% and 1.03%, respectively. During discharge, the RMS and pack SOC error are 0.78% and 1.94%, respectively.

Notes:

11) 10S1P refers to the battery configuration. There are 10 groups of 1 parallel cell connected in series.

12) The RMS error is equal to $\sqrt{\frac{\sum_{n=1}^N (\theta_n - \hat{\theta}_n)^2}{N}}$, where θ is the actual SOC, $\hat{\theta}$ is the estimated SOC, and N is the number of samples.

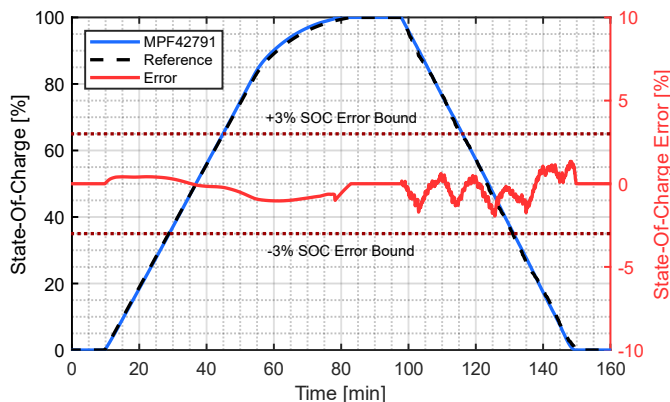


Figure 4: Combined MP2787 + MPF42791 Performance for a CC/CV Charge and Dynamic Discharge (Ambient Temperature = 25°C)

Figure 5 shows the performance of the combined MP2787 and MPF42791 for the CC/CV charge and dynamic discharge cycle at an ambient temperature of 0°C. During charge, the RMS and maximum pack SOC error are 0.68% and 1.22%, respectively. During discharge, the RMS and pack SOC error are 1.15% and 2.97%, respectively.

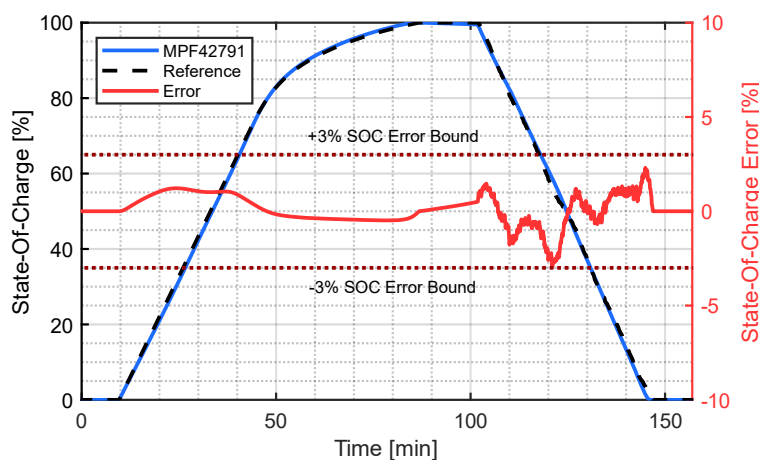


Figure 5: Combined MP2787+MPF42791 Performance for a CC/CV Charge and Dynamic Discharge (Ambient Temperature = 0°C)

Figure 6 shows the performance of the combined MP2787 and MPF42791 for the CC/CV charge and dynamic discharge cycle at an ambient temperature of 40°C. During charge, the RMS and maximum pack SOC error are 0.40% and 0.60%, respectively. During discharge, the RMS and pack SOC error are 0.77% and 1.89%, respectively.

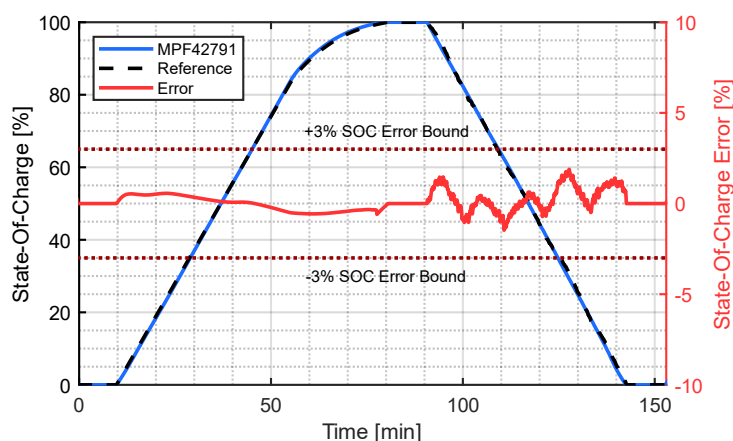


Figure 6: Combined MP2787+MPF42791 Performance for a CC/CV Charge and Dynamic Discharge (Ambient Temperature = 40°C)

Performance Summary

This section provides a summary the combined MP2787 and MPF42791 real-world performance. Table 1 shows a summary of the pack SOC performance metrics for a 10S1P battery.

Table 1: MPF42791 SOC Root-Mean-Squared (and Maximum) Error

Test Case	0°C	25°C	40°C
CC/CV charge	0.68% (1.22%)	0.61% (1.03%)	0.40% (0.60%)
Dynamic discharge	1.15% (2.97%)	0.78% (1.94%)	0.77% (1.89%)

FUNCTIONAL BLOCK DIAGRAM

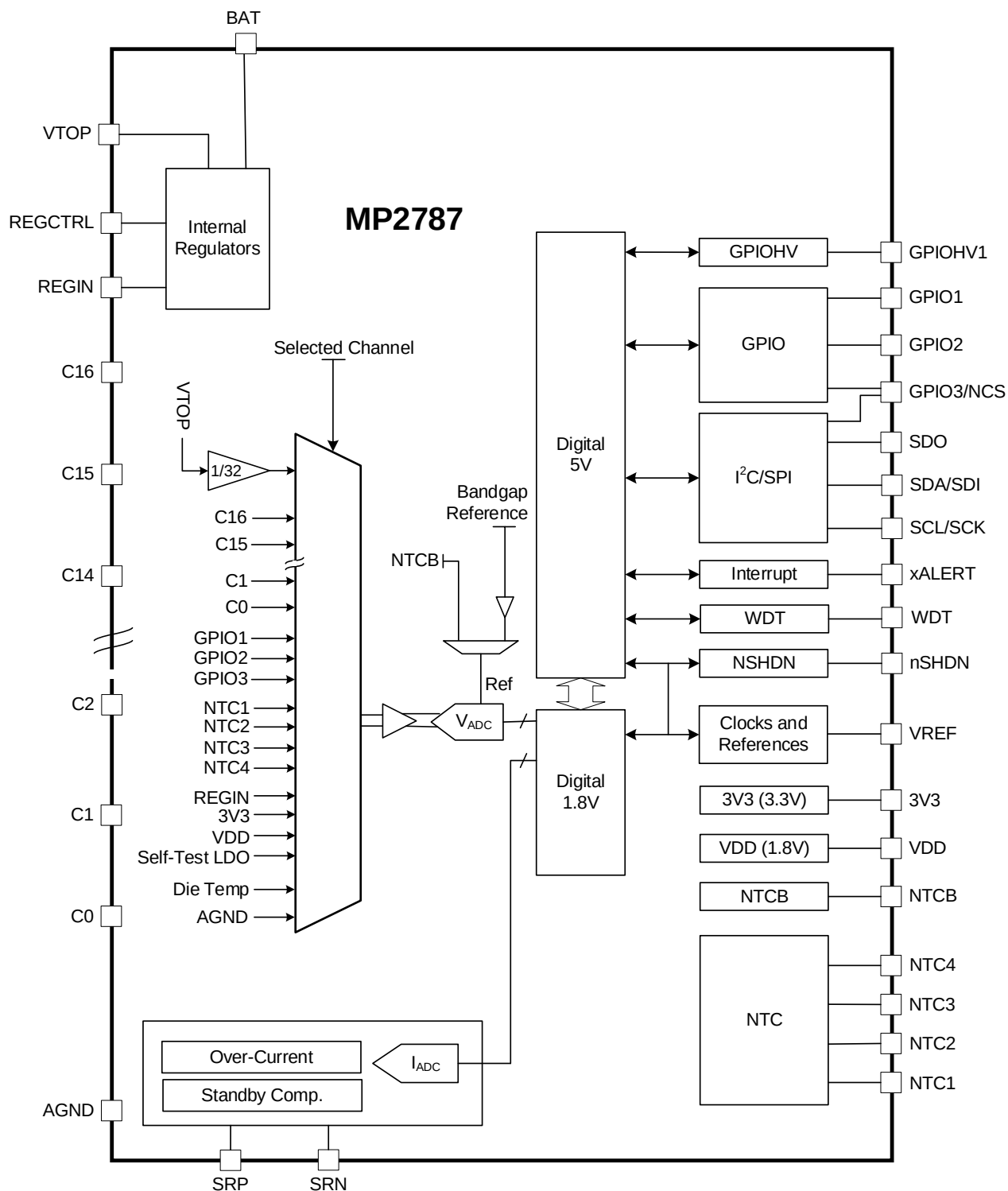


Figure 7: Functional Block Diagram

OPERATION

Main Modes

The MP2787 operates in two main modes, which are listed and described below.

Shutdown Mode

It is vital to preserve a battery's capacity. Shutdown mode minimizes the amount of leakage from the battery pack, which extends the battery pack's shelf storage life.

In shutdown mode, the communication interface is unavailable and REGIN is loosely regulated, which means its voltage is below the normal voltage without any load capacity. However, it is possible to keep 3.3V active in shutdown mode with a slight margin for current consumption.

Pull the NSHDN pin to ground to enter shutdown mode.

Active Mode

To enter active mode from shutdown mode, pull up the NSHDN pin and wait at least 5ms before issuing an I²C or SPI command. If any functional commands are required (e.g. high-resolution voltage scanning, Coulomb counting, hardware monitoring, open wire, cell-balancing), then the I²C or SPI bus must be idle for at least 200μs after the functional command is enabled.

Active mode is characterized by the following conditions:

- The communication interface is enabled.
- The low-dropout (LDO) regulators (REGIN, 3.3V and 1.8V) are activated.
- The hardware monitoring alarm can be enabled.

Registers (Default and Lock)

Most of the configuration settings and options have a configurable default value. Certain registers can be configured independently, or they can be locked in read-only mode. This prevents critical safety features from being changed.

GPIO Pins

The GPIO pins can be directly controlled by the microcontroller unit (MCU) through registers, or they can be assigned to special functions. The GPIO pins have the following functions:

- Push-pull output: The pull-up voltage can be configured to REGIN or 3V3.
- Pull-up capability: Connect a 20kΩ pull-up resistor to REGIN or 3V3.
- Digital input.
- Analog input: Can act as a buffered analog-to-digital converter (ADC) input with a 0V to 3.3V range.

In addition to the GPIO capability, the GPIO3 pin can be configured to initiate automatic cell-balancing (as an input), or indicate over-current (OC) alarm status (as an output).

GPIOHV1 Pin

The GPIOHV1 pin can drive the output to VTOP or ground, or be placed in a high-impedance (Hi-Z) mode. GPIOHV1 can also be set as an input and can read the logic value.

WDT Pin

The WDT pin is controlled by the watchdog timer (WDT), and provides the following functions:

- Toggles with a high pulse when a watchdog event occurs to reset an external IC (e.g. the host MCU).
- Triggers a self-reset to the IC by bringing the device back to its default values.
- Can be pulled high externally to reset the device.

Alert (xALERT)

The xALERT pin can be configured to be an active high (3.3V) or active low interrupt pin. When set to active low, xALERT goes low if there is a pending interrupt. When set to active high, this pin goes high if there is a pending interrupt.

Alarms and other events trigger this pin, but certain bits must be enabled to ensure that only the relevant sources generate an interrupt.

Monitoring and Alarms

If the working conditions of the battery pack in active mode exceed the cell's safe operating area (SOA), then the hardware monitoring function is activated, and the MP2787 generates an alarm.

Over-Current (OC) Alarms

All OC monitoring is performed on a single OC analog comparator. Each OC limit has a configurable deglitch time between 100 μ s and 25.5ms.

Use `CURR_MONT_EN` to enable current monitoring, which includes OC monitoring. `OC_DCHG_EN_CTRL` and `OC_CHG_EN_CTRL` are the enable bits for discharge OC and charge OC monitoring, respectively.

Once OC monitoring is enabled, the interrupt and OC alarm polices can be set.

Discharge Over-Current (OC)

Table 2 shows the resolution and range for the discharge OC limit.

Table 2: Discharge Over-Current Limit

Discharge OC	Limit		Deglitch
	RNG = 0	RNG = 1	
LSB	5.5mV	16.5mV	200 μ s
FSR	176mV	528mV	25.4ms

Where LSB is the least significant bit, and FSR is the full-scale range.

Charge Over-Current (OC)

Table 3 shows the resolution and range for the charge OC limit.

Table 3: Charge Over-Current Limit

Charge OC	Limit		Deglitch
	RNG = 0	RNG = 1	
LSB	2.5mV	7.5mV	200 μ s
FSR	80mV	240mV	25.4ms

Voltage Alarms

Use `VOLT_MONT_EN` to enable voltage monitoring. When enabled, an autonomous hardware state machine periodically schedules the conversion for all relevant channels, and the results are internally checked and compared to the limits.

The deglitch filters described refers to the number of consecutive readings, during which the relevant channel exceeds its thresholds. The interval between readings is typically 968ms, which can be configured via `MONITOR_CFG`.

Cell Under-Voltage (UV) and Over-Voltage (OV) Thresholds

When fewer than 16 cells are used, only the cells enabled by the `CELL_S_CTRL` register are monitored for under-voltage (UV) and over-voltage (OV) conditions.

These thresholds are available in the multiple-time programmable (MTP) memory, which means they can be customized during battery pack assembly for different projects. Table 4 lists the values for cell OV and UV thresholds.

Table 4: Cell OV and UV Thresholds

Cell OV/UV	Limit	Hysteresis	Deglitch
LSB	19.5mV	19.5mV	1 reading
FSR	4.98V	292.5mV	16

Pack Under-Voltage (UV) and Over-Voltage (OV) Thresholds

The pack OV and UV thresholds are monitored by the VTOP pin. These thresholds are available in the MTP, which means they can be customized during battery pack assembly for different projects (see Table 5).

Table 5: Pack OV and UV Thresholds

Pack OV/UV	Limit	Hysteresis	Deglitch
LSB	19.5mV	78mV	1 reading
FSR	80V	4.922V	16

Die Temperature

The die temperature alarm has two thresholds. The first is a digital threshold that can be configured between 70°C and 120°C. The second is an analog threshold that forces the device to shut down at 140°C.

The digital die temperature threshold is typically used as an early warning, since the analog threshold forces the IC to shut down. Until the analog OT condition is resolved, the IC remains in a forced shutdown state, and the communication interface is unavailable.

Dead Cells

The pack is monitored at a cell level for dead cells. A dead cell is defined as any cell whose voltage falls below the manufacturer's specified final discharge voltage, in which continued charging should be avoided. If dead cell detection is enabled, the MP2787 checks if any cell is below the configurable threshold (CELL_DEAD_LIMIT), which is typically much lower than the cell's UV threshold.

Once a dead cell condition is detected, the persistent dead cell flag (CELL_DEAD_LOG_STS) is set to 1. When the default values for CELL_DEAD_EN and

VLOT_MONT_EN are enabled via the one-time programmable (OTP) memory, this flag remains at 1 even if the device enters and exits shutdown mode. This flag can be cleared by the CELL_DEAD_DET_CLEAR command.

Negative Temperature Coefficient (NTC) Temperature

The negative temperature coefficient (NTC) voltages are checked during the automatic hardware voltage monitoring sequence.

Figure 8 shows the block diagram of the NTC-sensing architecture.

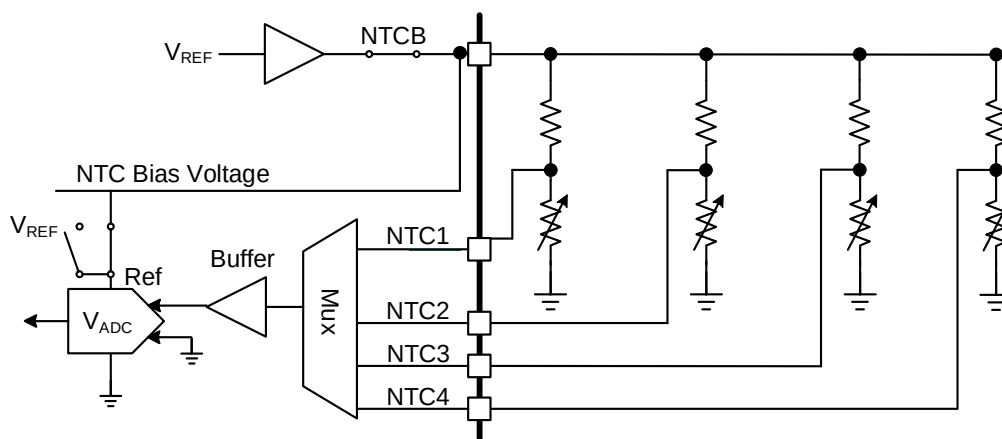


Figure 8: Architecture for NTC ADC Acquisition

All NTC channels are monitored in a ratiometric conversion. This means that the ADC reference is switched to NTCB for the NTC channels. In this scenario, all effects related to NTCB drifts due to the temperature and operating conditions are removed from the reading.

NTC1~4 have limits that can be set according to two modes:

1. **Cell monitor mode:** The NTC monitors the temperature of the lithium cell in the pack. The MP2787 provides independent cell temperature limits (hot and cold) for the charging and discharging currents.

These thresholds are available in the MTP, and can be customized during battery pack assembly for different projects.

2. **PCB monitor mode:** The NTC monitors the PCB temperature. A standard hot temperature limit is provided for both the charge and discharge current.

Note that using NTCs results in a lower voltage threshold for OT (hot) conditions and a higher voltage threshold for under-temperature (cold) conditions.

Mismatched Cells

Excessively mismatched or unbalanced cells can be detected by monitoring the maximum and minimum cell voltages.

If the difference between the maximum cell voltage and the minimum cell voltage exceeds a defined threshold, then a cell mismatch status is triggered. The mismatched cell with the higher voltage is reported as an individual flag. The cell with the minimum voltage is reported in the CELL_MSMT_LOWER register.

Diagnostics and Integrity

LDO (REGIN, 3.3V and 1.8V) Rail Monitoring

The LDO UV threshold is provided for the REGIN, 3.3V and 1.8V rails. This UV threshold is compared to the ADC readings with the same

refresh interval as the other voltage monitors, thus it measures the nominal rail voltage rather than detecting spike voltage.

Analog-to-Digital Converter (ADC) Self-Test Conversion

The ADC converts a known voltage value and determines whether the conversion result is within a predefined window that accounts for tolerance and temperature shifts.

One-Time Programmable (OTP) Cyclic Redundancy Check (CRC)

An OTP cyclic redundancy check (CRC) is executed when the device exits shutdown mode. If the CRC fails, the device can be configured to trigger the OTP CRC alarm.

Communication Cyclic Redundancy Check (CRC)

Typically, the CRC detects errors in exchanged data. The MP2787 supports CRC for exchanged data. When this functionality is enabled, the following is possible:

- Write a transaction to 1 register (2 bytes) to append a 1-byte CRC. A correct CRC is required for a successful write transaction.
- Read a transaction for 1 register (2 bytes) to append a 1-byte CRC.
- The read transaction length can be temporarily increased to 126 bytes using a dedicated register. The CRC is appended at the end of the transaction.

The communication CRC is implemented by the CRC-8 algorithm. The CRC can detect 1 incorrect bit. However, all CRC techniques encounter limitations when there are more incorrect bits.

If more than 1 bit is corrupted, the error detection depends on the specific pattern. Statistically, detection can occur, but it is not guaranteed. In general, the ability to detect errors decreases when there is more data to check. For more details, see the I²C Read section on page 29 and the SPI Read section on page 31.

Watchdog Timer

The watchdog timer monitors communication with the I²C or SPI interface. If there is no write to the WDT_RST register within a certain interval, the timer is triggered. The watchdog has a bark-bite style.

The bark notifies the device when the MCU has failed to clear the watchdog timer. This results in a bark timeout, which can be configured to trigger the xALERT pin. The WDT bark counter setting is 25ms per LSB, with up to 3.2s maximum.

After a bite timeout, there is an optional feature to reset the device's registers to the default values. The WDT bite counter setting is 25ms per LSB, with up to 3.2s maximum.

Using the bidirectional, configurable WDT, it is possible to set up the IC to self-reset and/or to reset an external IC if its reset trigger is connected to the WDT pin.

Interrupts

Interrupt statuses are split between two register addresses (RD_INT0 and RD_INT1). Each interrupt source has a matching enable bit and a dedicated clear bit to clear the interrupt. Table 6 on page 22 shows the main interrupts. Table 7 on page 23 shows additional interrupts.

Table 6: Main Interrupts (RD_INT0)

Condition	Flag	Description	Related Status Register(s)
Cell OV	CELL_OV_INT_STS	Unified flag for cell OV conditions. The related register reports each cell's OV flag, which is checked to determine which cell is generating the interrupt.	Flags in address, RD_CELL_OV
Cell UV	CELL_UV_INT_STS	Unified flag for cell UV conditions. The related register reports each cell's UV flag, which is checked to determine which cell is generating the interrupt.	Flags in address, RD_CELL_UV
VTOP pack OV	VTOP_OV_INT_STS	Dedicated flag for battery pack OV conditions detected on the VTOP pin.	-
VTOP pack UV	VTOP_UV_INT_STS	Dedicated flag for battery pack UV conditions detected on the VTOP pin. This is unrelated to the VTOP UV analog threshold.	-
OC	OVER_CURR_INT_STS	OC interrupt for both charge and discharge.	OC_CHG_STS, OC_DCHG_STS
NTC cold	NTC_COLD_INT_STS	Unified flag for NTC cold conditions for all NTC channels configured in cell mode. To identify which NTC is the source of the interrupt, the additional individual flags should be checked.	NTC1_CELL_COLD_STS, NTC2_CELL_COLD_STS, NTC3_CELL_COLD_STS, NTC4_CELL_COLD_STS
NTC hot	NTC_HOT_INT_STS	Unified flag for NTC hot conditions for all NTC channels configured in cell mode. To identify which NTC is the source of the interrupt, the additional individual flags should be checked.	NTC1_CELL_HOT_STS, NTC2_CELL_HOT_STS, NTC3_CELL_HOT_STS, NTC4_CELL_HOT_STS
Watchdog interrupt	WDT_INT_STS	Notification for either a bite or bark event from the communication watchdog timer.	WDT_BARKED, WDT_BITE
Scan complete	VSCAN_DONE_INT_STS	Notification that the high-resolution voltage ADC scan has finished converting of all the channels on its list.	-
CC done	CC_ACC_INT_STS	Flag reporting that a new Coulomb counting accumulation value has been updated to CC_ACC_LSBS and CC_ACC_MSBS.	-
Pack current	PACK_CURRENT_INT_STS	Interrupt indicating a change in the battery pack current range (discharge, standby, or charge).	PACK_CURRENT_STATUS

Table 7: Additional Interrupts (RD_INT1)

Condition	Flag	Description	Related Status Register(s)
Balancing complete	BAL_DONE_INT_STS	Balancing has been completed.	Flags in address, BAL_STS
Self-test fail	SELF_TEST_INT_STS	The ADC is failing its self-diagnostic test and/or the conversion of this value is outside the specified boundaries.	SELF_TEST_STS_OV, SELF_TEST_STS_UV
Scheduler error	FSM_ERROR_INT_STS	The scheduler was busy when a new feature command was requested, and concurrent operation is not supported. For example, the device may report that an MCU conversion command is ignored since open-wire detection is running, or it may report that a cell-balancing command is ignored since open-wire detection is running.	-
PCB temperature hot	PCB_MNTR_HOT_INT_STS	Unified flag for NTC hot conditions for all NTC channels configured in PCB mode. To identify which NTC is the source of the interrupt, the individual flags should be checked.	NTC1_PCB_MNTR_HOT_STS, NTC2_PCB_MNTR_HOT_STS, NTC3_PCB_MNTR_HOT_STS, NTC4_PCB_MNTR_HOT_STS
Die temperature	DIE_TEMP_INT_STS	The die temperature is too high, and is reported by the digital die temperature check.	-
Mismatched cells	CELL_MISMATCH_INT_STS	The voltage difference between cells is too great.	Flags in address, RD_CELL_MSMT
Dead cell	CELL_DEAD_INT_STS	The dead cell threshold has been reached.	Flags in address, RD_CELL_DEAD
Open wire	OPEN_WIRE_INT_STS	Open-wire detection is complete. The open-wire interrupt status register must be checked to identify if there are any disconnected wires.	Flags in address, RD_OPENH, RD_OPENL
VDD	VDD_INT_STS	The VDD rail is below the defined UV threshold.	-
3V3	3V3_INT_STS	The 3.3V rail is below the defined UV threshold.	-
REGIN	REGIN_INT_STS	The REGIN rail is below the defined UV threshold.	-
OTP CRC	OTP_CRC_EVENT_INT_STS	The CRC stored in the OTP does not match the computed CRC value from OTP memory readback. The manual CRC check is completed.	-

The interrupt source can be configured using a TYPE selector. Depending on the interrupt, the TYPE selector provides a few options:

- Level (high)
- Rising edge
- Falling edge
- Rising and falling edge

Each interrupt that is related to an alarm flag follows the processes described above.

Regulators and Power Domains

There are three main power rails:

- REGIN (5V): Supplies power to the analog front-end (AFE) analog circuitry
- 3V3 (3.3V): Powers an external MCU
- VDD (1.8V): Internal use only; powers the digital domain

The 3.3V rail can be configured to be on or off in shutdown mode.

Open-Wire Detection

The open wire state machine automatically controls the pull-up and pull-down currents sources during open-wire detection, and detects the voltage changes on each cell. The cells that are detected can be set by the CELLS_CTRL register.

The open-wire detection current uses a pair of 100µA pull-up/-down currents. The interval can be configured to be between 1ms and 16ms, with a 1ms resolution. The default is 8ms.

The threshold that determines how much change can be accepted for an open circuit is a configurable voltage threshold ranging between 39.06mV and 625mV, with 39.06mV steps. The default is 195mV. The host MCU can trigger open-wire detection with a dedicated command register. If multiple open wires are present simultaneously, then the detection logic reports at least one open wire, which tells the device to avoid using the overall battery pack (see Figure 9).

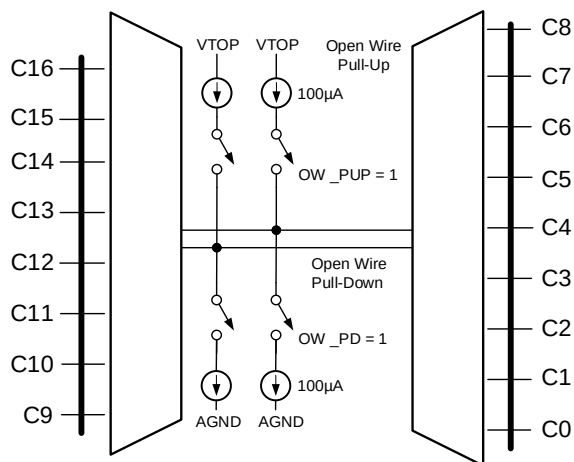


Figure 9: Open-Wire Detection Architecture

Cell-Balancing

The battery reaches its end of life (EOL) when the capacity of the weakest cell is too low to be usable. Unbalanced cells yield the same effect, with one cell at a higher state-of-charge (SOC), and another at a lower SOC. To reduce the effects of the issue, it is critical to balance lithium cells.

Unbalanced cells can be caused by cell temperature differences, different self-discharge rates, and general production tolerances that

affect cell chemistry and alter aging rates among strings of parallel cells.

Cell-balancing can extend the usable life of the battery pack by keeping the voltage of each cell within the defined SOA.

Internal MOSFET Balancing

The MP2787 supports direct cell balancing, without the use of external MOSFETs or bipolar junction transistors (BJTs), for currents up to about 58mA (with a 4V cell voltage). The balance current for each cell is dictated by the combined filter resistor and the on resistance of the internal balancing MOSFET ($R_{(DS)ON}$). The MP2787 can simultaneously balance only even or odd cells. Internal logic handles the sequencing between even and odd cells.

External Balancing

A balancing current exceeding 58mA can be achieved by adding external balancing MOSFETs or BJTs. This balancing current is limited by board thermals and the wiring resistance. For external MOSFETs, it is also limited by the MOSFETs' on resistance ($R_{(DS)ON}$) at the appropriate gate voltage (V_{GS}). For external BJTs, it may be limited by the DC current gain (h_{FE}). For external MOSFETs, a minimum gate threshold voltage of 1.8V is required (assuming a 4.2V lithium cell) to turn on the balancing MOSFETs.

Manual Cell-Balancing

The cells that must be balanced are manually marked by writing to the appropriate register (BAL_LIST).

The balancing time for marked cells can be configured via BAL_REPETITION, with 0 to 31 repetitions. When BAL_REPETITION is set to 0, there is only one 30ms execution.

The host MCU should periodically read the AFE die temperature to ensure it is within the operating range, as an excessive temperature can lead to over-temperature (OT) shutdown.

Automatic Cell-Balancing

Automatic cell-balancing offers the following configurations:

1. Enable automatic cell balancing by setting register BALANCE_MODE_REG = 1.

2. Configure the method for initiating automatic cell-balancing via register `BALANCE_MODE_CTRL`. The MP2787 can be configured to use register control (`BALANCE_GO`) or GPIO3 control to initiate automatic cell-balancing.
3. Disable constant automatic balancing by setting register `AUTO_BAL_ALWAYS` = 0. Register `BAL_REPETITION` sets the number of cell-balancing iterations. The cell-balancing repetition number is ignored if `AUTO_BAL_ALWAYS` = 1, and the device constantly balances the cells until the `AUTO_BAL_ALWAYS` is set to 0, or until the balancing list is empty.
4. Set the balancing threshold via register `BAL_MSM_TH`. This is the minimum voltage difference between the current cell voltage and the lowest cell voltage to be eligible for cell balancing. This value ranges between 19.5mV and 87.855mV, with 9.765mV steps.
5. Set the minimum balancing voltage via register `CELL_BAL_MIN`. This is the

minimum cell voltage for a cell to be eligible for balancing. This value ranges from 2500mV to 4961mV with 39mV steps.

6. Set `ABAL_ON_CHARGE` = 1 to enable automatic cell-balancing when the state of the current is in charge mode. Set `ABAL_ON_STBY` = 1 to enable automatic cell-balancing when the state of the current is in standby mode.

If the state of the current is in discharge mode, the automatic cell-balancing will be skipped.

The die temperature threshold can be used to prevent or suspend balancing. The MP2787 can be configured to suspend constant automatic cell-balancing (`AUTO_BAL_ALWAYS` = 1), when the die temperature threshold has been exceeded (see `STOP_ON_HOT` on page 89). Once the OT condition is removed, the constant automatic cell-balancing resumes.

Figure 10 shows the cell-balancing list update for automatic cell-balancing. Figure 11 on page 26 shows the cell-balancing sequence for the cell-balancing list.

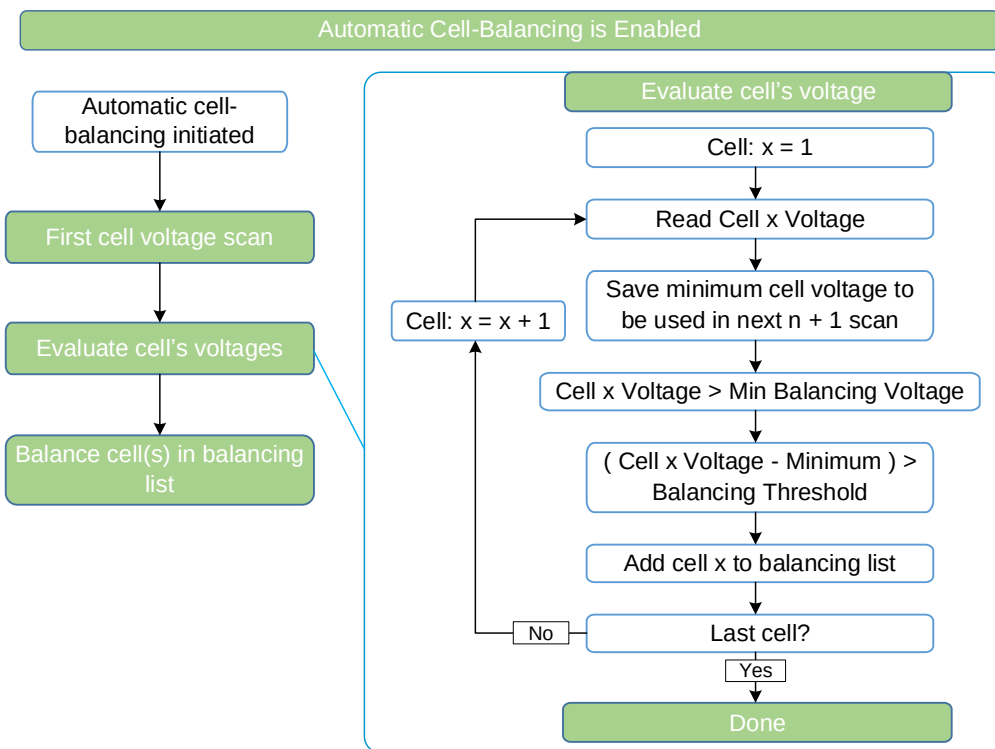


Figure 10: Automatic Cell-Balancing (Cell Voltage Evaluation)

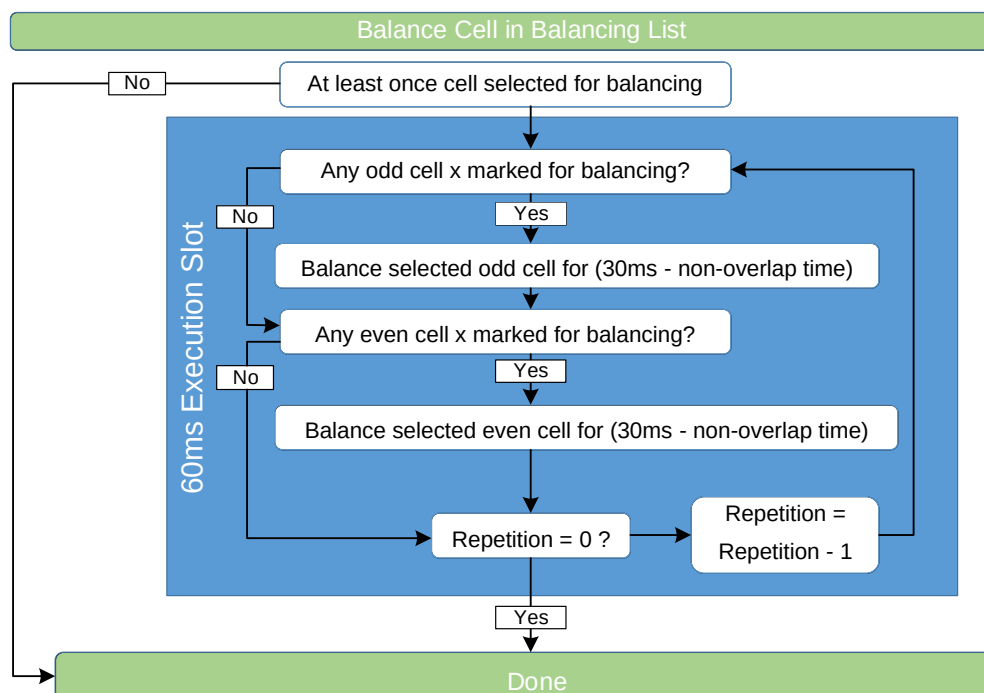


Figure 11: Executing Balancing for Cell in Balancing List

Supported Concurrency

Table 8 shows which features can be enabled once a feature is active. When multiple features are enabled simultaneously, some remaining features can still be activated.

It is possible to use a high-resolution voltage scan during cell-balancing, but care should be taken, as voltage readings on the cell being balanced and adjacent cells are affected.

Table 8: Supported Concurrencies

Active Task	Triggered By	Concurrent Features
High-resolution voltage scan	ADC_SCAN_GO	- Cell-balancing - Coulomb counter
Cell balancing	BALANCE_GO	- High-resolution voltage scan - Coulomb counter
Open wire	OPEN_WIRE_GO	N/A
Coulomb counter	CC_EN	- High-resolution voltage scan - Cell-balancing

The command sequence to the MP2787 should be controlled in the following manner:

- During high-resolution voltage scanning, wait to set `ADC_SCAN_GO = 0` until `SCAN_DONE_STS` or `SCAN_ERROR_STS` updates to 1.
- During cell-balancing, wait to set `BALANCE_GO = 0` until `BAL_DONE_STS` or `BAL_ERROR_STS` updates to 1. To stop automatic cell-balancing, set `AUTO_BAL_ALWAYS = 0`, wait until `BAL_DONE_STS` or `BAL_ERROR_STS` updates to 1, and then set `BALANCE_GO = 0`.
- When conducting open wire checks, wait to set `OPEN_WIRE_GO = 0` until `OPEN_WIRE_DONE_STS` or `OPEN_WIRE_ERR_STS` updates to 1.
- For Coulomb counting, wait to set `CC_EN = 0` until `CC_DONE` or `CC_ERROR_STS` updates to 1. To stop the back-to-back Coulomb counter, set `CC_B2B_ACC_CTRL = 0`. Wait until `CC_DONE` or `CC_ERROR_STS` updates to 1, and then set `CC_EN = 0`.

For these command sequences, the status can be determined via polling or via interrupts (VSCAN_DONE_INT_STS, BAL_DONE_INT_STS, OPEN_WIRE_INT_STS, and CC_ACC_INT_STS). See the Interrupts section on page 21 for more details.

Coulomb Counter

The MP2787 is capable of accumulating up to 2s worth of the current ADC reading.

When the accumulation is complete, the value is stored in the CC_ACC_LSBS (lower 2 bytes) and CC_ACC_MSBS (upper 2 bytes) registers for MCU readback, and an interrupt is posted. The sequence below provides the typical steps to handle a Coulomb counting conversion:

1. Set CC_B2B_ACC_CTRL to enable back-to-back conversions, and set CC_ACC_INT_EN to enable interrupt.
2. Set CC_EN to enable starting the conversion.
3. Accumulation reaches the amount specified in CC_INT_CNT.
4. CC_ACC_LSBS and CC_ACC_MSBS are updated to the values reported in CC_RT_ACC_LSBS and CC_RT_ACC_MSBS.
5. CC_ACC_INT_STS reports a detected interrupt.
6. When back-to-back Coulomb counting accumulation is disabled, the CC_DONE flag can be checked to verify that accumulation has been completed. Readings are available in CC_ACC_LSBS and CC_ACC_MSBS.
7. If back-to-back conversion is enabled, the CC_RT_ACC_xSBS registers are automatically cleared and Coulomb counting restarts. The integrated value for completed accumulation can be read from CC_ACC_xSBS. If CC_ACC_INT_EN is enabled, a Coulomb counting done interrupt is triggered when CC_ACC_xSBS updates.
8. If back-to-back Coulomb counting accumulation is disabled, CC_EN can be set to disable clearing CC_DONE. To stop Coulomb counting if back-to-back conversions are enabled, the back-to-back flag should be cleared to prevent the ongoing conversion from triggering a new conversion.

9. If an CC_ACC_INT_STS interrupt is detected, it should be cleared using the CC_ACC_STS_INT_CLEAR command.

High-Resolution Voltage ADC Scan for the Microcontroller Unit (MCU)

A high-resolution voltage scan can convert the following classes of inputs:

- Die temperature voltage
- NTC1, NTC2, NTC3, and NTC4 voltages
- Cell voltages
- VTOP pin voltage
- GPIO voltages (GPIO1, GPIO2, and GPIO3)
- Regulators (VDD, 3.3V, and REGIN)

If a class of inputs is not enabled, the scan skips to the next class. Control bits are provided to enable individual channels in each class. The device is notified when the scan is complete, and the results are available for readback.

Synchronized Voltage and Current Reading

For cell and pack voltages, the exact, synchronized current readings can be read in dedicated registers paired with the matching voltage reading. Synchronous current readings can be enabled using CELL_SYNC_EN and VTOP_SYNC_EN.

The synchronized voltage and current reading can be requested concurrently with Coulomb counting monitoring.

Non-Volatile Memory (NVM) Configuration

The default values of most registers can be configured through the non-volatile memory (NVM). Some registers are OTP, while others are MTP. The MTP registers can be configured up to three times. These values can be locked to allow the same version of the IC to be adjusted for similar projects with minor differences.

To configure the MTP, apply 7.5V to the NSHDN pin and follow the steps below:

1. Ensure that the NSHDN pin is set to 7.5V.
2. Write the appropriate value to the register that allows MTP.
3. Write prior to the store command with the following command access code: 0xA5B6 (REG0xB9 = 0xA5B6).

4. Send the command to store the register's current value to the NVM by writing 1 to STORE_MTP_CMD (0xB8, bit[3]).
5. Wait for STORE_IN_PROGRESS to go back to 0.
6. Recover the NSHDN pin to 3.3V.

I²C Interface

The MP2787 can use an I²C interface to flexibly set parameters and report device statuses instantaneously. The I²C is a two-wire serial interface with two bus lines: a serial data line (SDA) and a serial clock line (SCL). Both SDA and SCL are open drains, and they must be connected to the positive supply voltage via a pull-up resistor.

The IC operates as a slave device that receives control inputs from the master device, such as a MCU. The SCL is always driven by the master device. The I²C interface supports both standard mode (up to 100kbit/s) and fast mode (up to 400kbit/s).

All transactions begin with a start (S) command and are terminated by a stop (P) command. Start and stop commands are always generated by the master. A start command is defined as a high-to-low transition on the SDA while the SCL is high. A stop command is defined as a low-to-high transition on the SDA while the SCL is high (see Figure 12).

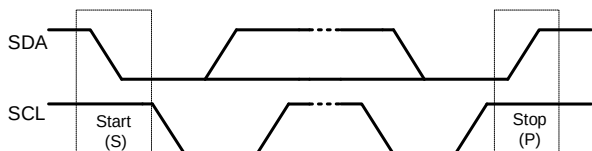


Figure 12: Start and Stop Commands

For data validity, the data on the SDA must be stable during the clock's high period. The high or low state of the SDA can only change when the clock signal on the SCL is low (see Figure 13).

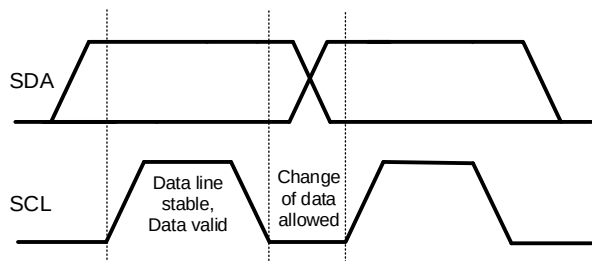


Figure 13: Bit Transfer on the I²C Bus

Every byte on the SDA must be 8 bits long. The number of bytes that can be transmitted per transfer is unrestricted. Data is transferred with the most significant bit (MSB) first.

Each byte must be followed by an acknowledge (ACK) bit, which is generated by the receiver to signal to the transmitter that the byte was successfully received.

The ACK signal occurs when the transmitter releases the SDA line during the acknowledge clock pulse. This allows the receiver to pull the SDA line low. The SDA line stays low during the high period of the ninth clock.

If the SDA line is high during the 9th clock pulse, this is defined as a not acknowledge (NACK) signal. The master can then generate either a stop command to abort the transfer, or a repeated start (Sr) command to start a new transfer.

After the start command, a slave address is sent. This address is 7 bits long, followed by an 8th data direction bit (R/W). A 0 indicates a transmission (write), and a 1 indicates a request for data (read). Figure 14 shows the address bit arrangement.

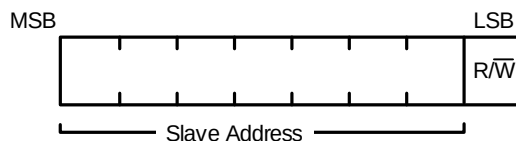


Figure 14: 7-Bit Address

Figure 15 on page 29 shows a data transfer on the I²C bus.

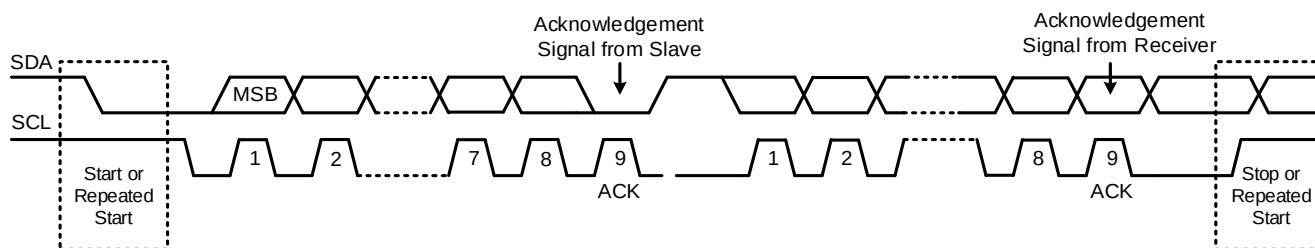


Figure 15: Data Transfer on the I²C Bus

I²C Read

When CRC is enabled, the payload is limited to 2 bytes by default (see Figure 16 on page 30). This value can be increased to 63 words (126 bytes) using the XFR_NUM_RD_WORDS register. Since XFR_NUM_RD_WORDS retains the new value only for the next transaction, ensure that the next transaction uses the increased payload count.

CRC in an I²C Read Transaction

After completing the request for a read transaction, the host MCU must ensure that the CRC matches the value provided in the read transaction to confirm that no bits were corrupted during the transmission.

The CRC-8 algorithm follows the polynomial $(1 + x^1 + x^2 + x^8)$, and is applied byte-wise to the bytes, following the order in which the bytes are transmitted or received. If it is bit-wise, the MSB is first processed in each byte. An example of this sequence is shown below:

1. Slave address byte + write (0x02)
2. Register address byte (0x00)
3. Slave address byte + read (0x03)
4. Register address byte (0x00)
5. Word 1 - byte 1 / bits[7:0] (0x7C)
6. Word 1 - byte 2 / bits[15:8] (0x00)

This sequence results in CRC = 0x36 being appended as the last CRC byte.

Although this byte of the fourth sequence does not exist in the actual read sequence, it must be added in the CRC calculation.

I²C Write

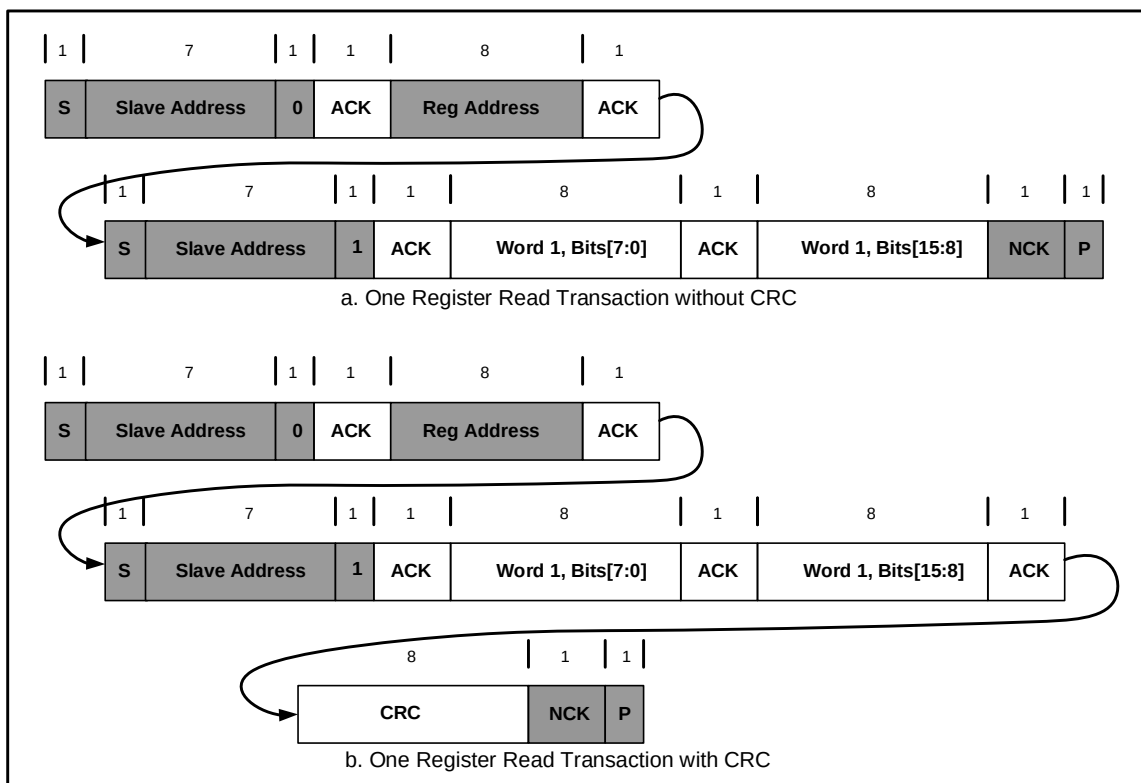
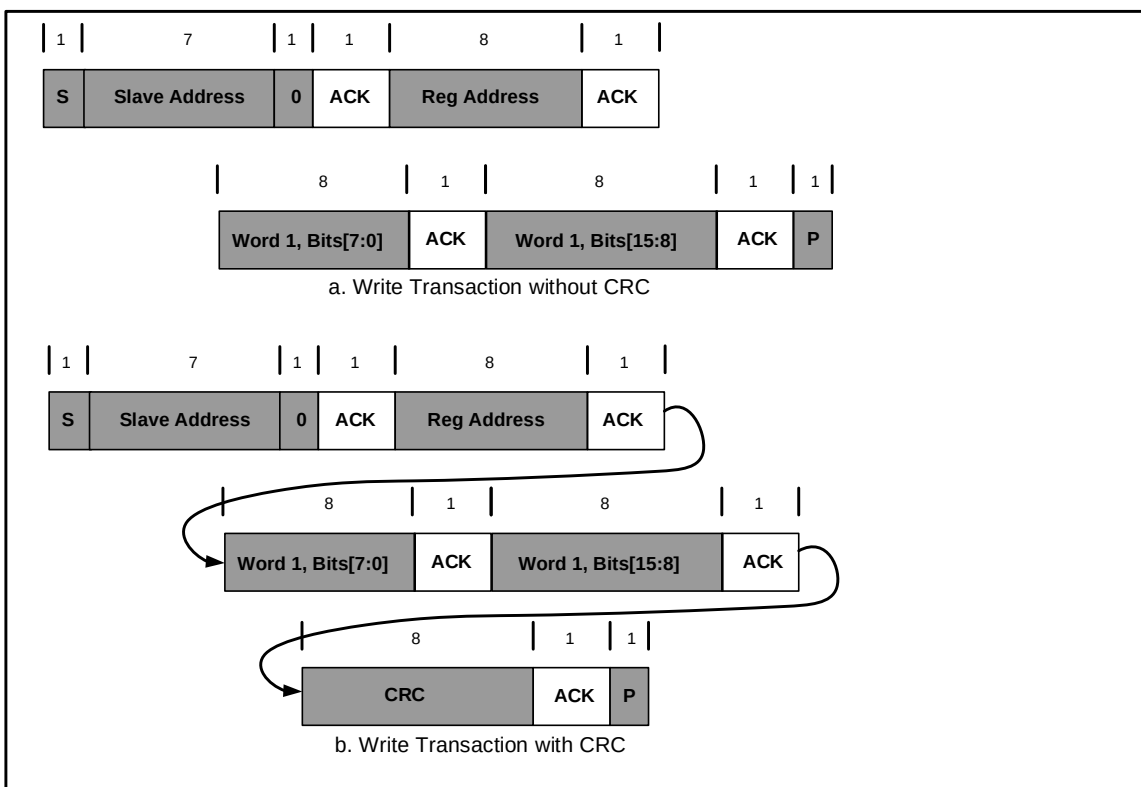
Figure 17 on page 30 shows how the byte is ordered in a write transaction. When CRC is enabled, the targeted register address is modified according to the payload only if the CRC results match.

CRC in an I²C Write Transaction

The CRC-8 algorithm follows the polynomial $(1 + x^1 + x^2 + x^8)$, and is applied byte-wise to the bytes following the order in which they are transmitted or received. If it is bit-wise, the MSB is first processed in each byte. An example of this sequence is shown below:

1. Slave address byte + write (0x02)
2. Register address byte (0x00)
3. Word 1 - byte 1 / bits[7:0] (0x7C)
4. Word 1 - byte 2 / bits[15:8] (0x00)

This sequence results in CRC = 0x72, which the host MCU should append as the last CRC byte to ensure a successful transaction.


Figure 16: I²C Single Address Read

Figure 17: I²C Write Transaction

Serial Peripheral Interface (SPI)

The MP2787 has an interface that is compatible with serial peripheral interfaces (SPIs). This interface is configured to operate using clock phase (CPHA) = 1 and clock polarity (CPOL) = 1. Consequently, data on SDI must be stable during SCK's rising edge. Words are transferred with the MSB first.

During a write transaction, the data value on SDI is latched to the device on SCK's rising edge.

During a read transaction, the bit stream is presented on SDO with the MSB first, and is valid during SCK's rising edge, while the SDO transitions on SCK's falling edge.

nCS must remain low for the entire command sequence duration, including the time between the command byte and subsequent data. During a write command, data is latched on nCS's rising edge (see Figure 18).

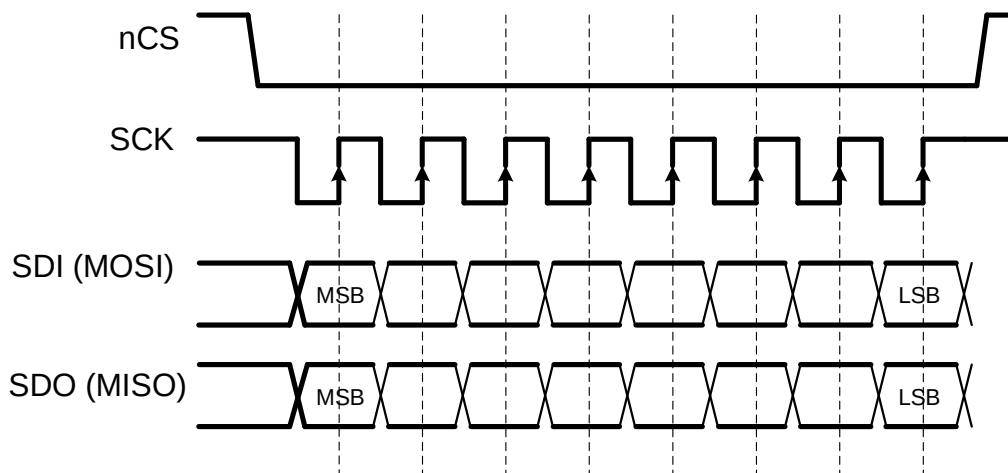


Figure 18: SPI Signal Sequencing

SPI Data Protocol

For a successful SPI transaction, nCS must go low, and there must be a successful match between the transaction slave address and the internally configured slave address (located in register `DEVICE_ADD`). The MP2787's SPI requires a specific transaction structure.

SPI Read

Read transactions should have the fields arranged with a matching order for the slave address, read bit, register address, and data payload (see Figure 19 on page 32).

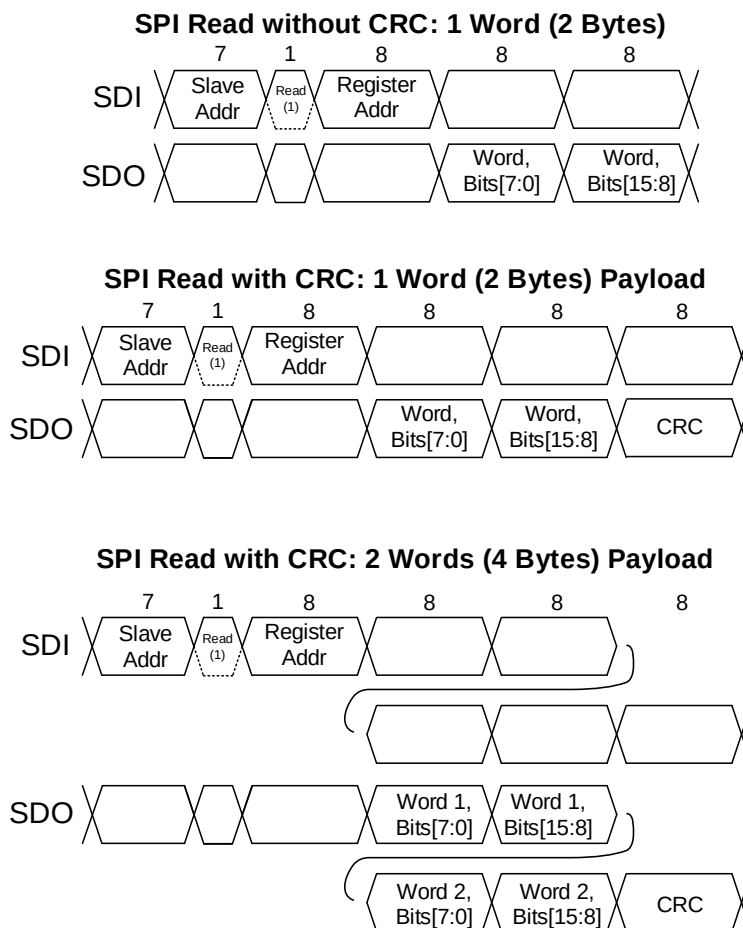


Figure 19: SPI Read Transaction

When CRC is enabled, the payload is limited to two bytes by default. This value can be increased to 63 words (126 bytes) using the XFR_NUM_RD_WORDS register. Since XFR_NUM_RD_WORDS retains the new value only for the next transaction, ensure that the next transaction uses the increased payload count.

CRC in an SPI Read Transaction

After completing the request for a read transaction, the host MCU must ensure that the CRC matches the value provided in the read transaction to confirm that no bits were corrupted during the transmission.

The CRC-8 algorithm follows the polynomial $(1 + x^1 + x^2 + x^8)$, and is applied byte-wise to the bytes, following the order in which bytes are transmitted or received. If it is bit-wise, the MSB is first processed in each byte. An example of this sequence is shown below:

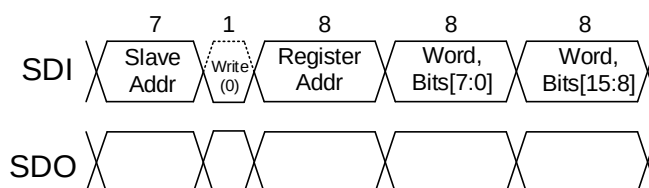
1. Slave address byte + read (0x03)
2. Register address byte (0x00)
3. Word 1 - byte 1 / bits[7:0] (0x7C)
4. Word 1 - byte 2 / bits[15:8] (0x00)

This sequence results in CRC = 0x64 being appended as the last CRC byte.

SPI Write

Write transactions should have the fields arranged with a slave address, write bit, register address, and data payload (see Figure 20 on page 33). When CRC is enabled, the SPI write payload is limited to 2 bytes and targeted register addresses are modified according to the payload only if the CRC matches.

SPI Write without CRC: 1 Word (2 Bytes)



SPI Write with CRC: 1 Word (2 Bytes)

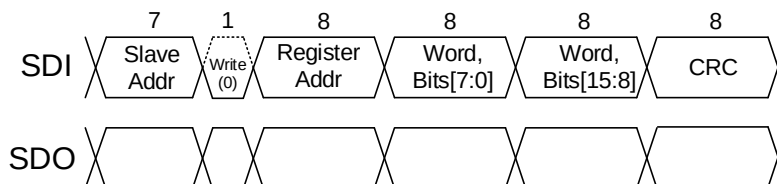


Figure 20: SPI Write Transaction

CRC in an SPI Write Transaction

The CRC-8 algorithm follows the polynomial $(1 + x^1 + x^2 + x^8)$, and is applied byte-wise to the bytes following the order in which they are transmitted or received. If it is bit-wise, the MSB is first processed in each byte. An example of this sequence is shown below:

1. Slave address byte + write (0x02)
2. Register address byte (0x00)
3. Word 1 - byte 1 / bits[7:0] (0x7C)
4. Word 1 - byte 2 / bits[15:8] (0x00)

This sequence results in CRC = 0x72, which the host MCU should append as the last CRC byte to ensure a successful transaction.

REGISTER MAP

The default slave address is 01h. The address can be configured by modifying DEVICE_ADD (register 0xA3, bits[14:8]). Once the value has been changed, the next communication should use the new address. Note that the default address may be different for the chips with different -xxxx suffixes.

All the bits in an address that are not populated by a register should be treated as reserved bits. Their values should not be modified during write operations that are meant to change bits in the same address.

Refer to the MP279x User Guide for register details and how to configure and control the MP2787.

Cell Configuration

Address Name and Location	Field Bit Position and Name	Description	Type	Reset Value (OTP/MTP)	Encoding, LSB, and Range
CELLS_CTRL: 0x00	Bits[3:0]: CELL_S_CTRL	Sets the number of stacked cells in use. 0x0: Cells 7~1 enabled 0x1: Cells 7~1 enabled 0x2: Cells 7~1 enabled 0x3: Cells 7~1 enabled 0x4: Cells 7~1 enabled 0x5: Cells 7~1 enabled 0x6: Cells 7~1 enabled 0x7: Cells 8~1 enabled 0x8: Cells 9~1 enabled 0x9: Cells 10~1 enabled 0xA: Cells 11~1 enabled 0xB: Cells 12~1 enabled 0xC: Cells 13~1 enabled 0xD: Cells 14~1 enabled 0xE: Cells 15~1 enabled 0xF: Cells 16~1 enabled	R/W (can lock to read-only)	0xF (MTP)	N/A

IC State Control

Address Name and Location	Field Bit Position and Name	Description	Type	Reset Value (OTP/MTP)	Encoding, LSB, and Range
PWR_STATUS: 0x01	Bits[9:7]: PACK_CURRENT_STATUS	Returns the pack current status. 0x1: The battery pack current is discharging 0x2: The battery pack current is in standby range, meaning it is within $\pm$ STBY_CUR_TH 0x4: The battery pack current is charging	Read-only	0x2	N/A
ACT_CFG: 0x06	Bits[2:1]: STBY_CUR_TH	Sets the standby comparator current threshold.	R/W	0x0 (OTP)	0x0: 250 μ V 0x1: 375 μ V 0x2: 500 μ V 0x3: 625 μ V
ACT_CFG: 0x06	Bit[3]: STBY_HYS	Enables the digital hysteresis on the standby current comparator. 1: Enabled 0: Disabled	R/W	0x0 (OTP)	N/A

ACT_CFG: 0x06	Bits[5:4]: MONITOR_CFG	Selects the interval used for voltage monitoring 0x0: Voltage protection readings are refreshed every 254ms 0x1: Voltage protection readings are refreshed every 492ms 0x2: Voltage protection readings are refreshed every 968ms 0x3: Not allowed	R/W (can lock to read-only)	0x2 (OTP)	N/A
HARD_CFG: 0x07	Bit[0]: VOLT_MONT_EN	1: Enables voltage monitoring 0: Disables voltage monitoring	R/W	0x0 (OTP)	0: Disabled 1: Enabled
HARD_CFG: 0x07	Bit[1]: CURR_MONT_EN	1: Enables standby current, and over-current (OC) monitoring 2: Disables standby current, and OC monitoring	R/W	0x0	0: Disabled 1: Enabled
RGL_CFG: 0x08	Bit[2]: V3P3_SHDN_EN	Enables the 3.3V regulator during shutdown mode.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
RGL_CFG: 0x08	Bit[3]: V3P3_SHOFF_STS	Reports the internal enable/disable control for the 3.3V regulator during shutdown mode. When enabled, the 3.3V rails stay on even in shutdown mode.	Read-only	0x1	0: Disabled 1: Enabled

Pins and GPIO

Address Name and Location	Field Bit Position and Name	Description	Type	Value Reset (OTP/MTP)	Encoding, LSB and Range
GPIO_STATUS: 0x0A	Bit[0]: GPIO1	Reports the GPIO1 pin status (high/low).	Read-only	0x1	0: Low 1: High
GPIO_STATUS: 0x0A	Bit[1]: GPIO2	Reports the GPIO2 pin status (high/low).	Read-only	0x1	0: Low 1: High
GPIO_STATUS: 0x0A	Bit[2]: GPIO3	Reports the GPIO3 pin status (high/low).	Read-only	0x1	0: Low 1: High
GPIO_STATUS: 0x0A	Bit[3]: GPIOHV1	Reports the GPIOHV1 pin status (high/low).	Read-only	0x0	0: Low 1: High
GPIO_OUT: 0x0B	Bit[0]: GPIO1_O	Sets the target output level for GPIO1 (high or low). This bit is effective only when GPIO1 is used as a digital output (GPIO1_IO is set to output).	R/W	0x0 (OTP)	0: Low 1: High
GPIO_OUT: 0x0B	Bit[1]: GPIO2_O	Sets the target output level for GPIO2 (high or low). This bit is effective only when GPIO2 is used as a digital output (GPIO2_IO is set to output).	R/W	0x0 (OTP)	0: Low 1: High
GPIO_OUT: 0x0B	Bit[2]: GPIO3_O	Sets the target output level for GPIO3 (high or low). This bit is effective only when GPIO3 is used as a digital output (GPIO3_IO is set to output).	R/W	0x0 (OTP)	0: Low 1: High

GPIO_OUT: 0x0B	Bit[3]: GPIOHV1_O	Sets the target level for GPIOHV1 (high or low). This bit is effective only when GPIOHV1 is used as a digital output (GPIOHV1_CFG is set to output) and GPIOHV1_HZ = 0.	R/W	0x0 (OTP)	0: Low 1: High
GPIO_OUT: 0x0B	Bit[4]: GPIOHV1_HZ	0: GPIOHV1 is controlled in output mode, following GPIOHV1_O 1: GPIOHV1 is in high-impedance (Hi-Z), and GPIOHV1_O is ignored	R/W	0x1 (OTP)	N/A
GPIO_CFG: 0x0C	Bit[0]: GPIO1_IO	Defines the direction for GPIO1. 0: Output 1: Input	R/W (can lock to read-only)	0x1 (OTP)	N/A
GPIO_CFG: 0x0C	Bit[1]: GPIO1_TYPE	Defines the type of input for GPIO1. 0: Digital input 1: Buffered ADC input, 3.3V range	R/W (can lock to read-only)	0x0 (OTP)	N/A
GPIO_CFG: 0x0C	Bit[2]: GPIO1_PUP	Enables the GPIO1 pull-up capability. When enabled, a 20kΩ pull-up resistor is applied to GPIO1.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
GPIO_CFG: 0x0C	Bit[4]: GPIO2_IO	Defines the direction for GPIO2. 0: Output 1: Input	R/W (can lock to read-only)	0x1 (OTP)	N/A
GPIO_CFG: 0x0C	Bit[5]: GPIO2_TYPE	Defines the input type for GPIO2. 0: Digital input 1: Buffered ADC input, 3.3V range	R/W (can lock to read-only)	0x0 (OTP)	N/A
GPIO_CFG: 0x0C	Bit[6]: GPIO2_PUP	Enables the GPIO2 pull-up capability. When enabled, a 20kΩ pull-up resistor is applied to GPIO2.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
GPIO_CFG: 0x0C	Bit[8]: GPIO3_IO	Defines the direction for GPIO3. 0: Output 1: Input	R/W (can lock to read-only)	0x1 (OTP)	N/A
GPIO_CFG: 0x0C	Bit[9]: GPIO3_TYPE	Defines the input type for GPIO3. 0: Digital input 1: Buffered ADC input, 3.3V range	R/W (can lock to read-only)	0x0 (OTP)	N/A
GPIO_CFG: 0x0C	Bit[10]: GPIO3_PUP	Enables the GPIO3 pull-up capability. When enabled, a 20kΩ pull-up resistor is applied to GPIO3.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
GPIO_CFG: 0x0C	Bit[11]: GPIO3_FSEL	0: GPIO 1: OC alarm indicator. Valid when GPIO3 is set to be an output	R/W (can lock to read-only)	0x0 (OTP)	N/A
PINS_CFG: 0x0D	Bit[0]: ALERT_POL	0: Active low. xALERT goes low when an interrupt is pending 1: Active high. xALERT goes high when an interrupt is pending	R/W (can lock to read-only)	0x1 (OTP)	N/A
PINS_CFG: 0x0D	Bit[5]: WDT_RPT	When enabled, a bite event triggers the WDT pin to toggle a high pulse.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
PINS_CFG: 0x0D	Bit[6]: WDT_RST_EN	Enables the WDT pin to reset the MP2787 back to its factory settings. When disabled, a WDT pulse caused by a watchdog bite does not trigger a reset.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled

PINS_CFG: 0x0D	Bit[8]: GPIO_LV_CFG	Sets the GPIO1~3 pull-up voltage. 0: 3V3 1: REGIN	R/W (can lock to read-only)	0x0 (OTP)	N/A
PINS_CFG: 0x0D	Bit[10]: GPIOHV1_CFG	Defines the direction of GPIOHV1.	R/W	0x0 (OTP)	0: Output 1: Input

Watchdog

Address Name and Location	Field Bit Position and Name	Description	Type	Value Reset (OTP/MTP)	Encoding, LSB and Range
WDT_STATUS: 0x0E	Bit[0]: WDT_BARKED	Indicates whether the bark timer has expired and a bark event has occurred.	Read-only	0x0	0: False 1: True
WDT_STATUS: 0x0E	Bit[1]: WDT_BITE	Indicates whether the bite timer has expired and a bite event has occurred. This event toggles the WDT pin, if WDT_RPT is set to enable.	Read-only	0x0	0: False 1: True
WDT_RST: 0x0F	Bit[0]: WDT_RST	Writing 1 to this bit resets the watchdog timer counter, and clears WDT_BITE and WDT_BARKED. This is a self-clearing register.	Write-only	0x0	1: Reset the counter and clear the bits
WDT_CFG: 0x10	Bit[0]: WDT_COM_ CTRL	Enables watchdog communication.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled
WDT_CFG: 0x10	Bits[8:2]: WDT_BARK_ CFG	Configures the watchdog bark timeout. The bark setting defines the delay from the last watchdog reset to the bark.	R/W (can lock to read-only)	0x4F (OTP)	LSB: 25ms RNG: 25ms to 3200ms 0x00: 25ms
WDT_CFG: 0x10	Bits[15:9]: WDT_BITE_CFG	Configures the watchdog bite timeout. The bite setting defines the delay from the bark to the bite.	R/W (can lock to read-only)	0x77 (OTP)	LSB: 25ms RNG: 25ms to 3200ms 0x00: 25ms

Interrupt

Address Name and Location	Field Bit Position and Name	Description	Type	Value Reset (OTP/MTP)	Encoding, LSB and Range
RD_INT0: 0x15	Bit[0]: CELL_OV_INT_ STS	Reports whether an interrupt related to a cell over-voltage (OV) event has been detected. The flags located in RD_CELL_OV should be checked to identify which cell caused the interrupt.	Read-only	0x0	0: Not detected 1: Detected
RD_INT0: 0x15	Bit[1]: CELL_UV_INT_ STS	Reports whether an interrupt related to a cell under-voltage (UV) event has been detected. The flags located in RD_CELL_UV should be checked to identify which cell caused the interrupt.	Read-only	0x0	0: Not detected 1: Detected
RD_INT0: 0x15	Bit[2]: VTOP_UV_INT_ STS	Reports whether a UV interrupt related to the battery stack positive terminal (VTOP) has been detected.	Read-only	0x0	0: Not detected 1: Detected

RD_INT0: 0x15	Bit[3]: VTOP_OV_INT_ STS	Reports whether an OV interrupt related to the battery stack positive terminal (VTOP) has been detected.	Read-only	0x0	0: Not detected 1: Detected
RD_INT0: 0x15	Bit[5]: OVER_CURR_ INT_ STS	Reports whether an interrupt related to an OC event has been detected. This signal is generated by combining the interrupt detection for discharge OC and charge OC. OC_CHG_STS and OC_DCHG_STS should be checked to identify the source of the interrupt.	Read-only	0x0	0: Not detected 1: Detected
RD_INT0: 0x15	Bit[6]: NTC_COLD_ INT_ STS	Reports whether an interrupt related to an NTC cold event has been detected. This signal is generated combining the interrupt detection for all 4 NTC channels. NTC1_CELL_COLD_STS, NTC2_CELL_COLD_STS, NTC3_CELL_COLD_STS, and NTC4_CELL_COLD_STS should be checked to identify the source of the interrupt.	Read-only	0x0	0: Not detected 1: Detected
RD_INT0: 0x15	Bit[7]: NTC_HOT_INT_ STS	Reports whether an interrupt related to an NTC hot event has been detected. This signal is generated by combining the interrupt detection for all 4 NTC channels. NTC1_CELL_HOT_STS, NTC2_CELL_HOT_STS, NTC3_CELL_HOT_STS, and NTC4_CELL_HOT_STS should be checked to identify the source of the interrupt.	Read-only	0x0	0: Not detected 1: Detected
RD_INT0: 0x15	Bit[8]: WDT_INT_STS	Reports whether an interrupt related to watchdog communication has been detected. A watchdog bark and a watchdog bite event can both trigger a watchdog event interrupt.	Read-only	0x0	0: Not detected 1: Detected
RD_INT0: 0x15	Bit[11]: VSCAN_DONE_ INT_ STS	Reports whether an interrupt related to a high-resolution voltage scan has been detected.	Read-only	0x0	0: Not detected 1: Detected
RD_INT0: 0x15	Bit[12]: CC_ACC_INT_ STS	Reports whether an interrupt related to a new Coulomb counting accumulation value has occurred, and updated CC_ACC_LSBS and CC_ACC_MSBS.	Read-only	0x0	0: Not detected 1: Detected
RD_INT0: 0x15	Bit[14]: PACK_ CURRENT_INT_ STS	Reports whether an interrupt related to a battery pack current change has been detected. The pack current changes when switching between discharge, standby, and charge mode, which is reported in PACK_CURRENT_STATUS.	Read-only	0x0	0: Not detected 1: Detected

RD_INT1: 0x16	Bit[0]: OTP_CRC_ EVENT_INT_ STS	Reports whether an interrupt related to a one-time-programmable (OTP) memory CRC fault has been detected. When enabled, this interrupt can be triggered by the following conditions: - The OTP CRC calculation fails while restoring the OTP - OTP CRC is manually triggered using OTP_CRC_DO. In this scenario, an interrupt event is generated when a CRC check is completed (for either passing or failing)	Read-only	0x0	0: Not detected 1: Detected
RD_INT1: 0x16	Bit[1]: REGIN_INT_ STS	Reports whether an interrupt related to a UV condition on REGIN has been detected.	Read-only	0x0	0: Not detected 1: Detected
RD_INT1: 0x16	Bit[2]: 3V3_INT_STS	Reports whether an interrupt related to a UV condition on 3V3 has been detected. This is triggered by monitoring the 3V3 analog-to-digital converter (ADC) or the 3.3V analog comparator. Since the 3.3V UV analog comparator is always enabled, enabling 3V3_INT_EN can trigger an interrupt even if 3V3_EN is disabled.	Read-only	0x0	0: Not detected 1: Detected
RD_INT1: 0x16	Bit[3]: VDD_INT_STS	Reports whether an interrupt related to a UV condition on VDD has been detected. This is triggered by monitoring the VDD ADC or the VDD analog comparator. Since the VDD UV analog comparator is always enabled, enabling VDD_INT_EN can trigger an interrupt even if VDD_EN is disabled.	Read-only	0x0	0: Not detected 1: Detected
RD_INT1: 0x16	Bit[4]: OPEN_WIRE_ INT_STS	Reports whether an interrupt related to an open-wire event has been detected. The open wire status registers (located at the RD_OPENH and RD_OPENL addresses) must be checked to determine whether there are disconnected wires.	Read-only	0x0	0: Not detected 1: Detected
RD_INT1: 0x16	Bit[5]: CELL_DEAD_ INT_STS	Reports whether an interrupt related to a dead cell has been detected.	Read-only	0x0	0: Not detected 1: Detected
RD_INT1: 0x16	Bit[6]: CELL_ MISMATCH_ INT_STS	Reports whether an interrupt related to a cell mismatch has been detected. Flags in the RD_CELL_MSMT and CELL_MSMT_LOWER registers should be read to identify the cells triggering the event.	Read-only	0x0	0: Not detected 1: Detected

RD_INT1: 0x16	Bit[7]: DIE_TEMP_INT_ STS	Reports whether an interrupt related to a digital die temperature event has been detected.	Read-only	0x0	0: Not detected 1: Detected
RD_INT1: 0x16	Bit[8]: PCB_MNTR_ HOT_INT_STS	Reports whether an interrupt related to the NTC hot PCB monitor has been detected. NTC1_PCB_MNTR_HOT_STS, NTC2_PCB_MNTR_HOT_STS, NTC3_PCB_MNTR_HOT_STS, and NTC4_PCB_MNTR_HOT_STS should be checked to identify which NTC caused the interrupt.	Read-only	0x0	0: Not detected 1: Detected
RD_INT1: 0x16	Bit[9]: FSM_ERROR_ INT_STS	Reports whether an interrupt related to a scheduler error event has been detected.	Read-only	0x0	0: Not detected 1: Detected
RD_INT1: 0x16	Bit[10]: SELF_TEST_ INT_STS	Reports whether an interrupt related to an ADC self-diagnostic event has been detected. The SELF_TEST_STS_OV and SELF_TEST_STS_UV flags should be checked to identify the cause of the interrupt.	Read-only	0x0	0: Not detected 1: Detected
RD_INT1: 0x16	Bit[11]: BAL_DONE_ INT_STS	Reports whether an interrupt related to a cell-balancing done event has been detected.	Read-only	0x0	0: Not detected 1: Detected
INT0_CLR: 0x17	Bit[0]: CELL_OV_INT_ CLEAR	Write 1 to this bit to clear a cell OV interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT0_CLR: 0x17	Bit[1]: CELL_UV_INT_ CLEAR	Write 1 to this bit to clear a cell UV interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT0_CLR: 0x17	Bit[2]: VTOP_UV_INT_ CLEAR	Write 1 to this bit to clear a battery pack UV interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT0_CLR: 0x17	Bit[3]: VTOP_OV_INT_ CLEAR	Write 1 to this bit to clear a battery pack OV interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT0_CLR: 0x17	Bit[5]: OVER_CURR_ INT_CLEAR	Write 1 to this bit to clear an over-current interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT0_CLR: 0x17	Bit[6]: NTC_COLD_ INT_CLEAR	Write 1 to this bit to clear an NTC cold cell monitor interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT0_CLR: 0x17	Bit[7]: NTC_HOT_INT_ CLEAR	Write 1 to this bit to clear an NTC hot cell monitor interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT0_CLR: 0x17	Bit[8]: WDT_INT_ CLEAR	Write 1 to this bit to clear a watchdog event interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT0_CLR: 0x17	Bit[11]: VSCAN_DONE_ INT_CLEAR	Write 1 to this bit to clear a completed high-resolution voltage scan interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt

INT0_CLR: 0x17	Bit[12]: CC_ACC_STS_ INT_CLEAR	Write 1 to this bit to clear a CC completed interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT0_CLR: 0x17	Bit[14]: PACK_ CURRENT_INT_ CLEAR	Write 1 to this bit to clear a battery pack current change interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT1_CLR: 0x18	Bit[0]: OTP_CRC_ EVENT_INT_ CLEAR	Write 1 to this bit to clear an OTP CRC interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT1_CLR: 0x18	Bit[1]: REGIN_INT_ CLEAR	Write 1 to this bit to clear a REGIN interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT1_CLR: 0x18	Bit[2]: 3V3_INT_ CLEAR	Write 1 to this bit to clear a 3V3 UV interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT1_CLR: 0x18	Bit[3]: VDD_INT_ CLEAR	Write 1 to this bit to clear a VDD UV interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT1_CLR: 0x18	Bit[4]: OPEN_WIRE_ INT_CLEAR	Write 1 to this bit to clear an open wire interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT1_CLR: 0x18	Bit[5]: CELL_DEAD_IN T_CLEAR	Write 1 to this bit to clear a dead cell interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT1_CLR: 0x18	Bit[6]: CELL_ MISMATCH_ INT_CLEAR	Write 1 to this bit to clear a mismatched cell interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT1_CLR: 0x18	Bit[8]: PCB_MNTR_ TEMP_INT_ CLEAR	Write 1 to this bit to clear an NTC hot PCB monitor interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT1_CLR: 0x18	Bit[9]: FSM_ERROR_ INT_CLEAR	Write 1 to this bit to clear a schedule error interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT1_CLR: 0x18	Bit[10]: SELF_TEST_ INT_CLEAR	Write 1 to this bit to clear a self-diagnostic interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT1_CLR: 0x18	Bit[11]: BAL_DONE_ INT_CLEAR	Write 1 to this bit to clear a balancing complete interrupt. This is a self-clearing register.	Write-only	0x0	1: Clear the interrupt
INT0_EN: 0x19	Bit[0]: CELL_OV_INT_ EN	Enables interrupt reporting for cell OV conditions.	R/W	0x0	0: Disabled 1: Enabled
INT0_EN: 0x19	Bit[1]: CELL_UV_INT_ EN	Enables interrupt reporting for cell UV conditions.	R/W	0x0	0: Disabled 1: Enabled
INT0_EN: 0x19	Bit[2]: VTOP_UV_INT_ EN	Enables interrupt reporting for battery pack UV conditions.	R/W	0x0	0: Disabled 1: Enabled
INT0_EN: 0x19	Bit[3]: VTOP_OV_INT_ EN	Enables interrupt reporting for battery pack OV conditions.	R/W	0x0	0: Disabled 1: Enabled

INT0_EN: 0x19	Bit[5]: OVER_CURR_ INT_EN	Enables interrupt reporting for over-current conditions. Note that there are individual enable/disable control bits: OC_CHG_INT_EN and OC_DCHG_INT_EN. OVER_CURR_INT_EN acts as a master enabler, so it must be enabled for the charge and discharge interrupts to work.	R/W	0x0	0: Disabled 1: Enabled
INT0_EN: 0x19	Bit[6]: NTC_COLD_ INT_EN	Enables interrupt reporting for NTC cold cell monitoring conditions.	R/W	0x0	0: Disabled 1: Enabled
INT0_EN: 0x19	Bit[7]: NTC_HOT_INT_ EN	Enables interrupt reporting for NTC hot cell monitoring conditions.	R/W	0x0	0: Disabled 1: Enabled
INT0_EN: 0x19	Bit[8]: WDT_INT_EN	Enables interrupt reporting for watchdog communication.	R/W	0x0	0: Disabled 1: Enabled
INT0_EN: 0x19	Bit[11]: VSCAN_DONE_ INT_EN	Enables interrupt reporting for high-resolution voltage ADC scan completion.	R/W	0x0	0: Disabled 1: Enabled
INT0_EN: 0x19	Bit[12]: CC_ACC_INT_ EN	Enables interrupt reporting for Coulomb counting accumulation.	R/W	0x0	0: Disabled 1: Enabled
INT0_EN: 0x19	Bit[14]: PACK_ CURRENT_INT_ EN	Enables interrupt reporting for battery pack current direction changes.	R/W	0x0	0: Disabled 1: Enabled
INT0_EN: 0x19	Bit[15]: INT_ALERT_ CTRL	Enables interrupt reporting to the xALERT pin.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
INT1_EN: 0x1A	Bit[0]: OTP_CRC_ EVENT_INT_EN	Enables interrupt reporting for an OTP CRC event.	R/W	0x1	0: Disabled 1: Enabled
INT1_EN: 0x1A	Bit[1]: REGIN_INT_EN	Enables interrupt reporting for a REGIN UV event.	R/W	0x0	0: Disabled 1: Enabled
INT1_EN: 0x1A	Bit[2]: 3V3_INT_EN	Enables interrupt reporting for a 3V3 UV event.	R/W	0x0	0: Disabled 1: Enabled
INT1_EN: 0x1A	Bit[3]: VDD_INT_EN	Enables interrupt reporting for a VDD UV event.	R/W	0x0	0: Disabled 1: Enabled
INT1_EN: 0x1A	Bit[4]: OPEN_WIRE_ INT_EN	Enables interrupt reporting for completed open-wire detection.	R/W	0x0	0: Disabled 1: Enabled
INT1_EN: 0x1A	Bit[5]: CELL_DEAD_ INT_EN	Enables interrupt reporting for a dead cell condition.	R/W	0x0	0: Disabled 1: Enabled
INT1_EN: 0x1A	Bit[6]: CELL_ MISMATCH_ INT_EN	Enables interrupt reporting for a cell mismatch condition.	R/W	0x0	0: Disabled 1: Enabled
INT1_EN: 0x1A	Bit[7]: DIE_TEMP_INT_ EN	Enables interrupt reporting for a die over-temperature (OT) event.	R/W	0x0	0: Disabled 1: Enabled
INT1_EN: 0x1A	Bit[8]: PCB_MNTR_ TEMP_INT_EN	Enables interrupt reporting for an NTC hot PCB condition.	R/W	0x0	0: Disabled 1: Enabled

INT1_EN: 0x1A	Bit[9]: FSM_ERROR_ INT_EN	Enables interrupt reporting for a scheduled error.	R/W	0x0	0: Disabled 1: Enabled
INT1_EN: 0x1A	Bit[10]: SELF_TEST_ INT_EN	Enables interrupt reporting for ADC self-diagnostics OV/UV conditions.	R/W	0x0	0: Disabled 1: Enabled
INT1_EN: 0x1A	Bit[11]: BAL_DONE_ INT_EN	Enables interrupt reporting for completed cell-balancing.	R/W	0x0	0: Disabled 1: Enabled
INT_TYPE0: 0x1B	Bits[1:0]: VTOP_UV_INT_ TYPE	Controls the triggering logic for a battery pack UV interrupt. 0x0: Interrupt generated on input status signal high 0x1: Interrupt generated on the rising edge of the input status signal 0x2: Interrupt generated on the falling edge of the input status signal 0x3: Interrupt generated on the rising and falling edges of the input status signal	R/W	0x0	N/A
INT_TYPE0: 0x1B	Bits[3:2]: VTOP_OV_INT_ TYPE	Controls the triggering logic for a battery pack OV interrupt. 0x0: Interrupt generated on input status signal high 0x1: Interrupt generated on the rising edge of the input status signal 0x2: Interrupt generated on the falling edge of the input status signal 0x3: Interrupt generated on the rising and falling edges of the input status signal	R/W	0x0	N/A
INT_TYPE0: 0x1B	Bits[5:4]: NTC_COLD_ INT_TYPE	Controls the triggering logic for an NTC cold cell monitor interrupt. 0x0: Interrupt generated on input status signal high 0x1: Interrupt generated on the rising edge of the input status signal 0x2: Interrupt generated on the falling edge of the input status signal 0x3: Interrupt generated on the rising and falling edges of the input status signal	R/W	0x0	N/A
INT_TYPE0: 0x1B	Bits[7:6]: NTC_HOT_INT_ TYPE	Controls the triggering logic for an NTC hot cell monitor interrupt. 0x0: Interrupt generated on input status signal high 0x1: Interrupt generated on the rising edge of the input status signal 0x2: Interrupt generated on the falling edge of the input status signal 0x3: Interrupt generated on the rising and falling edges of the input status signal	R/W	0x0	N/A

INT_TYPE1: 0x1C	Bits[9:8]: REGIN_INT_ TYPE	Controls the triggering logic for a REGIN interrupt. 0x0: Interrupt generated on input status signal high 0x1: Interrupt generated on the rising edge of the input status signal 0x2: Interrupt generated on the falling edge of the input status signal 0x3: Interrupt generated on the rising and falling edges of the input status signal	R/W	0x0	N/A
INT_TYPE1: 0x1C	Bits[11:10]: 3V3_INT_TYPE	Controls the triggering logic for a 3V3 interrupt. 0x0: Interrupt generated on input status signal high 0x1: Interrupt generated on the rising edge of the input status signal 0x2: Interrupt generated on the falling edge of the input status signal 0x3: Interrupt generated on the rising and falling edges of the input status signal	R/W	0x0	N/A
INT_TYPE1: 0x1C	Bits[13:12]: VDD_INT_TYPE	Controls the triggering logic for a VDD interrupt. 0x0: Interrupt generated on input status signal high 0x1: Interrupt generated on the rising edge of the input status signal 0x2: Interrupt generated on the falling edge of the input status signal 0x3: Interrupt generated on the rising and falling edges of the input status signal	R/W	0x0	N/A
INT_TYPE2: 0x1D	Bits[1:0]: CELL_DEAD_ INT_TYPE	Controls the triggering logic for a dead cell interrupt. 0x0: Interrupt generated on input status signal high 0x1: Interrupt generated on the rising edge of the input status signal 0x2: Interrupt generated on the falling edge of the input status signal 0x3: Interrupt generated on the rising and falling edges of the input status signal	R/W	0x0	N/A
INT_TYPE2: 0x1D	Bits[3:2]: CELL_ MISMATCH_ INT_TYPE	Controls the triggering logic for a mismatched cell interrupt. 0x0: Interrupt generated on input status signal high 0x1: Interrupt generated on the rising edge of the input status signal 0x2: Interrupt generated on the falling edge of the input status signal 0x3: Interrupt generated on the rising and falling edges of the input status signal	R/W	0x0	N/A

INT_TYPE2: 0x1D	Bits[5:4]: DIE_TEMP_INT_ TYPE	Controls the triggering logic for a die temperature interrupt. 0x0: Interrupt generated on input status signal high 0x1: Interrupt generated on the rising edge of the input status signal 0x2: Interrupt generated on the falling edge of the input status signal 0x3: Interrupt generated on the rising and falling edges of the input status signal	R/W	0x0	N/A
INT_TYPE2: 0x1D	Bits[7:6]: PCB_MNTR_ TEMP_INT_ TYPE	Controls the triggering logic for an NTC hot PCB monitor interrupt. 0x0: Interrupt generated on input status signal high 0x1: Interrupt generated on the rising edge of the input status signal 0x2: Interrupt generated on the falling edge of the input status signal 0x3: Interrupt generated on the rising and falling edges of the input status signal	R/W	0x0	N/A
INT_TYPE2: 0x1D	Bits[11:10]: SELF_TEST_ INT_TYPE	Controls the triggering logic for a self-diagnostic interrupt. 0x0: Interrupt generated on input status signal high 0x1: Interrupt generated on the rising edge of the input status signal 0x2: Interrupt generated on the falling edge of the input status signal 0x3: Interrupt generated on the rising and falling edges of the input status signal	R/W	0x0	N/A
MASK_INT0: 0x1E	Bits[0]: CELL_OV_ MASK	Enables clearing the cell OV interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again.	R/W	0x0	0: Disabled 1: Enabled
MASK_INT0: 0x1E	Bit[1]: CELL_UV_ MASK	Enables clearing the cell UV interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again.	R/W	0x0	0: Disabled 1: Enabled
MASK_INT0: 0x1E	Bit[2]: VTOP_UV_ MASK	Enables clearing the VTOP UV interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again.	R/W	0x0	0: Disabled 1: Enabled
MASK_INT0: 0x1E	Bit[3]: VTOP_OV_ MASK	Enables clearing the VTOP OV interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again.	R/W	0x0	0: Disabled 1: Enabled

MASK_INT0: 0x1E	Bit[5]: OC_MASK	Enables clearing the OC interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again.	R/W	0x0	0: Disabled 1: Enabled
MASK_INT0: 0x1E	Bit[6]: NTCS_CELL_COLD_MASK	Enables clearing the NTC cold cell interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again.	R/W	0x0	0: Disabled 1: Enabled
MASK_INT0: 0x1E	Bit[7]: NTCS_CELL_HOT_MASK	Enables clearing the NTC hot cell interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again.	R/W	0x0	0: Disabled 1: Enabled
MASK_INT1: 0x1F	Bit[0]: SELF_TEST_MASK	Enables clearing the self-test interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again.	R/W	0x0	0: Disabled 1: Enabled
MASK_INT1: 0x1F	Bit[1]: REGIN_MASK	Enables clearing the REGIN interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again.	R/W	0x0	0: Disabled 1: Enabled
MASK_INT1: 0x1F	Bit[2]: 3V3_MASK	Enables clearing the 3V3 interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again.	R/W	0x0	0: Disabled 1: Enabled
MASK_INT1: 0x1F	Bit[3]: VDD_MASK	Enables clearing the VDD interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again.	R/W	0x0	0: Disabled 1: Enabled
MASK_INT1: 0x1F	Bit[6]: CELL_MSMT_MASK	Enables clearing the mismatched cell interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again.	R/W	0x0	0: Disabled 1: Enabled
MASK_INT1: 0x1F	Bit[7]: DIE_TEMP_DIG_MASK	Enables clearing the die temperature interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again.	R/W	0x0	0: Disabled 1: Enabled
MASK_INT1: 0x1F	Bit[8]: PCB_MNTR_HOT_MASK	Enables clearing the NTC hot PCB monitor interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again.	R/W	0x0	0: Disabled 1: Enabled

Over-Current (OC)

Address Name and Location	Field Bit Position and Name	Description	Type	Value Reset (OTP/MTP)	Encoding, LSB and Range
OC_STATUS: 0x27	Bit[0]: OC_DCHG_STS	Reports whether a latched discharge OC event has been detected. The latched value depends on the interrupt type (e.g. rising, falling, or rising and falling) selected by OC_DCHG_INT_TYPE. For a rising edge type, the bit is set to 1 if a rising edge is detected. For a falling edge type, the bit is set to 1 if a falling edge is detected. For a rising and falling edge type, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high. This status bit can be cleared with OVER_CURR_INT_CLEAR.	Read-only	0x0	0: Not detected 1: Detected
OC_STATUS: 0x27	Bit[1]: OC_CHG_STS	Reports whether a latched charge OC event has been detected. The latched value depends on the interrupt type (e.g. rising, falling, or rising and falling) selected by OC_CHG_INT_TYPE. For a rising edge type, the bit is set to 1 if a rising edge is detected. For a falling edge type, the bit is set to 1 if a falling edge is detected. For a rising and falling edge type, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high. This status bit can be cleared with OVER_CURR_INT_CLEAR.	Read-only	0x0	0: Not detected 1: Detected
OC_STATUS: 0x27	Bit[8]: OC_DCHG_RT_STS	Reports whether a discharge OC event has been detected in real time.	Read-only	0x0	0: Not detected 1: Detected
OC_STATUS: 0x27	Bit[9]: OC_CHG_RT_STS	Reports whether a charge OC event has been detected in real time.	Read-only	0x0	0: Not detected 1: Detected
OCFT_CTRL: 0x2A	Bit[0]: OC_DCHG_EN_CTRL	Enables discharge OC monitoring. When disabled, this interrupt or alarm cannot be triggered, regardless of whether those bits are enabled.	R/W (can lock to read-only)	0x1 (MTP)	0: Disabled 1: Enabled
OCFT_CTRL: 0x2A	Bit[1]: OC_CHG_EN_CTRL	Enables charge OC monitoring. When disabled, this interrupt or alarm cannot be triggered, regardless of whether those bits are enabled.	R/W (can lock to read-only)	0x1 (MTP)	0: Disabled 1: Enabled
OCFT_CTRL: 0x2A	Bit[2]: OC_DCHG_INT_EN	Enables the interrupt for a discharge OC. OC_DCHG_EN_CTRL must be enabled for this bit to be effective.	R/W	0x0	0: Disabled 1: Enabled

OCFT_CTRL: 0x2A	Bit[3]: OC_CHG_INT_EN	Enables the interrupt for a charge OC. OC_CHG_EN_CTRL must be enabled for this bit to be effective.	R/W	0x0	0: Disabled 1: Enabled
SCFT_CTRL: 0x2A	Bit[4]: OC_DCHG_ALARM_EN	Enables the alarm for a discharge OC event. OC_DCHG_EN_CTRL must be enabled for this bit to be effective.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled
SCFT_CTRL: 0x2A	Bit[5]: OC_CHG_ALARM_EN	Enables the alarm for a charge OC event. OC_CHG_EN_CTRL must be enabled for this bit to be effective.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled
OCFT_CTRL: 0x2A	Bits[13:12]: OC_DCHG_INT_TYPE	Controls the triggering logic for discharge OC interrupt. 0x0: Interrupt generated on input status signal high 0x1: Interrupt generated on the rising edge of the input status signal 0x2: Interrupt generated on the falling edge of the input status signal 0x3: Interrupt generated on the rising and falling edges of the input status signal	R/W	0x0	N/A
OCFT_CTRL: 0x2A	Bits[15:14]: OC_CHG_INT_TYPE	Controls the triggering logic for charge OC interrupt. 0x0: Interrupt generated on input status signal high 0x1: Interrupt generated on the rising edge of the input status signal 0x2: Interrupt generated on the falling edge of the input status signal 0x3: Interrupt generated on the rising and falling edges of the input status signal	R/W	0x0	N/A
DSGOC_CFG: 0x2B	Bits[4:0]: OC_DCHG_LIM	Configures the OC discharge threshold. With a 1x range (OC_DCHG_RNG = 0), the smallest available code is 5.5mV. With a 3x range (OC_DCHG_RNG = 1), the minimum code is 16.5mV. This threshold has an offset equal to the LSB, meaning 0x00 = 5.5mV with a 1x range and 0x00 = 16.5mV with a 3x range. The full-scale range depends on SC_DCHG_RNG (176mV for 1x, and 528mV for 3x).	R/W (can lock to read-only)	0x11 (MTP)	N/A
DSGOC_CFG: 0x2B	Bit[5]: OC_DCHG_RNG	Selects the range and the LSB discharge OC threshold.	R/W (can lock to read-only)	N/A (MTP)	0: 5.5mV/ LSB, 176mV FSR 1: 16.5mV/ LSB, 528mV FSR

DSGOC_CFG: 0x2B	Bits[14:8]: OC_DCHG_DG	Configures the discharge OC deglitch time. When OC_DCHG_DG = 0x00, there is no deglitch. This means an OC event is flagged as soon as it is detected. The response time is about 100µs.	R/W (can lock to read-only)	0x01 (MTP)	LSB: 200µs, RNG: 100µs to 25500µs, 0x00: 100µs
CHGOC_CFG: 0x2C	Bits[4:0]: OC_CHG_LIM	Configures the charge OC threshold. With a 1x range (OC_CHG_RNG = 0), the smallest available code is 2.5mV. With a 3x range (OC_CHG_RNG = 1), the minimum code is 7.5mV. The limit has an offset equal to the LSB, meaning 0x00 = 2.5mV with a 1x range and 0x00 = 7.5mV with a 3x range. The full-scale range depends on the OC_CHG_RNG (80mV for 1x, and 240mV for 3x).	R/W (can lock to read-only)	0x11 (MTP)	N/A
CHGOC_CFG: 0x2C	Bit[5]: OC_CHG_RNG	Selects the range and the LSB for the charge OC threshold.	R/W (can lock to read-only)	0x0 (MTP)	0: 2.5mV/ LSB, 80mV FSR 1: 7.5mV/ LSB, 240mV FSR
CHGOC_CFG: 0x2C	Bits[14:8]: OC_CHG_DG	Configures the charge OC deglitch time. When OC_CHG_DG = 0x00, there is no deglitch. This means an OC event is flagged as soon as it is detected. The response time is about 100µs.	R/W (can lock to read-only)	0x08 (MTP)	LSB: 200µs, RNG: 100µs to 25500µs, 0x00: 100µs

Over-Voltage (OV), Under-Voltage (UV), Dead Cell, and Mismatched Cell

Address Name and Location	Field Bit Position and Name	Description	Type	Value Reset (OTP/MTP)	Encoding, LSB and Range
RD_CELL_UV: 0x2D	Bit[0]: CELL_1_UV_STS	Reports whether an UV condition has been detected on cell 1. Send an interrupt UV clear to clear this bit. 0: No UV event has been detected 1: A UV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_UV: 0x2D	Bit[1]: CELL_2_UV_STS	Reports whether a UV condition has been detected on cell 2. Send an interrupt UV clear to clear this bit. 0: No UV event has been detected 1: A UV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_UV: 0x2D	Bit[2]: CELL_3_UV_STS	Reports whether a UV condition has been detected on cell 3. Send an interrupt UV clear to clear this bit. 0: No UV event has been detected 1: A UV event has been detected	Read-only	0x0	0: Not detected 1: Detected

RD_CELL_UV: 0x2D	Bit[3]: CELL_4_UV_ STS	Reports whether a UV condition has been detected on cell 4. Send an interrupt UV clear to clear this bit. 0: No UV event has been detected 1: A UV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_UV: 0x2D	Bit[4]: CELL_5_UV_ STS	Reports whether a UV condition has been detected on cell 5. Send an interrupt UV clear to clear this bit. 0: No UV event has been detected 1: A UV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_UV: 0x2D	Bit[5]: CELL_6_UV_ STS	Reports whether a UV condition has been detected on cell 6. Send an interrupt UV clear to clear this bit. 0: No UV event has been detected 1: A UV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_UV: 0x2D	Bit[6]: CELL_7_UV_ STS	Reports whether a UV condition has been detected on cell 7. Send an interrupt UV clear to clear this bit. 0: No UV event has been detected 1: A UV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_UV: 0x2D	Bit[7]: CELL_8_UV_ STS	Reports whether a UV condition has been detected on cell 8. Send an interrupt UV clear to clear this bit. 0: No UV event has been detected 1: A UV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_UV: 0x2D	Bit[8]: CELL_9_UV_ STS	Reports whether a UV condition has been detected on cell 9. Send an interrupt UV clear to clear this bit. 0: No UV event has been detected 1: A UV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_UV: 0x2D	Bit[9]: CELL_10_UV_ STS	Reports whether a UV condition has been detected on cell 10. Send an interrupt UV clear to clear this bit. 0: No UV event has been detected 1: A UV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_UV: 0x2D	Bit[10]: CELL_11_UV_ STS	Reports whether a UV condition has been detected on cell 11. Send an interrupt UV clear to clear this bit. 0: No UV event has been detected 1: A UV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_UV: 0x2D	Bit[11]: CELL_12_UV_ STS	Reports whether a UV condition has been detected on cell 12. Send an interrupt UV clear to clear this bit. 0: No UV event has been detected 1: A UV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_UV: 0x2D	Bit[12]: CELL_13_UV_ STS	Reports whether a UV condition has been detected on cell 13. Send an interrupt UV clear to clear this bit. 0: No UV event has been detected 1: A UV event has been detected	Read-only	0x0	0: Not detected 1: Detected

RD_CELL_UV: 0x2D	Bit[13]: CELL_14_UV_ STS	Reports whether a UV condition has been detected on cell 14. Send an interrupt UV clear to clear this bit. 0: No UV event has been detected 1: A UV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_UV: 0x2D	Bit[14]: CELL_15_UV_ STS	Reports whether a UV condition has been detected on cell 15. Send an interrupt UV clear to clear this bit. 0: No UV event has been detected 1: A UV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_UV: 0x2D	Bit[15]: CELL_16_UV_ STS	Reports whether a UV condition has been detected on cell 16. Send an interrupt UV clear to clear this bit. 0: No UV event has been detected 1: A UV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_OV: 0x2E	Bit[0]: CELL_1_OV_ STS	Reports whether an OV condition has been detected on cell 1. Send an interrupt OV clear to clear this bit. 0: No OV event has been detected 1: An OV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_OV: 0x2E	Bit[1]: CELL_2_OV_ STS	Reports whether an OV condition has been detected on cell 2. Send an interrupt OV clear to clear this bit. 0: No OV event has been detected 1: An OV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_OV: 0x2E	Bit[2]: CELL_3_OV_ STS	Reports whether an OV condition has been detected on cell 3. Send an interrupt OV clear to clear this bit. 0: No OV event has been detected 1: An OV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_OV: 0x2E	Bit[3]: CELL_4_OV_ STS	Reports whether an OV condition has been detected on cell 4. Send an interrupt OV clear to clear this bit. 0: No OV event has been detected 1: An OV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_OV: 0x2E	Bit[4]: CELL_5_OV_ STS	Reports whether an OV condition has been detected on cell 5. Send an interrupt OV clear to clear this bit. 0: No OV event has been detected 1: An OV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_OV: 0x2E	Bit[5]: CELL_6_OV_ STS	Reports whether an OV condition has been detected on cell 6. Send an interrupt OV clear to clear this bit. 0: No OV event has been detected 1: An OV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_OV: 0x2E	Bit[6]: CELL_7_OV_ STS	Reports whether an OV condition has been detected on cell 7. Send an interrupt OV clear to clear this bit. 0: No OV event has been detected 1: An OV event has been detected	Read-only	0x0	0: Not detected 1: Detected

RD_CELL_OV: 0x2E	Bit[7]: CELL_8_OV_ STS	Reports whether an OV condition has been detected on cell 8. Send an interrupt OV clear to clear this bit. 0: No OV event has been detected 1: An OV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_OV: 0x2E	Bit[8]: CELL_9_OV_ STS	Reports whether an OV condition has been detected on cell 9. Send an interrupt OV clear to clear this bit. 0: No OV event has been detected 1: An OV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_OV: 0x2E	Bit[9]: CELL_10_OV_ STS	Reports whether an OV condition has been detected on cell 10. Send an interrupt OV clear to clear this bit. 0: No OV event has been detected 1: An OV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_OV: 0x2E	Bit[10]: CELL_11_OV_ STS	Reports whether an OV condition has been detected on cell 11. Send an interrupt OV clear to clear this bit. 0: No OV event has been detected 1: An OV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_OV: 0x2E	Bit[11]: CELL_12_OV_ STS	Reports whether an OV condition has been detected on cell 12. Send an interrupt OV clear to clear this bit. 0: No OV event has been detected 1: An OV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_OV: 0x2E	Bit[12]: CELL_13_OV_ STS	Reports whether an OV condition has been detected on cell 13. Send an interrupt OV clear to clear this bit. 0: No OV event has been detected 1: An OV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_OV: 0x2E	Bit[13]: CELL_14_OV_ STS	Reports whether an OV condition has been detected on cell 14. Send an interrupt OV clear to clear this bit. 0: No OV event has been detected 1: An OV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_OV: 0x2E	Bit[14]: CELL_15_OV_ STS	Reports whether an OV condition has been detected on cell 15. Send an interrupt OV clear to clear this bit. 0: No OV event has been detected 1: An OV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_OV: 0x2E	Bit[15]: CELL_16_OV_ STS	Reports whether an OV condition has been detected on cell 16. Send an interrupt OV clear to clear this bit. 0: No OV event has been detected 1: An OV event has been detected	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_MSMT: 0x2F	Bit[0]: CELL_1_MSMT_ STS	Reports the cell 1 mismatch interrupt status. Clear a mismatch with CELL_MISMATCH_INT_CLEAR. 0: No mismatch 1: A mismatch has been detected	Read-only	0x0	0: False 1: True

RD_CELL_MSMT: 0x2F	Bit[1]: CELL_2_MSMT_STS	Reports the cell 2 mismatch interrupt status. Clear a mismatch with CELL_MISMATCH_INT_CLEAR. 0: No mismatch 1: A mismatch has been detected	Read-only	0x0	0: False 1: True
RD_CELL_MSMT: 0x2F	Bit[2]: CELL_3_MSMT_STS	Reports the cell 3 mismatch interrupt status. Clear a mismatch with CELL_MISMATCH_INT_CLEAR. 0: No mismatch 1: A mismatch has been detected	Read-only	0x0	0: False 1: True
RD_CELL_MSMT: 0x2F	Bit[3]: CELL_4_MSMT_STS	Reports the cell 4 mismatch interrupt status. Clear a mismatch with CELL_MISMATCH_INT_CLEAR. 0: No mismatch 1: A mismatch has been detected	Read-only	0x0	0: False 1: True
RD_CELL_MSMT: 0x2F	Bit[4]: CELL_5_MSMT_STS	Reports the cell 5 mismatch interrupt status. Clear a mismatch with CELL_MISMATCH_INT_CLEAR. 0: No mismatch 1: A mismatch has been detected	Read-only	0x0	0: False 1: True
RD_CELL_MSMT: 0x2F	Bit[5]: CELL_6_MSMT_STS	Reports the cell 6 mismatch interrupt status. Clear a mismatch with CELL_MISMATCH_INT_CLEAR. 0: No mismatch 1: A mismatch has been detected	Read-only	0x0	0: False 1: True
RD_CELL_MSMT: 0x2F	Bit[6]: CELL_7_MSMT_STS	Reports the cell 7 mismatch interrupt status. Clear a mismatch with CELL_MISMATCH_INT_CLEAR. 0: No mismatch 1: A mismatch has been detected	Read-only	0x0	0: False 1: True
RD_CELL_MSMT: 0x2F	Bit[7]: CELL_8_MSMT_STS	Reports the cell 8 mismatch interrupt status. Clear a mismatch with CELL_MISMATCH_INT_CLEAR. 0: No mismatch 1: A mismatch has been detected	Read-only	0x0	0: False 1: True
RD_CELL_MSMT: 0x2F	Bit[8]: CELL_9_MSMT_STS	Reports the cell 9 mismatch interrupt status. Clear a mismatch with CELL_MISMATCH_INT_CLEAR. 0: No mismatch 1: A mismatch has been detected	Read-only	0x0	0: False 1: True

RD_CELL_MSMT: 0x2F	Bit[9]: CELL_10_MSMT_STS	Reports the cell 10 mismatch interrupt status. Clear a mismatch with CELL_MISMATCH_INT_CLEAR. 0: No mismatch 1: A mismatch has been detected	Read-only	0x0	0: False 1: True
RD_CELL_MSMT: 0x2F	Bit[10]: CELL_11_MSMT_STS	Reports the cell 11 mismatch interrupt status. Clear a mismatch with CELL_MISMATCH_INT_CLEAR. 0: No mismatch 1: A mismatch has been detected	Read-only	0x0	0: False 1: True
RD_CELL_MSMT: 0x2F	Bit[11]: CELL_12_MSMT_STS	Reports the cell 12 mismatch interrupt status. Clear a mismatch with CELL_MISMATCH_INT_CLEAR. 0: No mismatch 1: A mismatch has been detected	Read-only	0x0	0: False 1: True
RD_CELL_MSMT: 0x2F	Bit[12]: CELL_13_MSMT_STS	Reports the cell 13 mismatch interrupt status. Clear a mismatch with CELL_MISMATCH_INT_CLEAR. 0: No mismatch 1: A mismatch has been detected	Read-only	0x0	0: False 1: True
RD_CELL_MSMT: 0x2F	Bit[13]: CELL_14_MSMT_STS	Reports the cell 14 mismatch interrupt status. Clear a mismatch with CELL_MISMATCH_INT_CLEAR. 0: No mismatch 1: A mismatch has been detected	Read-only	0x0	0: False 1: True
RD_CELL_MSMT: 0x2F	Bit[14]: CELL_15_MSMT_STS	Reports the cell 15 mismatch interrupt status. Clear a mismatch with CELL_MISMATCH_INT_CLEAR. 0: No mismatch 1: A mismatch has been detected	Read-only	0x0	0: False 1: True
RD_CELL_MSMT: 0x2F	Bit[15]: CELL_16_MSMT_STS	Reports the cell 16 mismatch interrupt status. Clear a mismatch with CELL_MISMATCH_INT_CLEAR. 0: No mismatch 1: A mismatch has been detected.	Read-only	0x0	0: False 1: True
RD_CELL_DEAD: 0x30	Bit[0]: CELL_1_DEAD_STS	Reports cell 1's dead interrupt status. If CELL_DEAD_STS_SEL = 0, this bit indicates the interrupt status; if CELL_DEAD_STS_SEL = 1, this bit indicates the real-time status. 0: This cell is not dead 1: This cell is dead	Read-only	0x0	0: Not detected 1: Detected

RD_CELL_ DEAD: 0x30	Bit[1]: CELL_2_DEAD_ STS	Reports cell 2's dead interrupt status. If CELL_DEAD_STS_SEL = 0, this bit indicates the interrupt status; if CELL_DEAD_STS_SEL = 1, this bit indicates the real-time status. 0: This cell is not dead 1: This cell is dead	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_ DEAD: 0x30	Bit[2]: CELL_3_DEAD_ STS	Reports cell 3's dead interrupt status. If CELL_DEAD_STS_SEL = 0, this bit indicates the interrupt status; if CELL_DEAD_STS_SEL = 1, this bit indicates the real-time status. 0: This cell is not dead 1: This cell is dead	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_ DEAD: 0x30	Bit[3]: CELL_4_DEAD_ STS	Reports cell 4's dead interrupt status. If CELL_DEAD_STS_SEL = 0, this bit indicates the interrupt status; if CELL_DEAD_STS_SEL = 1, this bit indicates the real-time status. 0: This cell is not dead 1: This cell is dead	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_ DEAD: 0x30	Bit[4]: CELL_5_DEAD_ STS	Reports cell 5's dead interrupt status. If CELL_DEAD_STS_SEL = 0, this bit indicates the interrupt status; if CELL_DEAD_STS_SEL = 1, this bit indicates the real-time status. 0: This cell is not dead 1: This cell is dead	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_ DEAD: 0x30	Bit[5]: CELL_6_DEAD_ STS	Reports cell 6's dead interrupt status. If CELL_DEAD_STS_SEL = 0, this bit indicates the interrupt status; if CELL_DEAD_STS_SEL = 1, this bit indicates the real-time status. 0: This cell is not dead 1: This cell is dead	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_ DEAD: 0x30	Bit[6]: CELL_7_DEAD_ STS	Reports cell 7's dead interrupt status. If CELL_DEAD_STS_SEL = 0, this bit indicates the interrupt status; if CELL_DEAD_STS_SEL = 1, this bit indicates the real-time status. 0: This cell is not dead 1: This cell is dead	Read-only	0x0	0: Not detected 1: Detected

RD_CELL_ DEAD: 0x30	Bit[7]: CELL_8_DEAD_ STS	Reports cell 8's dead interrupt status. If CELL_DEAD_STS_SEL = 0, this bit indicates the interrupt status; if CELL_DEAD_STS_SEL = 1, this bit indicates the real-time status. 0: This cell is not dead 1: This cell is dead	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_ DEAD: 0x30	Bit[8]: CELL_9_DEAD_ STS	Reports cell 9's dead interrupt status. If CELL_DEAD_STS_SEL = 0, this bit indicates the interrupt status; if CELL_DEAD_STS_SEL = 1, this bit indicates the real-time status. 0: This cell is not dead 1: This cell is dead	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_ DEAD: 0x30	Bit[9]: CELL_10_ DEAD_STS	Reports cell 10's dead interrupt status. If CELL_DEAD_STS_SEL = 0, this bit indicates the interrupt status; if CELL_DEAD_STS_SEL = 1, this bit indicates the real-time status. 0: This cell is not dead 1: This cell is dead	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_ DEAD: 0x30	Bit[10]: CELL_11_ DEAD_STS	Reports cell 11's dead interrupt status. If CELL_DEAD_STS_SEL = 0, this bit indicates the interrupt status; if CELL_DEAD_STS_SEL = 1, this bit indicates the real-time status. 0: This cell is not dead 1: This cell is dead	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_ DEAD: 0x30	Bit[11]: CELL_12_ DEAD_STS	Reports cell 12's dead interrupt status. If CELL_DEAD_STS_SEL = 0, this bit indicates the interrupt status; if CELL_DEAD_STS_SEL = 1, this bit indicates the real-time status. 0: This cell is not dead 1: This cell is dead	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_ DEAD: 0x30	Bit[12]: CELL_13_ DEAD_STS	Reports cell 13's dead interrupt status. If CELL_DEAD_STS_SEL = 0, this bit indicates the interrupt status; if CELL_DEAD_STS_SEL = 1, this bit indicates the real-time status. 0: This cell is not dead 1: This cell is dead	Read-only	0x0	0: Not detected 1: Detected

RD_CELL_DEAD: 0x30	Bit[13]: CELL_14_DEAD_STS	Reports cell 14's dead interrupt status. If CELL_DEAD_STS_SEL = 0, this bit indicates the interrupt status; if CELL_DEAD_STS_SEL = 1, this bit indicates the real-time status. 0: This cell is not dead 1: This cell is dead	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_DEAD: 0x30	Bit[14]: CELL_15_DEAD_STS	Reports cell 15's dead interrupt status. If CELL_DEAD_STS_SEL = 0, this bit indicates the interrupt status; if CELL_DEAD_STS_SEL = 1, this bit indicates the real-time status. 0: This cell is not dead 1: This cell is dead	Read-only	0x0	0: Not detected 1: Detected
RD_CELL_DEAD: 0x30	Bit[15]: CELL_16_DEAD_STS	Reports cell 16's dead interrupt status. If CELL_DEAD_STS_SEL = 0, this bit indicates the interrupt status; if CELL_DEAD_STS_SEL = 1, this bit indicates the real-time status. 0: This cell is not dead 1: This cell is dead	Read-only	0x0	0: Not detected 1: Detected
CELL_MSMT_STS: 0x33	Bits[9:0]: CELL_MSMT_DELTA	Reports the voltage difference between the cell with the lowest voltage and the cell with the highest voltage. If the interrupt is disabled, this bit indicates the real-time status; otherwise, this bit is latched at the interrupt and returns to the real-time status after being cleared.	Read-only	0x000	LSB: 4.88mV, RNG: 0mV to 4997.12mV, 000: 0mV
CELL_MSMT_STS: 0x33	Bits[13:10]: CELL_MSMT_LOWER	Reports the cell with the lowest voltage. If the interrupt is disabled, this bit indicates the real-time status; otherwise, this bit is latched at the interrupt and returns to the real-time status after being cleared. 0x0: Cell 1 ... 0xF: Cell 16	Read-only	0x0	N/A
CELL_MSMT_STS: 0x33	Bit[14]: CELL_MSMT_RT_STS	Reports the real-time cell mismatch status. 0: There is no mismatch 1: A mismatch has been detected	Read-only	0x0	N/A
PACKFT_CTRL: 0x34	Bit[0]: VTOP_UV_EN_CTRL	Enables the monitoring for the battery pack (VTOP) UV alarm.	R/W	0x0 (OTP)	0: Disabled 1: Enabled
PACKFT_CTRL: 0x34	Bit[4]: VTOP_OV_EN_CTRL	Enables the monitoring for the battery pack (VTOP) OV alarm	R/W	0x0 (OTP)	0: Disabled 1: Enabled
PACKFT_CTRL: 0x34	Bit[8]: CELL_DEAD_EN	Enables dead cell monitoring. This bit must be enabled to trigger dead cell interrupts.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled

PACKFT_CTRL: 0x34	Bit[10]: CELL_DEAD_STS_SEL	0: Show the latched dead cell status going to the interrupt controller 1: Show the real-time dead cell status	R/W	0x0	N/A
PACKFT_CTRL: 0x34	Bit[11]: CELL_MSMT_EN	0: Disable cell mismatch logic 1: Enable cell mismatch logic	R/W	0x0 (OTP)	0: Disabled 1: Enabled
PACKFT_CTRL: 0x34	Bit[14]: CELL_DEAD_LOG_STS	Reports whether a dead cell event has been detected. This status is not cleared by shutdown. It must be cleared with CELL_DEAD_DET_CLEAR.	Read-only	0x0	0: Not detected 1: Detected
PACKFT_CTRL: 0x34	Bit[15]: CELL_DEAD_DET_CLEAR	Set this bit to 1 to clear the dead cell status. This is a self-clearing bit.	Write-only	0x0	N/A
CELLFT_CTRL: 0x35	Bit[0]: SYNC_RT_STATUS	When this bit is turned on, it synchronizes the status bit to allow the voltage ADC monitoring input status to be cleared after it has been disabled.	R/W	0x0	0: Off 1: On
CELLFT_CTRL: 0x35	Bit[1]: CELL_UV_EN_CTRL	Enables cell UV. This bit must be enabled to trigger UV interrupt.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled
CELLFT_CTRL: 0x35	Bit[4]: CELL_OV_EN_CTRL	Enables cell OV. This bit must be enabled to trigger OV interrupt.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled
CELLFT_CTRL: 0x35	Bits[8:7]: CELL_UV_INT_TYPE	Controls the triggering logic for cell UV interrupt. 0x0: Interrupt generated on input status signal high 0x1: Interrupt generated on the rising edge of the input status signal 0x2: Interrupt generated on the falling edge of the input status signal 0x3: Interrupt generated on the rising and falling edges of the input status signal	R/W	0x1	N/A
CELLFT_CTRL: 0x35	Bits[10:9]: CELL_OV_INT_TYPE	Controls the triggering logic for cell OV interrupt. 0x0: Interrupt generated on input status signal high 0x1: Interrupt generated on the rising edge of the input status signal 0x2: Interrupt generated on the falling edge of the input status signal 0x3: Interrupt generated on the rising and falling edges of the input status signal	R/W	0x1	N/A
CELLFT_CTRL: 0x35	Bit[11]: CELL_UV_STS_SEL	Selects what each of the CELL_x_UV_STS bits report. 0: Report the latched cell x status going to the interrupt controller 1: Report the latest outcome of the cell x UV check	R/W	0x0	N/A

CELLFT_CTRL: 0x35	Bit[12]: CELL_OV_STS_SEL	Selects what each of the CELL_x_OV_STS bits report. 0: Report the latched cell x status going to the interrupt controller 1: Report the latest outcome of the cell x OV check	R/W	0x0	N/A
CELL_HYST: 0x36	Bits[7:4]: CELL_UV_HYST	Sets the cell UV threshold hysteresis.	R/W	0xA (OTP)	LSB: 19.5mV RNG: 0mV to 292.5mV 0: 0mV
CELL_HYST: 0x36	Bits[11:8]: CELL_OV_HYST	Sets the cell OV threshold hysteresis.	R/W	0xA (OTP)	LSB: 19.5mV RNG: 0mV to 292.5mV 0: 0mV
PACK_UV_OV: 0x37	Bit[0]: VTOP_UV_RT_STS	Reports the real-time result of the VTOP vs. UV threshold comparison. Depending on the settings, a hysteresis is applied to the threshold for comparison. 0: VTOP ≥ UV threshold 1: VTOP < UV threshold	Read-only	0x0	0: Not detected 1: Detected
PACK_UV_OV: 0x37	Bits[7:2]: VTOP_UV_HYST	Sets the battery pack UV threshold hysteresis.	R/W	0x20 (MTP)	LSB: 78.125mV RNG: 0mV to 4921.875mV 00: 0mV
PACK_UV_OV: 0x37	Bit[8]: VTOP_OV_RT_STS	Reports the real-time result of the VTOP vs. OV threshold comparison. Depending on the settings, a hysteresis is applied to the threshold for comparison. 0: VTOP ≤ OV threshold 1: VTOP > OV threshold	Read-only	0x0	0: Not detected 1: Detected
PACK_UV_OV: 0x37	Bits[15:10]: VTOP_OV_HYST	Sets the battery pack OV threshold hysteresis.	R/W	0x20 (MTP)	LSB: 78.125mV RNG: 0mV to 4921.875mV 0x00: 0mV
CELL_UV: 0x38	Bits[7:0]: CELL_UV	Configures the cell UV threshold.	R/W (can lock to read-only)	0x98 (MTP)	LSB: 19.53mV RNG: 0mV to 4980.15mV 0x00: 0mV
CELL_UV: 0x38	Bits[11:8]: CELL_UV_DG	Sets the cell UV deglitch delay, which can be set between 1 reading (no deglitch) and 16 readings. 0x0: A UV condition is reported as soon as the UV threshold is reached 0x1: A UV condition is reported when the current (and previous) protection readings violate the cell UV threshold ... 0xF: A UV condition is reported when a total of 16 protection cycles violate the cell UV threshold	R/W (can lock to read-only)	0x0 (OTP)	LSB: 1 update RNG: 1 to 16 update(s) 0: 1 update

CELL_OV: 0x39	Bits[7:0]: CELL_OV	Configures the cell OV threshold.	R/W (can lock to read-only)	0XD7 (MTP)	LSB: 19.53mV RNG: 0mV to 4980.15mV 0x00: 0mV
CELL_OV: 0x39	Bits[11:8]: CELL_OV_DG	Sets the cell OV deglitch delay, which can be set between 1 reading (no deglitch) and 16 readings. 0x0: An OV condition is reported as soon as the OV threshold is reached 0x1: An OV condition is reported when the current (and previous) protection readings violate the cell OV threshold ... 0xF: An OV condition is reported when a total of 16 protection cycles violate the cell UV threshold	R/W (can lock to read-only)	0x0 (OTP)	LSB: 1 update RNG: 1 to 16 update(s) 0: 1 update
PACK_UV: 0x3A	Bits[11:0]: VTOP_UV_LIMIT	Configures the pack UV threshold.	R/W	0x948 (MTP)	LSB: 19.531mV RNG: 0mV to 79979.5mV 0x000: 0mV
PACK_UV: 0x3A	Bits[15:12]: VTOP_UV_DG	Sets the VTOP UV deglitch delay, which can be set between 1 reading (no deglitch) and 16 readings. 0x0: A UV condition is reported as soon as the UV threshold is reached 0x1: A UV condition is reported when the current (and previous) protection readings violate the VTOP UV threshold ... 0xF: A UV condition is reported when a total of 16 protection cycles violate the VTOP UV threshold	R/W	0x0 (OTP)	LSB: 1 update RNG: 1 to 16 update(s) 0: 1 update
PACK_OV: 0x3B	Bits[11:0]: VTOP_OV_LIMIT	Configures the pack OV threshold.	R/W	0xD71 (MTP)	LSB: 19.531mV RNG: 0mV to 79979.5mV, 0x000: 0mV
PACK_OV: 0x3B	Bits[15:12]: VTOP_OV_DG	Sets the VTOP OV deglitch delay, which can be set between 1 reading (no deglitch) and 16 readings. 0x0: An OV condition is reported as soon as the OV threshold is reached 0x1: An OV condition is reported when the current (and previous) protection readings violate the VTOP OV threshold ... 0xF: An OV condition is reported when a total of 16 protection cycles violate the VTOP OV threshold	R/W	0x0 (OTP)	LSB: 1 update RNG: 1 to 16 update(s) 0: 1 update

CELL_DEAD_THR: 0x3C	Bits[6:0]: CELL_DEAD_LIMIT	Configures the dead cell threshold.	R/W	0x68 (MTP)	LSB: 19.53mV RNG: 0mV to 2480.31mV 0x00: 0mV
CELL_DEAD_THR: 0x3C	Bits[10:7]: CELL_DEAD_DGL_N	Sets the dead cell deglitch delay, which can be set between 1 reading (no deglitch) and 16 readings. 0x0: A dead cell condition is reported as soon as the dead cell threshold is reached 0x1: A dead cell condition is reported when the current (and previous) protection readings violate the dead cell threshold ... 0xF: A dead cell condition is reported when a total of 16 protection cycles violate the dead cell threshold	R/W	0x0 (OTP)	LSB: 1 update RNG: 1 to 16 update(s) 0: 1 update
CELL_MSMT: 0x3D	Bits[4:0]: MSMT_TH	Sets the mismatched cell threshold. The weight of each bit is below: Bit[4]: 625mV Bit[3]: 312.5mV Bit[2]: 156.25mV Bit[1]: 78.1mV Bit[0]: 39mV	R/W	0x02 (OTP)	N/A
CELL_MSMT: 0x3D	Bits[8:5]: CELL_MSMT_DGL_N	Sets the mismatched cell deglitch delay, which can be set between 1 reading (no deglitch) and 16 readings. 0x0: A mismatched cell condition is reported as soon as the dead cell threshold is reached 0x1: A mismatched cell condition is reported when the current (and previous) protection readings violate the mismatched cell threshold ... 0xF: A mismatched cell condition is reported when a total of 16 protection cycles violate the mismatched cell threshold	R/W	0x0 (OTP)	LSB: 1 update RNG: 1 to 16 update(s) 0: 1 update

NTC and Die Temperature

Address Name and Location	Field Bit Position and Name	Description	Type	Value Reset (OTP/MTP)	Encoding, LSB and Range
RD_NTC_DIE: 0x3E	Bit[0]: NTC1_CELL_ HOT_STS	Reports the NTC1 hot cell interrupt status. 0: No interrupt detected 1: An interrupt was detected The latched value depends on the selected interrupt type (e.g. rising, falling, or rising and falling). For a rising edge, the bit is set to 1 if a rising edge is detected. For a falling edge, the bit is set to 1 if a falling edge is detected. For a rising and falling edge, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high.	Read-only	0x0	N/A
RD_NTC_DIE: 0x3E	Bit[1]: NTC2_CELL_ HOT_STS	Reports the NTC2 hot cell interrupt status. 0: No interrupt detected 1: An interrupt was detected The latched value depends on the selected interrupt type (e.g. rising, falling, or rising and falling). For a rising edge, the bit is set to 1 if a rising edge is detected. For a falling edge, the bit is set to 1 if a falling edge is detected. For a rising and falling edge, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high.	Read-only	0x0	N/A
RD_NTC_DIE: 0x3E	Bit[2]: NTC3_CELL_ HOT_STS	Reports the NTC3 hot cell interrupt status. 0: No interrupt detected 1: An interrupt was detected The latched value depends on the selected interrupt type (e.g. rising, falling, or rising and falling). For a rising edge, the bit is set to 1 if a rising edge is detected. For a falling edge, the bit is set to 1 if a falling edge is detected. For a rising and falling edge, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high.	Read-only	0x0	N/A

RD_NTC_DIE: 0x3E	Bit[3]: NTC4_CELL_ HOT_STS	Reports the NTC4 hot cell interrupt status. 0: No interrupt detected 1: An interrupt was detected The latched value depends on the selected interrupt type (e.g. rising, falling, or rising and falling). For a rising edge, the bit is set to 1 if a rising edge is detected. For a falling edge, the bit is set to 1 if a falling edge is detected. For a rising and falling edge, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high.	Read-only	0x0	N/A
RD_NTC_DIE: 0x3E	Bit[4]: NTC1_CELL_ COLD_STS	Reports the NTC1 cold cell interrupt status. 0: No interrupt detected 1: An interrupt was detected The latched value depends on the selected interrupt type (e.g. rising, falling, or rising and falling). For a rising edge, the bit is set to 1 if a rising edge is detected. For a falling edge, the bit is set to 1 if a falling edge is detected. For a rising and falling edge, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high.	Read-only	0x0	N/A
RD_NTC_DIE: 0x3E	Bit[5]: NTC2_CELL_ COLD_STS	Reports the NTC2 cold cell interrupt status. 0: No interrupt detected 1: An interrupt was detected The latched value depends on the selected interrupt type (e.g. rising, falling, or rising and falling). For a rising edge, the bit is set to 1 if a rising edge is detected. For a falling edge, the bit is set to 1 if a falling edge is detected. For a rising and falling edge, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high.	Read-only	0x0	N/A

RD_NTC_DIE: 0x3E	Bit[6]: NTC3_CELL_ COLD_STS	Reports the NTC3 cold cell interrupt status. 0: No interrupt detected 1: An interrupt was detected The latched value depends on the selected interrupt type (e.g. rising, falling, or rising and falling). For a rising edge, the bit is set to 1 if a rising edge is detected. For a falling edge, the bit is set to 1 if a falling edge is detected. For a rising and falling edge, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high.	Read-only	0x0	N/A
RD_NTC_DIE: 0x3E	Bit[7]: NTC4_CELL_ COLD_STS	Reports the NTC4 cold cell interrupt status. 0: No interrupt detected 1: An interrupt was detected The latched value depends on the selected interrupt type (e.g. rising, falling, or rising and falling). For a rising edge, the bit is set to 1 if a rising edge is detected. For a falling edge, the bit is set to 1 if a falling edge is detected. For a rising and falling edge, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high.	Read-only	0x0	N/A
RD_NTC_DIE: 0x3E	Bit[8]: NTC1_PCB_ MNTR_HOT_ STS	Reports the NTC1 hot PCB interrupt status. Depending on PCB_MNTR_STS_SEL, this bit may report the latched status or the real-time status. The latched value depends on the selected interrupt type (e.g. rising, falling, or rising and falling). For a rising edge, the bit is set to 1 if a rising edge is detected. For a falling edge, the bit is set to 1 if a falling edge is detected. For a rising and falling edge, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high. If PCB_MNTR_STS_SEL reports the real-time status: 0: The NTC1 PCB temperature is within its normal range 1: The NTC1 PCB temperature is too hot If PCB_MNTR_STS_SEL reports the latched status: 0: No interrupt has been detected 1: An interrupt has been detected	Read-only	0x0	N/A

RD_NTC_DIE: 0x3E	Bit[9]: NTC2_PCB_ MNTR_HOT_ STS	Reports the NTC2 hot PCB interrupt status. Depending on PCB_MNTR_STS_SEL, this bit may report the latched status or the real-time status. The latched value depends on the selected interrupt type (e.g. rising, falling, or rising and falling). For a rising edge, the bit is set to 1 if a rising edge is detected. For a falling edge, the bit is set to 1 if a falling edge is detected. For a rising and falling edge, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high. If PCB_MNTR_STS_SEL reports the real-time status: 0: The NTC2 PCB temperature is within its normal range 1: The NTC2 PCB temperature is too hot If PCB_MNTR_STS_SEL reports the latched status: 0: No interrupt has been detected 1: An interrupt has been detected	Read-only	0x0	N/A
RD_NTC_DIE: 0x3E	Bit[10]: NTC3_PCB_ MNTR_HOT_ STS	Reports the NTC3 hot PCB interrupt status. Depending on PCB_MNTR_STS_SEL, this bit may report the latched status or the real-time status. The latched value depends on the selected interrupt type (e.g. rising, falling, or rising and falling). For a rising edge, the bit is set to 1 if a rising edge is detected. For a falling edge, the bit is set to 1 if a falling edge is detected. For a rising and falling edge, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high. If PCB_MNTR_STS_SEL reports the real-time status: 0: The NTC3 PCB temperature is within its normal range 1: The NTC3 PCB temperature is too hot If PCB_MNTR_STS_SEL reports the latched status: 0: No interrupt has been detected 1: An interrupt has been detected	Read-only	0x0	N/A

RD_NTC_DIE: 0x3E	Bit[11]: NTC4_PCB_ MNTR_HOT_ STS	Reports the NTC4 hot PCB interrupt status. Depending on PCB_MNTR_STS_SEL, this bit may report the latched status or the real-time status. The latched value depends on the selected interrupt type (e.g. rising, falling, or rising and falling). For a rising edge, the bit is set to 1 if a rising edge is detected. For a falling edge, the bit is set to 1 if a falling edge is detected. For a rising and falling edge, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high. If PCB_MNTR_STS_SEL reports the real-time status: 0: The NTC4 PCB temperature is within its normal range 1: The NTC4 PCB temperature is too hot If PCB_MNTR_STS_SEL reports the latched status: 0: No interrupt has been detected 1: An interrupt has been detected	Read-only	0x0	N/A
RD_NTC_DIE: 0x3E	Bit[13]: DIE_TEMP_ DIG_RT_STS	Reports the real-time digital die temperature status, based on the result of the latest comparison between the digital die temperature reading and the associated hot threshold. 0: The die temperature is within its normal range 1: The digital die temperature is too hot	Read-only	0x0	N/A
RD_NTC_DIE: 0x3E	Bit[14]: DIE_TEMP_ DIG_STS	Reports the die temperature interrupt status. The latched value depends on the selected interrupt type (e.g. rising, falling, or rising and falling). For a rising edge, the bit is set to 1 if a rising edge is detected. For a falling edge, the bit is set to 1 if a falling edge is detected. For a rising and falling edge, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high. This bit is cleared by DIE_TEMP_DIG_CLEAR. 0: No digital die temperature event has been detected 1: The digital die temperature event has been detected	Read-only	0x0	N/A

RD_NTC_DIE: 0x3E	Bit[15]: DIE_TEMP_ ANA_EVENT	Obtains the status from analog circuit recordings after a hot die temperature event. This bit cannot be read in real time, since digital logic is shutdown when this event occurs. This bit is cleared by DIE_TEMP_ANA_CLEAR. 0: The die temperature is within its normal range 1: The analog die temperature is too hot	Read-only	0x0	N/A
RD_V_NTC4_ LR: 0x3F	Bits[9:0]: NTC4_VALUE	Reports the NTC4 autonomous hardware monitoring ADC reading. NTC4 monitoring is controlled by NTC4_EN, thus this register updates when hardware voltage monitoring is requested and NTC4_EN is enabled.	Read-only	0x000	LSB: 0.09766% of NTCB RNG: 0% to 99.906% of NTCB 0x000: 0% of NTCB
RD_V_NTC4_ LR: 0x3F	Bit[12]: NTC1_CELL_ COLD_RT_STS	Reports the NTC1 cold cell real-time status. 0: The NTC1 temperature is within its normal range 1: The NTC1 temperature is too cold	Read-only	0x0	N/A
RD_V_NTC4_ LR: 0x3F	Bit[13]: NTC2_CELL_ COLD_RT_STS	Reports the NTC2 cold cell real-time status. 0: The NTC2 temperature is within its normal range 1: The NTC2 temperature is too cold	Read-only	0x0	N/A
RD_V_NTC4_ LR: 0x3F	Bit[14]: NTC3_CELL_ COLD_RT_STS	Reports the NTC3 cold cell real-time status. 0: The NTC3 temperature is within its normal range 1: The NTC3 temperature is too cold	Read-only	0x0	N/A
RD_V_NTC4_ LR: 0x3F	Bit[15]: NTC4_CELL_ COLD_RT_STS	Reports the NTC4 cold cell real-time status. 0: The NTC4 temperature is within its normal range 1: The NTC4 temperature is too cold	Read-only	0x0	N/A
RD_V_NTC3_ LR: 0x40	Bits[9:0]: NTC3_VALUE	Reports the NTC3 autonomous hardware monitoring ADC reading. NTC3 monitoring is controlled by NTC3_EN, thus this register updates when hardware voltage monitoring is requested and NTC3_EN is enabled.	Read-only	0x000	LSB: 0.09766% of NTCB RNG: 0% to 99.906% of NTCB 0x000: 0% of NTCB
RD_V_NTC3_ LR: 0x40	Bit[12]: NTC1_CELL_ HOT_RT_STS	Reports the NTC1 hot cell real-time status. 0: The NTC1 temperature is within its normal range 1: The NTC1 temperature is too hot	Read-only	0x0	N/A

RD_V_NTC3_ LR: 0x40	Bit[13]: NTC2_CELL_ HOT_RT_STS	Reports the NTC2 hot cell real-time status. 0: The NTC2 temperature is within its normal range 1: The NTC2 temperature is too hot	Read-only	0x0	N/A
RD_V_NTC3_ LR: 0x40	Bit[14]: NTC3_CELL_ HOT_RT_STS	Reports the NTC3 hot cell real-time status. 0: The NTC3 temperature is within its normal range 1: The NTC3 temperature is too hot	Read-only	0x0	N/A
RD_V_NTC3_ LR: 0x40	Bit[15]: NTC4_CELL_ HOT_RT_STS	Reports the NTC4 hot cell real-time status. 0: The NTC4 temperature is within its normal range 1: The NTC4 temperature is too hot	Read-only	0x0	N/A
RD_V_NTC2_ LR: 0x41	Bits[9:0]: NTC2_VALUE	Reports the NTC2 autonomous hardware monitoring ADC reading. NTC2 monitoring is controlled by NTC2_EN, so this register updates when hardware voltage monitoring is requested and NTC2_EN is enabled.	Read-only	0x000	LSB: 0.09766% of NTCB RNG: 0% to 99.906% of NTCB 0x000: 0% of NTCB
RD_V_NTC1_ LR: 0x42	Bits[9:0]: NTC1_VALUE	Reports the NTC1 autonomous hardware monitoring ADC reading. NTC1 monitoring is controlled by NTC1_EN, so this register updates when hardware voltage monitoring is requested and NTC1_EN is enabled.	Read-only	0x000	LSB: 0.09766% of NTCB RNG: 0% to 99.906% of NTCB 0x000: 0% of NTCB
RD_T_DIE: 0x43	Bits[9:0]: DIE_TEMP_V	Reports the die temperature autonomous hardware monitoring ADC reading, which can be calculated with the following equation: $T = \text{Reading} \times 0.474 - 269.12^{\circ}\text{C}$	Read-only	0x000	LSB: 0.474°C RNG: -269.12°C to +215.78°C 0x2E3: 81.18°C
NTC_CLR: 0x44	Bit[2]: PCB_MNTR_ STS_SEL	Controls the NTCx_PCB_MNTR_HOT_STS register. 0: Show the latched status going to the interrupt controller 1: Show the real-time status	R/W	0x0	N/A
NTC_CLR: 0x44	Bit[14]: DIE_TEMP_ DIG_CLEAR	Write 1 to clear a digital die temperature too hot event. This bit is a self-clearing register.	Write-only	0x0	N/A
NTC_CLR: 0x44	Bit[15]: DIE_TEMP_ ANA_CLEAR	Write 1 to clear an analog die too hot event. This bit is a self-clearing register.	Write-only	0x0	N/A
DIE_CFG: 0x46	Bit[1]: DIE_TEMP_ DIG_EN	0: Disable 1: Enable the digital die temperature sensor	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
NTC_CFG: 0x47	Bit[0]: NTC1_EN	Enables NTC1 monitoring if the hardware voltage monitoring is enabled by VOLT_MONT_EN.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled

NTC_CFG: 0x47	Bit[1]: NTC1_TYPE_SEL	Configures the NTC1 monitor type. 0: Cell monitor 1: PCB monitor	R/W (can lock to read-only)	0x0 (OTP)	N/A
NTC_CFG: 0x47	Bit[2]: NTC2_EN	Enables NTC2 monitoring if the hardware voltage monitoring is enabled by VOLT_MONT_EN.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled
NTC_CFG: 0x47	Bit[3]: NTC2_TYPE_SEL	Configures the NTC2 monitor type. 0: Cell monitor 1: PCB monitor	R/W (can lock to read-only)	0x0 (OTP)	N/A
NTC_CFG: 0x47	Bit[4]: NTC3_EN	Enables NTC3 monitoring if the hardware voltage monitoring is enabled by VOLT_MONT_EN.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled
NTC_CFG: 0x47	Bit[5]: NTC3_TYPE_SEL	Configures the NTC3 monitor type. 0: Cell monitor 1: PCB monitor	R/W (can lock to read-only)	0x0 (OTP)	N/A
NTC_CFG: 0x47	Bit[6]: NTC4_EN	Enables NTC4 monitoring if the hardware voltage monitoring is enabled by VOLT_MONT_EN.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled
NTC_CFG: 0x47	Bit[7]: NTC4_TYPE_SEL	Configures the NTC4 monitor type. 0: Cell monitor 1: PCB monitor	R/W (can lock to read-only)	0x1 (OTP)	N/A
NTC_CFG: 0x47	Bit[8]: NTC3_PD_EN	Enables the internal pull-down function on NTC3.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled
NTC_CFG: 0x47	Bit[9]: NTC4_PD_EN	Enables the internal pull-down function on NTC4.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled
NTC_CFG: 0x47	Bit[10]: NTCB_DYNAMIC_ON	Enables dynamic biasing on NTCB during ADC conversions of NTC channels. When disabled, NTCB is always enabled, which increases current consumption.	R/W (can lock to read-only)	0x1 (OTP)	N/A
NTCC_OTHR_DSG: 0x48	Bits[9:0]: NTC_CELL_HOT_DISCH	Sets the discharge NTC hot cell monitor voltage threshold. The alarm is triggered if the value is equal to or below the set threshold.	R/W (can lock to read-only)	0x12E (MTP)	LSB: 0.09756% of NTCB RNG: 0% to 99.9% of NTCB 0x000: 0% of NTCB
NTCC_UTHR_DSG: 0x49	Bits[9:0]: NTC_CELL_COLD_DISCH	Sets the discharge NTC cold cell monitor voltage threshold. The alarm is triggered if the value exceeds the set threshold.	R/W (can lock to read-only)	0x294 (MTP)	LSB: 0.09756% of NTCB RNG: 0% to 99.9% of NTCB 0x000: 0% of NTCB

NTCC_OTHR_CHG: 0x4A	Bits[9:0]: NTC_CELL_HOT_CHG	Sets the charge NTC hot cell monitor voltage threshold. The alarm is triggered if the value is equal to or below the set threshold.	R/W (can lock to read-only)	0x12E (MTP)	LSB: 0.0975% of NTCB RNG: 0% to 99.9% of NTCB 0x000: 0% of NTCB
NTCC_UTHR_CHG: 0x4B	Bits[9:0]: NTC_CELL_COLD_CHG	Sets the charge NTC cold cell monitor voltage threshold. The alarm is triggered if the value exceeds the set threshold.	R/W (can lock to read-only)	0x294 (MTP)	LSB: 0.09756% of NTCB RNG: 0% to 99.9% of NTCB 0x000: 0% of NTCB
NTCC_UTHR_CHG: 0x4B	Bits[15:11]: NTC_CELL_HYST	Sets the NTC cell monitor hysteresis.	R/W (can lock to read-only)	0x11 (OTP)	LSB: 0.1953% of NTCB RNG: 0% to 6.055% of NTCB, 00: 0% of NTCB
NTCM_OTHR: 0x4C	Bits[9:0]: PCB_MNTR_HOT	Sets the NTC voltage threshold for NTC hot PCB monitor alarm. The alarm is triggered if the NTC reading is equal to or below the set threshold.	R/W	0x0EB (OTP)	LSB: 0.09756% of NTCB RNG: 0% to 99.9% of NTCB 0x000: 0% of NTCB
NTCM_OTHR: 0x4C	Bits[15:11]: PCB_MNTR_HYST	Sets the NTC hot PCB monitor hysteresis.	R/W	0x10 (OTP)	LSB: 0.1953% of NTCB RNG: 0% to 6.055% of NTCB, 00: 0% of NTCB
DIE_OT: 0x4D	Bits[9:0]: DIE_TEMP_HOT	Sets the digital die temperature threshold.	R/W	0x2E3 (OTP)	LSB: 0.474°C RNG: -269.12°C to +215.78°C 0x2E3: 81.18°C
DIE_OT: 0x4D	Bits[15:11]: DIE_TEMP_HYST	Sets the digital die temperature hysteresis.	R/W	0x15 (OTP)	LSB: 0.474°C RNG: 0°C to 14.68°C, 0x15: 10°C

Diagnosis

Address Name and location	Field Bit Position and Name	Description	Type	Value Reset (OTP/MTP)	Encoding, LSB and Range
SELF_STS: 0x4E	Bit[0]: SELF_TEST_STS_UV	0: No failure has been detected 1: A failure has been detected	Read-only	0x0	N/A
SELF_STS: 0x4E	Bit[1]: SELF_TEST_STS_OV	0: No failure has been detected 1: A failure has been detected	Read-only	0x0	N/A
SELF_STS: 0x4E	Bit[2]: REGIN_STS	0: No failure has been detected 1: A failure has been detected	Read-only	0x0	N/A
SELF_STS: 0x4E	Bit[3]: 3V3_STS	0: No failure has been detected 1: A failure has been detected	Read-only	0x0	N/A
SELF_STS: 0x4E	Bit[4]: VDD_STS	0: No failure has been detected 1: A failure has been detected	Read-only	0x0	N/A
SELF_STS: 0x4E	Bit[5]: OTPCHK_DONE_STS	When true, OTP CRC check is complete. This bit can be cleared by setting OTP_CRC_DO = 0.	Read-only	0x0	0: False 1: True
SELF_STS: 0x4E	Bit[6]: OTP_CRC_OUTCOME	Reports the errors in the outcome CRC OTP check. 0: No errors detected 1: One or more error(s) detected The result of the previous check is retained until OTP_CRC_DO is set to 1 (from 0). Then OTP_CRC_OUTCOME remains at 0 until the check is completed and a new result is available.	Read-only	0x0	0: False 1: True
RD_VA1P8: 0x4F	Bits[9:0]: VDD_VALUE	Reports the latest hardware monitoring ADC reading for the VDD regulator.	Read-only	0x000	LSB: 3.2227mV RNG: 0mV to 3296.82mV, 0x000: 0mV
RD_VA3P3: 0x50	Bits[9:0]: 3V3_VALUE	Reports the latest hardware monitoring ADC reading for the 3V3 regulator.	Read-only	0x000	LSB: 6.4453mV RNG: 0mV to 6593.55mV, 0x000: 0mV
RD_VA5: 0x51	Bits[9:0]: REGIN_VALUE	Reports the latest hardware monitoring ADC reading for REGIN.	Read-only	0x000	LSB: 6.4453mV RNG: 0mV to 6593.55mV, 0x000: 0mV
RD_VASELF: 0x52	Bits[9:0]: SELF_TEST_VALUE	Reports the latest 10-bit ADC reading for the internal fixed voltage reference, which acts as an ADC self-test.	Read-only	0x000	LSB: 3.2227mV RNG: 0mV to 3296.82mV, 0x000: 0mV
RD_OPENH: 0x53	Bit[0]: CELL_0_OPW_STS	Reports whether C0 has as an open wire after a detection has been executed.	Read-only	0x0	0: Not detected 1: Detected

RD_OPENH: 0x53	Bit[1]: CELL_1_OPW_ STS	Reports whether C1 has as an open wire after a detection has been executed.	Read-only	0x0	0: Not detected 1: Detected
RD_OPENH: 0x53	Bit[2]: CELL_2_OPW_ STS	Reports whether C2 has as an open wire after a detection has been executed.	Read-only	0x0	0: Not detected 1: Detected
RD_OPENH: 0x53	Bit[3]: CELL_3_OPW_ STS	Reports whether C3 has as an open wire after a detection has been executed.	Read-only	0x0	0: Not detected 1: Detected
RD_OPENH: 0x53	Bit[4]: CELL_4_OPW_ STS	Reports whether C4 has as an open wire after a detection has been executed.	Read-only	0x0	0: Not detected 1: Detected
RD_OPENH: 0x53	Bit[5]: CELL_5_OPW_ STS	Reports whether C5 has as an open wire after a detection has been executed.	Read-only	0x0	0: Not detected 1: Detected
RD_OPENH: 0x53	Bit[6]: CELL_6_OPW_ STS	Reports whether C6 has as an open wire after a detection has been executed.	Read-only	0x0	0: Not detected 1: Detected
RD_OPENH: 0x53	Bit[7]: CELL_7_OPW_ STS	Reports whether C7 has as an open wire after a detection has been executed.	Read-only	0x0	0: Not detected 1: Detected
RD_OPENH: 0x53	Bit[8]: CELL_8_OPW_ STS	Reports whether C8 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function.	Read-only	0x0	0: Not detected 1: Detected
RD_OPENH: 0x53	Bit[9]: CELL_9_OPW_ STS	Reports whether C9 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function.	Read-only	0x0	0: Not detected 1: Detected
RD_OPENH: 0x53	Bit[10]: CELL_10_OPW_ STS	Reports whether C10 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function.	Read-only	0x0	0: Not detected 1: Detected
RD_OPENH: 0x53	Bit[11]: CELL_11_OPW_ STS	Reports whether C11 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function.	Read-only	0x0	0: Not detected 1: Detected
RD_OPENH: 0x53	Bit[12]: CELL_12_OPW_ STS	Reports whether C12 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function.	Read-only	0x0	0: Not detected 1: Detected
RD_OPENH: 0x53	Bit[13]: CELL_13_OPW_ STS	Reports whether C13 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function.	Read-only	0x0	0: Not detected 1: Detected

RD_OPENH: 0x53	Bit[14]: CELL_14_OPW_ STS	Reports whether C14 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function.	Read-only	0x0	0: Not detected 1: Detected
RD_OPENH: 0x53	Bit[15]: CELL_15_OPW_ STS	Reports whether C15 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function.	Read-only	0x0	0: Not detected 1: Detected
RD_OPENL: 0x54	Bit[0]: CELL_16_OPW_ STS	Reports whether C16 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function.	Read-only	0x0	0: Not detected 1: Detected
SFT_GO: 0x55	Bit[0]: OTP_CRC_DO	Write 0 to this bit to clear the results. Write 1 to this bit to schedule an OTP CRC.	R/W	0x0	N/A
SFT_GO: 0x55	Bit[8]: OPEN_WIRE_ GO	Write 1 to this bit to schedule an open-wire check. Write 0 to this bit to clear the results of a completed open-wire detection.	R/W	0x0	1: Execute this function
SFT_GO: 0x55	Bit[9]: OPEN_WIRE_ DONE_STS	Reports whether open-wire detection is complete. When true, open-wire detection is complete. Write 0 to OPEN_WIRE_GO to clear this flag.	Read-only	0x0	0: False 1: True
SFT_GO: 0x55	Bit[10]: OPEN_WIRE_ ERR_STS	Reports whether an error has occurred during open-wire detection (e.g. cell-balancing is already operating).	Read-only	0x0	0: False 1: True
SELF_CFG: 0x56	Bit[0]: REGIN_EN	Enables REGIN monitoring to ensure that the REGIN supply is above the defined UV level. 0: Disable REGIN check 1: Enable REGIN check	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
SELF_CFG: 0x56	Bit[1]: 3V3_EN	Enables 3V3 UV monitoring to ensure that the 3.3V supply is above the defined UV level.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
SELF_CFG: 0x56	Bit[2]: VDD_EN	Enables the VDD monitoring check to ensure that the digital 1.8V supply is above the defined UV level. 0: Disable VDD check 1: Enable VDD check	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
SELF_CFG: 0x56	Bit[3]: ADC_SELF_ TEST_EN	Enables the ADC self-test check that indicates whether the applied voltage is within the allowed range. 0: Disabled 1: Enabled	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
SELF_CFG: 0x56	Bit[6]: OTP_CRC_EN	Enables the OTP CRC.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled

OPEN_CFG: 0x57	Bits[3:0]: OPW_CHECK_LEN	Sets the length of each pull-up and pull-down phase.	R/W	0x7 (OTP)	LSB: 1ms RNG: 1ms to 16ms 0: 1ms
OPEN_CFG: 0x57	Bits[11:8]: OPW_CHECK_TH	Sets the open-wire threshold used during the detection sequence.	R/W	0x4 (OTP)	LSB: 39.06mV RNG: 39.06mV to 624.96mV 0: 39.06mV
REGIN_UV: 0x58	Bits[8:0]: REGIN_UV_LIMIT	Sets the threshold for REGIN UV conditions, which is applied to the ADC reading reported in REGIN_VALUE.	Read-only	0x16D (OTP)	LSB: 12.891mV RNG: 0mV to 6587.1mV 0x000: 0mV
V3P3_UV: 0x59	Bits[7:0]: 3V3_UV_LIMIT	Sets the threshold for 3V3 UV conditions, which is applied to the ADC reading reported in the 3V3_VALUE.	R/W	0xF0 (OTP)	LSB: 12.89mV RNG: 0mV to 3287.1mV 0x00: 0mV
VDD_UV: 0x5A	Bits[7:0]: VDD_UV_LIMIT	Sets the threshold for VDD 1.8V UV conditions, which is applied to the ADC reading reported in VDD_VALUE.	Read-only	0x84 (OTP)	LSB: 12.89mV RNG: 0mV to 3287.1mV 0x00: 0mV
SELF_THR: 0x5B	Bits[7:0]: SELF_TEST_UV_LIMIT	Sets the threshold for self-test UV conditions, which is applied to the SELF_TEST_VALUE ADC readings.	Read-only	0x55 (OTP)	LSB 12.89mV RNG: 0mV to 3287.1mV, 0x00: 0mV
SELF_THR: 0x5B	Bits[15:8]: SELF_TEST_OV_LIMIT	Sets the threshold for self-test OV conditions, which is applied to SELF_TEST_VALUE ADC readings.	Read-only	0x65 (OTP)	LSB: 12.89mV RNG: 0mV to 3287.1mV 0x00: 0mV

Recovery

Address Name and Location	Field Bit Position and Name	Description	Type	Value Reset (OTP/MTP)	Encoding, LSB and Range
FT_CLR: 0x5F	Bit[13]: OC_DCHG_ALARM_CLR	Write 1 to this bit to manually clear a discharge OC alarm. This is a self-clearing register.	Write-only	0x0	1: Execute this function
FT_CLR: 0x5F	Bit[14]: OC_CHG_ALARM_CLR	Write 1 to this bit to manually clear a charge OC alarm. This is a self-clearing register.	Write-only	0x0	1: Execute this function
FT_CFG: 0x61	Bit[2]: CELL_HOT_STBY_MODE	Defines the selection criteria for the NTC hot threshold when the battery pack current is within the STBY_CUR_TH range. 0: Aggressive. The hotter threshold (and lower voltage) is selected between NTC_CELL_HOT_CHG and NTC_CELL_HOT_DISCH, and applied when the battery pack current is within the standby region 1: Conservative. The colder threshold (and higher voltage) is selected between NTC_CELL_HOT_CHG and NTC_CELL_HOT_DISCH, and applied when the battery pack current is within the standby region	R/W	0x0	N/A
FT_CFG: 0x61	Bit[3]: CELL_COLD_STBY_MODE	Defines the selection criteria for the NTC cold threshold when the battery pack current is within the STBY_CUR_TH range. 0: Aggressive. The colder threshold (and higher voltage) is selected between NTC_CELL_COLD_DISCH and NTC_CELL_COLD_CHG, and applied when the battery pack current is within the standby region 1: Conservative. The hotter threshold (and lower voltage) is selected between NTC_CELL_COLD_DISCH and NTC_CELL_COLD_CHG, and applied when the battery pack current is within the standby region	R/W	0x0	N/A

Analog-to-Digital Converter (ADC) and Coulomb Counter (CC)

Address Name and Location	Field Bit Position and Name	Description	Type	Value Reset (OTP/MTP)	Encoding, LSB and Range
RD_CCIRQL: 0x65	Bits[15:0]: CC_ACC_LSBS	Accumulated Coulomb counter reading for the LSB. This content is latched from the CC_RT_ACC registers when a Coulomb counting accumulation has been completed. The reading is complement-signed and 26 bits long (combined with CC_ACC_MSBS). The value ranges from -33554432 to +33554431.	Read-only	0x0000	N/A
RD_CCIRQH: 0x66	Bits[9:0]: CC_ACC_MSBS	Accumulated Coulomb counter reading for the MSB. This content is latched from the CC_RT_ACC registers when a Coulomb counting accumulation has been completed. The reading is complement-signed and 26 bits long (combined with CC_ACC_LSBS). The value ranges from -33554432 to +33554431. The value can be calculated with the following equation: Value (mΩ x A x s) = Reading / 32768 / 5	Read-only	0x000	N/A
RD_CCACCQL: 0x67	Bits[15:0]: CC_RT_ACC_LSBS	Returns the real-time accumulated Coulomb counter reading for the LSB. This register is constantly updated when Coulomb counting is active. The reading is complement-signed and 26 bits long (combining with CC_RT_ACC_MSBS). The value ranges from -33554432 to +33554431.	Read-only	0x0000	N/A
RD_CCACCQH: 0x68	Bits[9:0]: CC_RT_ACC_MSBS	Returns the real-time accumulated Coulomb counter reading for the MSB. This register is constantly updated when Coulomb counting is active. The reading is complement-signed and 26 bits long (combining with CC_RT_ACC_LSBS). The value ranges from -33554432 to +33554431. The value can be calculated with the following equation: Value (mΩ x A x s) = Reading / 32768 / 5	Read-only	0x000	N/A
RD_VTOP: 0x6A	Bits[14:0]: VTOP_V	Returns the VTOP high-resolution voltage reading.	Read-only	0x0000	LSB: 0.002441V RNG: 0V to 79.9976V 0x0000: 0V

RD_ITOP: 0x6B	Bits[15:0]: VTOP_I	Returns the battery pack current reading (synchronous to VTOP_V). Signed short integer, and ranges from -32768 to +32767.	Read-only	0x0000	LSB: 0.00305176mV RNG: -100mV to +99.997mV 0x0000: 0mV
RD_VCELL1: 0x6C	Bits[14:0]: CELL_1_V	Returns the cell 1 high-resolution reading.	Read-only	0x0000	LSB: 0.15259mV RNG: 0mV to 4999.85mV 0x0000: 0mV
RD_ICELL1: 0x6D	Bits[15:0]: CELL_1_I_SYNC	Returns the cell 1 battery pack current reading (synchronous with CELL_1_V).	Read-only	0x0000	LSB: 0.00305176mV RNG: -100mV to +99.99695mV 0x0000: 0mV
RD_VCELL2: 0x6E	Bits[14:0]: CELL_2_V	Returns the cell 2 high-resolution reading.	Read-only	0x0000	LSB: 0.15259mV, RNG = 0mV to 4999.85mV, 0x0000 = 0mV
RD_ICELL2: 0x6F	Bits[15:0]: CELL_2_I_SYNC	Returns the cell 2 battery pack current reading (synchronous with CELL_2_V).	Read-only	0x0000	LSB: 0.00305176mV RNG: -100mV to +99.99695mV 0x0000: 0mV
RD_VCELL3: 0x70	Bits[14:0]: CELL_3_V	Returns the cell 3 high-resolution reading.	Read-only	0x0000	LSB: 0.15259mV RNG: 0mV to 4999.85mV 0x0000: 0mV
RD_ICELL3: 0x71	Bits[15:0]: CELL_3_I_SYNC	Returns the cell 3 battery pack current reading (synchronous with CELL_3_V).	Read-only	0x0000	LSB: 0.00305176mV RNG: -100mV to +99.99695mV 0x0000: 0mV
RD_VCELL4: 0x72	Bits[14:0]: CELL_4_V	Returns the cell 4 high-resolution reading.	Read-only	0x0000	LSB: 0.15259mV RNG: 0mV to 4999.85mV 0x0000: 0mV
RD_ICELL4: 0x73	Bits[15:0]: CELL_4_I_SYNC	Returns the cell 4 battery pack current reading (synchronous with CELL_4_V).	Read-only	0x0000	LSB: 0.00305176mV RNG: -100mV to +99.99695mV 0x0000: 0mV
RD_VCELL5: 0x74	Bits[14:0]: CELL_5_V	Returns the cell 5 high-resolution reading.	Read-only	0x0000	LSB: 0.15259mV RNG: 0mV to 4999.85mV 0x0000: 0mV
RD_ICELL5: 0x75	Bits[15:0]: CELL_5_I_SYNC	Returns the cell 5 battery pack current reading (synchronous with CELL_5_V).	Read-only	0x0000	LSB: 0.00305176mV RNG: -100mV to +99.99695mV 0x0000: 0mV

RD_VCELL6: 0x76	Bits[14:0]: CELL_6_V	Returns the cell 6 high-resolution reading.	Read-only	0x0000	LSB: 0.15259mV RNG: 0mV to 4999.85mV 0x0000: 0mV
RD_ICELL6: 0x77	Bits[15:0]: CELL_6_I_SYNC	Returns the cell 6 battery pack current reading (synchronous with CELL_6_V).	Read-only	0x0000	LSB: 0.00305176mV RNG: -100mV to +99.99695mV 0x0000: 0mV
RD_VCELL7: 0x78	Bits[14:0]: CELL_7_V	Returns the cell 7 high-resolution reading.	Read-only	0x0000	LSB: 0.15259mV RNG: 0mV to 4999.85mV 0x0000: 0mV
RD_ICELL7: 0x79	Bits[15:0]: CELL_7_I_SYNC	Returns the cell 7 battery pack current reading (synchronous with CELL_7_V).	Read-only	0x0000	LSB: 0.00305176mV RNG: -100mV to +99.99695mV 0x0000: 0mV
RD_VCELL8: 0x7A	Bits[14:0]: CELL_8_V	Returns the cell 8 high-resolution reading.	Read-only	0x0000	LSB: 0.15259mV RNG: 0mV to 4999.85mV 0x0000: 0mV
RD_ICELL8: 0x7B	Bits[15:0]: CELL_8_I_SYNC	Returns the cell 8 battery pack current reading (synchronous with CELL_8_V).	Read-only	0x0000	LSB: 0.00305176mV RNG: -100mV to +99.99695mV 0x0000: 0mV
RD_VCELL9: 0x7C	Bits[14:0]: CELL_9_V	Returns the cell 9 high-resolution reading.	Read-only	0x0000	LSB: 0.15259mV RNG: 0mV to 4999.85mV 0x0000: 0mV
RD_ICELL9: 0x7D	Bits[15:0]: CELL_9_I_SYNC	Returns the cell 9 battery pack current reading (synchronous with CELL_9_V).	Read-only	0x0000	LSB: 0.00305176mV RNG: -100mV to +99.99695mV 0x0000: 0mV
RD_VCELL10: 0x7E	Bits[14:0]: CELL_10_V	Returns the cell 10 high-resolution reading.	Read-only	0x0000	LSB: 0.15259mV RNG: 0mV to 4999.85mV 0x0000: 0mV
RD_ICELL10: 0x7F	Bits[15:0]: CELL_10_I_SYNC	Returns the cell 10 battery pack current reading (synchronous with CELL_10_V).	Read-only	0x0000	LSB: 0.00305176mV RNG: -100mV to +99.99695mV 0x0000: 0mV
RD_VCELL11: 0x80	Bits[14:0]: CELL_11_V	Returns the cell 11 high-resolution reading.	Read-only	0x0000	LSB: 0.15259mV RNG: 0mV to 4999.85mV 0x0000: 0mV

RD_ICELL11: 0x81	Bits[15:0]: CELL_11_I_ SYNC	Returns the cell 11 battery pack current reading (synchronous with CELL_11_V).	Read-only	0x0000	LSB: 0.00305176mV RNG: -100mV to +99.99695mV 0x0000: 0mV
RD_VCELL12: 0x82	Bits[14:0]: CELL_12_V	Returns the cell 12 high-resolution reading.	Read-only	0x0000	LSB: 0.15259mV RNG: 0mV to 4999.85mV 0x0000: 0mV
RD_ICELL12: 0x83	Bits[15:0]: CELL_12_I_ SYNC	Returns the cell 12 battery pack current reading (synchronous with CELL_12_V).	Read-only	0x0000	LSB: 0.00305176mV RNG: -100mV to +99.99695mV 0x0000: 0mV
RD_VCELL13: 0x84	Bits[14:0]: CELL_13_V	Returns the cell 13 high-resolution reading.	Read-only	0x0000	LSB: 0.15259mV RNG: 0mV to 4999.85mV 0x0000: 0mV
RD_ICELL13: 0x85	Bits[15:0]: CELL_13_I_ SYNC	Returns the cell 13 battery pack current reading (synchronous with CELL_13_V).	Read-only	0x0000	LSB: 0.00305176mV RNG: -100mV to +99.99695mV 0x0000: 0mV
RD_VCELL14: 0x86	Bits[14:0]: CELL_14_V	Returns the cell 14 high-resolution reading.	Read-only	0x0000	LSB: 0.15259mV RNG: 0mV to 4999.85mV 0x0000: 0mV
RD_ICELL14: 0x87	Bits[15:0]: CELL_14_I_ SYNC	Returns the cell 14 battery pack current reading (synchronous with CELL_10_V).	Read-only	0x0000	LSB: 0.00305176mV RNG: -100mV to +99.99695mV 0x0000: 0mV
RD_VCELL15: 0x88	Bits[14:0]: CELL_15_V	Returns the cell 15 high-resolution reading.	Read-only	0x0000	LSB: 0.15259mV RNG: 0mV to 4999.85mV 0x0000: 0mV
RD_ICELL15: 0x89	Bits[15:0]: CELL_15_I_ SYNC	Returns the cell 15 battery pack current reading (synchronous with CELL_15_V).	Read-only	0x0000	LSB: 0.00305176mV RNG: -100mV to +99.99695mV 0x0000: 0mV
RD_VCELL16: 0x8A	Bits[14:0]: CELL_16_V	Returns the cell 16 high-resolution reading.	Read-only	0x0000	LSB: 0.15259mV RNG: 0mV to 4999.85mV 0x0000: 0mV
RD_ICELL16: 0x8B	Bits[15:0]: CELL_16_I_ SYNC	Returns the cell 16 battery pack current reading (synchronous with CELL_16_V).	Read-only	0x0000	LSB: 0.00305176mV RNG: -100mV to +99.99695mV 0x0000: 0mV

RD_VNTC4: 0x8C	Bits[14:0]: NTC4_HIRES_V	Returns the NTC4 high-resolution voltage ADC reading. This reading is ratiometric to NTCB, which has a 3.3V nominal value.	Read-only	0x0000	LSB: 0.00305176% of NTCB RNG: 0% to 99.99695% of NTCB 0x0000: 0% of NTCB
RD_VNTC3: 0x8D	Bits[14:0]: NTC3_HIRES_V	Returns the NTC3 high-resolution voltage ADC reading. This reading is ratiometric to NTCB, which has a 3.3V nominal value.	Read-only	0x0000	LSB: 0.00305176% of NTCB RNG: 0% to 99.99695% of NTCB 0x0000: 0% of NTCB
RD_VNTC2: 0x8E	Bits[14:0]: NTC2_HIRES_V	Returns the NTC2 high-resolution voltage ADC reading. This reading is ratiometric to NTCB, which has a 3.3V nominal value.	Read-only	0x0000	LSB: 0.00305176% of NTCB RNG: 0% to 99.99695% of NTCB 0x0000: 0% of NTCB
RD_VNTC1: 0x8F	Bits[14:0]: NTC1_HIRES_V	Returns the NTC1 high-resolution voltage ADC reading. This reading is ratiometric to NTCB, which has a 3.3V nominal value.	Read-only	0x0000	LSB: 0.00305176% of NTCB RNG: 0% to 99.99695% of NTCB 0x0000: 0% of NTCB
RD_VGPIO3: 0x90	Bits[14:0]: GPIO3_ VOLTAGE	Returns the GPIO3 high-resolution voltage reading.	Read-only	0x0000	LSB: 0.1007mV RNG: 0mV to 3299.9mV 0x0000: 0mV
RD_VGPIO2: 0x91	Bits[14:0]: GPIO2_ VOLTAGE	Returns the GPIO2 high-resolution voltage reading.	Read-only	0x0000	LSB: 0.1007mV RNG: 0mV to 3299.9mV: 0x0000: 0mV
RD_VGPIO1: 0x92	Bits[14:0]: GPIO1_ VOLTAGE	Returns the GPIO1 high-resolution voltage reading.	Read-only	0x0000	LSB: 0.1007mV RNG: 0mV to 3299.9mV 0x0000: 0mV
RD_TDIE: 0x93	Bits[14:0]: DIE_T_ VOLTAGE	Returns the die temperature high-resolution reading, which is proportional to the internal die temperature. The temperature can be calculated with the following equation: $T = \text{Reading} \times 0.01481 - 269.12^{\circ}\text{C}$	Read-only	0x0000	LSB: 0.01481°C RNG: -269.12°C to +216.16°C 0x4D93: 25°C
RD_V1P8: 0x94	Bits[14:0]: VDD_ VOLTAGE	Returns the 1.8 regulator (VDD) high-resolution reading.	Read-only	0x0000	LSB: 0.10071mV RNG: 0mV to 3299.965mV, 0x0000: 0mV

RD_V3P3: 0x95	Bits[14:0]: 3V3_VOLTAGE	Returns the 3.3V regulator (3V3) high-resolution reading.	Read-only	0x0000	LSB: 0.20141mV RNG: 0mV to 6599.6mV, 0x0000: 0mV
RD_V5: 0x96	Bits[14:0]: REGIN_VOLTAGE	Returns the REGIN high-resolution reading.	Read-only	0x0000	LSB: 0.20141mV RNG: 0mV to 6599.6mV, 0x0000: 0mV
CC_STS: 0x97	Bit[0]: CC_STATUS	0: The Coulomb counter is not running 1: The Coulomb counter is running	Read-only	0x0	N/A
ADC_STS: 0x98	Bits[9:8]: FSM_STS	This field reports the status of the feature command scheduler. 0x0: Free 0x1: Executing voltage ADC scan 0x2: Executing cell-balancing 0x3: Executing open-wire detection	Read-only	0x0	N/A
ADC_STS: 0x98	Bits[12:11]: INT_SCHEDULER	This field reports the status of the internal scheduler. 0x0: Idle 0x1: Refreshing voltage protection reading 0x2: Ready for the feature command 0x3: Not allowed	Read-only	0x0	N/A
ADC_CTRL: 0x99	Bit[0]: ADC_SCAN_GO	Write 1 to this bit to start a high-resolution scan for all the selected channels.	R/W	0x0	1: Execute this function
ADC_CTRL: 0x99	Bit[1]: SCAN_DONE_STS	Reports whether the high-resolution voltage scan status is complete (true) or not (false).	Read-only	0x0	0: False 1: True
ADC_CTRL: 0x99	Bit[2]: SCAN_ERROR_STS	Reports whether an error has occurred when starting the high-resolution voltage scan (e.g. if open-wire detection is already running).	Read-only	0x0	0: False 1: True
CC_CFG: 0x9A	Bit[0]: CC_EN	Enables Coulomb counting accumulation.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled
CC_CFG: 0x9A	Bit[1]: CC_EN_CTRL	Controls whether Coulomb counting conversions can occur. When true, CC_EN can start a conversion.	R/W	0x0 (OTP)	0: False 1: True

CC_CFG: 0x9A	Bit[3]: CC_PWR_SAVE	Enables the Coulomb counter to operate in power-save mode. When disabled, power-save mode is not used. Power-save mode reduces overall current consumption at the expense of accuracy, since the current is effectively assumed to be constant during the interval of hardware monitoring scan. Power-save mode refreshes the IADC reading only during a refreshing voltage monitoring, and it accumulates the results of the last conversions without running the ADC to obtain a new reading during the feature command ready scheduler.	R/W	0x0 (OTP)	0: Disabled 1: Enabled
CC_CFG: 0x9A	Bit[4]: CC_DONE	Reports whether Coulomb counting accumulation has completed.	Read-only	0x0	0: False 1: True
CC_CFG: 0x9A	Bit[5]: CC_ERROR_STS	Reports whether a Coulomb counting state machine error has been detected.	Read-only	0x0	0: Not detected 1: Detected
CC_CFG: 0x9A	Bits[13:8]: CC_INT_CNT	Sets the Coulomb counter integration length. The length is set as the number of time slots (32ms each). This register should only be updated when Coulomb counting is not active. 0x00: 1 slot (32ms) 0x01: 2 slots (64ms) ... 0x3F: 64 slots (2048ms)	R/W	0x3F (OTP)	N/A
CC_CFG: 0x9A	Bit[14]: CC_B2B_ACC_CTRL	Controls back-to-back accumulation mode. When enabled, a new Coulomb counting conversion automatically restarts as soon as the current count is completed. With back-to-back conversions enabled, the CC_RT_ACC_LSB and CC_RT_ACC_MSB registers are automatically cleared and do not report the progress of the new accumulation.	R/W	0x0 (OTP)	0: Disabled 1: Enabled
TRIMG_IPCB: 0x9B	Bits[9:0]: I_PCB_GAIN_VALUE	Current-sense PCB gain correction can compensate for sense resistors and SMT variation. The correction is applied to both Coulomb counting and synchronous current ADC readings. It is not applied to OC detection. The encoding is in binary complement format, where LSB = 0.0244% and ranges between $\pm 12.5\%$.	R/W (can lock to read-only)	0x000 (MTP)	LSB: 0.0244% RNG -12.5% to +12.476%, 0x000: 0%

HR_SCAN0: 0x9C	Bit[0]: SCAN_VCELLS_EN	Enables the cell readings to be updated during the high-resolution voltage ADC scan, according to the enable setting of each individual cell. When disabled, the cell readings are excluded from the high-resolution voltage ADC scan.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN0: 0x9C	Bit[1]: SCAN_VTOP_EN	Enables the VTOP reading to be updated during the high-resolution voltage ADC scan.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN0: 0x9C	Bit[3]: SCAN_GPIO_EN	Enables GPIO readings to be updated during the high-resolution voltage ADC scan, according to each individual GPIO enable setting.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN0: 0x9C	Bit[4]: SCAN_NTCS_EN	Enables the NTCs scan to be updated during the high-resolution voltage ADC scan, according to the enable setting of each individual NTC channel.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN0: 0x9C	Bit[5]: SCAN_DIE_T	Enables the die temperature to be updated during the high-resolution voltage ADC scan.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN0: 0x9C	Bit[6]: SCAN_LDOS_EN	Enables VDD, REGIN, and 3.3V readings to be updated during the high-resolution voltage ADC scan.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled
HR_SCAN0: 0x9C	Bit[8]: CELL_SYNC_EN	Enables the battery pack current reading synchronous to each cell to be updated during the high-resolution voltage ADC scan, following the enable setting of each individual cell. When disabled, no synchronous cell current reading can be updated.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN0: 0x9C	Bit[9]: VTOP_SYNC_EN	Enables the battery pack current reading that is synchronous with the VTOP reading to be updated during the high-resolution voltage ADC scan. When disabled, the synchronous VTOP current reading cannot be updated.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN0: 0x9C	Bit[11]: CELL_1K_COMP	Enables ADC cell measurement compensation for the voltage drop caused by the cell input resistor, R _{CELL_FILTER} . This configuration should be enabled when the input filter resistance exceeds 500Ω. For R _{CELL_FILTER} values below 500Ω, this should be disabled.	R/W	0x0 (OTP)	0: Disabled 1: Enabled
HR_SCAN1: 0x9D	Bit[0]: CELL_1_VI_READ_EN	Enables the cell 1 high-resolution voltage. Effective when SCAN_VCELLS_EN is enabled. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 1 is also provided.	R/W	0x1 (OTP)	0: Disabled 1: Enabled

HR_SCAN1: 0x9D	Bit[1]: CELL_2_VI_ READ_EN	Enables the cell 2 high-resolution voltage. Effective when SCAN_VCELLS_EN is enabled. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 2 is also provided.	R/W	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN1: 0x9D	Bit[2]: CELL_3_VI_ READ_EN	Enables the cell 3 high-resolution voltage. Effective when SCAN_VCELLS_EN is enabled. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 3 is also provided.	R/W	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN1: 0x9D	Bit[3]: CELL_4_VI_ READ_EN	Enables the cell 4 high-resolution voltage. Effective when SCAN_VCELLS_EN is enabled. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 4 is also provided.	R/W	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN1: 0x9D	Bit[4]: CELL_5_VI_ READ_EN	Enables the cell 5 high-resolution voltage. Effective when SCAN_VCELLS_EN is enabled. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 5 is also provided.	R/W	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN1: 0x9D	Bit[5]: CELL_6_VI_ READ_EN	Enables the cell 6 high-resolution voltage. Effective when SCAN_VCELLS_EN is enabled. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 6 is also provided.	R/W	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN1: 0x9D	Bit[6]: CELL_7_VI_ READ_EN	Enables the cell 7 high-resolution voltage. Effective when SCAN_VCELLS_EN is enabled. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 7 is also provided.	R/W	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN1: 0x9D	Bit[7]: CELL_8_VI_ READ_EN	Enables the cell 8 high-resolution voltage. Effective when SCAN_VCELLS_EN is enabled. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 8 is also provided.	R/W	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN1: 0x9D	Bit[8]: CELL_9_VI_ READ_EN	Enables the cell 9 high-resolution voltage. Effective when SCAN_VCELLS_EN is enabled. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 9 is also provided.	R/W	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN1: 0x9D	Bit[9]: CELL_10_VI_ READ_EN	Enables the cell 10 high-resolution voltage. Effective when SCAN_VCELLS_EN is enabled. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 10 is also provided.	R/W	0x1 (OTP)	0: Disabled 1: Enabled

HR_SCAN1: 0x9D	Bit[10]: CELL_11_VI_READ_EN	Enables the cell 11 high-resolution voltage. Effective when SCAN_VCELLS_EN is enabled. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 11 is also provided.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN1: 0x9D	Bit[11]: CELL_12_VI_READ_EN	Enables the cell 12 high-resolution voltage. Effective when SCAN_VCELLS_EN is enabled. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 12 is also provided.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN1: 0x9D	Bit[12]: CELL_13_VI_READ_EN	Enables the cell 13 high-resolution voltage. Effective when SCAN_VCELLS_EN is enabled. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 13 is also provided.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN1: 0x9D	Bit[13]: CELL_14_VI_READ_EN	Enables the cell 14 high-resolution voltage. Effective when SCAN_VCELLS_EN is enabled. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 14 is also provided.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN1: 0x9D	Bit[14]: CELL_15_VI_READ_EN	Enables the cell 15 high-resolution voltage. Effective when SCAN_VCELLS_EN is enabled. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 15 is also provided.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN1: 0x9D	Bit[15]: CELL_16_VI_READ_EN	Enables the cell 16 high-resolution voltage. Effective when SCAN_VCELLS_EN is enabled. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 16 is also provided.	R/W (can lock to read-only)	0x1 (OTP)	0: Disabled 1: Enabled
HR_SCAN2: 0x9E	Bit[0]: GPIO1_READ_EN	Enables the GPIO1 high-resolution voltage. Effective when SCAN_GPIO_EN is enabled.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled
HR_SCAN2: 0x9E	Bit[1]: GPIO2_READ_EN	Enables the GPIO2 high-resolution voltage. Effective when SCAN_GPIO_EN is enabled.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled
HR_SCAN2: 0x9E	Bit[2]: GPIO3_READ_EN	Enables the GPIO3 high-resolution voltage. Effective when SCAN_GPIO_EN is enabled.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled
HR_SCAN2: 0x9E	Bit[5]: NTC1_READ_EN	Enables the NTC1 high-resolution voltage scan. Effective when SCAN_NTCS_EN is enabled.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled
HR_SCAN2: 0x9E	Bit[6]: NTC2_READ_EN	Enables the NTC2 high-resolution voltage scan. Effective when SCAN_NTCS_EN is enabled.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled

HR_SCAN2: 0x9E	Bit[7]: NTC3_READ_EN	Enables the NTC3 high-resolution voltage scan. Effective when SCAN_NTCS_EN is enabled.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled
HR_SCAN2: 0x9E	Bit[8]: NTC4_READ_EN	Enables the NTC4 high-resolution voltage scan. Effective when SCAN_NTCS_EN is enabled.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled

Communication

Address Name and Location	Field Bit Position and Name	Description	Type	Value Reset (OTP/MTP)	Encoding, LSB and Range
SILC_INFO1: 0xA0	Bits[5:0]: XFR_NUM_RD_WORDS	When CRC is enabled, this register defines the size of the readback in words before appending a CRC. After each read transaction, this bit self-resets back to 2 bytes. 6'b000001: 1 word / 2 bytes 6'b111111: 63 words / 126 bytes	R/W	0x01	LSB: 1 word, RNG: 1 to 63 words 0x00: 0 words
COMM_CFG: 0xA3	Bit[2]: USE_COMM_CRC	Enables the use of CRC over the communication protocol.	R/W (can lock to Read-only)	0x0 (OTP)	0: Disabled 1: Enabled
COMM_CFG: 0xA3	Bits[14:8]: DEVICE_ADD	Sets the configurable device address.	R/W (can lock to read-only)	0x01 (MTP)	N/A

Cell-Balancing

Address Name and Location	Field Bit Position and Name	Description	Type	Value Reset (OTP/MTP)	Encoding, LSB and Range
BAL_STS: 0xA4	Bit[0]: BALANCING_ACTIVE	Reports whether cell-balancing is currently ongoing.	Read-only	0x0	0: False 1: True
BAL_STS: 0xA4	Bits[6:1]: AUTO_BALANCING_COUNT_STS	Shows the number of balancing cycles that are actually performed. If any balancing cycle is skipped, the reason can be verified by reading AUTO_BAL_SKIPPED_HOT, AUTO_BAL_SKIPPED_DISCHARGE, AUTO_BAL_SKIPPED_STANDBY, or AUTO_BAL_SKIPPED_CHARGE. This register is not updated when AUTO_BAL_ALWAYS is enabled.	Read-only	0x00	N/A
BAL_STS: 0xA4	Bit[7]: AUTO_BAL_SKIPPED_CHARGE	When true, balancing cycles were skipped due to a detected charge current.	Read-only	0x0	0: False 1: True
BAL_STS: 0xA4	Bit[8]: AUTO_BAL_SKIPPED_STANDBY	When true, balancing cycles were skipped due to a detected standby current.	Read-only	0x0	0: False 1: True
BAL_STS: 0xA4	Bit[9]: AUTO_BAL_SKIPPED_DISCHARGE	When true, balancing cycles were skipped due to a detected discharge current.	Read-only	0x0	0: False 1: True

BAL_STS: 0xA4	Bit[10]: AUTO_BAL_ SKIPPED_HOT	When true, balancing cycles were skipped due to the die temperature being too hot.	Read-only	0x0	0: False 1: True
BAL_LIST: 0xA5	Bit[0]: CELL_1_TO_ BALANCE	When true, cell 1 is balanced during the next balancing session. When false, cell 1 is skipped.	R/W	0x0	0: False 1: True
BAL_LIST: 0xA5	Bit[1]: CELL_2_TO_ BALANCE	When true, cell 2 is balanced during the next balancing session. When false, cell 2 is skipped.	R/W	0x0	0: False 1: True
BAL_LIST: 0xA5	Bit[2]: CELL_3_TO_ BALANCE	When true, cell 3 is balanced during the next balancing session. When false, cell 3 is skipped.	R/W	0x0	0: False 1: True
BAL_LIST: 0xA5	Bit[3]: CELL_4_TO_ BALANCE	When true, cell 4 is balanced during the next balancing session. When false, cell 4 is skipped.	R/W	0x0	0: False 1: True
BAL_LIST: 0xA5	Bit[4]: CELL_5_TO_ BALANCE	When true, cell 5 is balanced during the next balancing session. When false, cell 5 is skipped.	R/W	0x0	0: False 1: True
BAL_LIST: 0xA5	Bit[5]: CELL_6_TO_ BALANCE	When true, cell 6 is balanced during the next balancing session. When false, cell 6 is skipped.	R/W	0x0	0: False 1: True
BAL_LIST: 0xA5	Bit[6]: CELL_7_TO_ BALANCE	When true, cell 7 is balanced during the next balancing session. When false, cell 7 is skipped.	R/W	0x0	0: False 1: True
BAL_LIST: 0xA5	Bit[7]: CELL_8_TO_ BALANCE	When true, cell 8 is balanced during the next balancing session. When false, cell 8 is skipped.	R/W	0x0	0: False 1: True
BAL_LIST: 0xA5	Bit[8]: CELL_9_TO_ BALANCE	When true, cell 9 is balanced during the next balancing session. When false, cell 9 is skipped.	R/W	0x0	0: False 1: True
BAL_LIST: 0xA5	Bit[9]: CELL_10_TO_ BALANCE	When true, cell 10 is balanced during the next balancing session. If false, cell 10 is skipped.	R/W	0x0	0: False 1: True
BAL_LIST: 0xA5	Bit[10]: CELL_11_TO_ BALANCE	When true, cell 11 is balanced during the next balancing session. When false, cell 11 is skipped.	R/W	0x0	0: False 1: True
BAL_LIST: 0xA5	Bit[11]: CELL_12_TO_ BALANCE	When true, cell 12 is balanced during the next balancing session. When false, cell 12 is skipped.	R/W	0x0	0: False 1: True
BAL_LIST: 0xA5	Bit[12]: CELL_13_TO_ BALANCE	When true, cell 13 is balanced during the next balancing session. When false, cell 13 is skipped.	R/W	0x0	0: False 1: True
BAL_LIST: 0xA5	Bit[13]: CELL_14_TO_ BALANCE	When true, cell 14 is balanced during the next balancing session. When false, cell 14 is skipped.	R/W	0x0	0: False 1: True
BAL_LIST: 0xA5	Bit[14]: CELL_15_TO_ BALANCE	When true, cell 15 is balanced during the next balancing session. When false, cell 15 is skipped.	R/W	0x0	0: False 1: True
BAL_LIST: 0xA5	Bit[15]: CELL_16_TO_ BALANCE	When true, cell 16 is balanced during the next balancing session. When false, cell 16 is skipped.	R/W	0x0	0: False 1: True

BAL_CTRL: 0xA6	Bit[0]: BALANCE_GO	Controls the start for both manual and automatic cell-balancing. When the bit is turned on, balancing begins.	R/W	0x0	0: Off 1: On
BAL_CTRL: 0xA6	Bit[1]: BAL_DONE_STS	Reports whether cell-balancing is completed.	Read-only	0x0	0: False 1: True
BAL_CTRL: 0xA6	Bit[2]: BAL_ERROR_STS	Reports whether an error occurs when balancing starts (e.g. if open-wire detection is already running).	Read-only	0x0	0: False 1: True
BAL_CFG: 0xA7	Bit[0]: BALANCE_MODE_CTRL	This register is used only when automatic cell-balancing is enabled. This feature is only available on I ² C versions of the MP2787. 0: Register control (BALANCE_GO) 1: GPIO3 control (the direction of GPIO3 is set as input, and the input type is set as digital input). When set to high, balancing begins.	R/W	0x0 (OTP)	N/A
BAL_CFG: 0xA7	Bit[1]: BALANCE_MODE_REG	Controls the cell-balancing mode. 0: Manual cell-balancing 1: Automatic cell-balancing	R/W (can lock to read-only)	0x0 (OTP)	N/A
BAL_CFG: 0xA7	Bit[2]: AUTO_BAL_ALWAYS	This register is used only when automatic cell-balancing is enabled. When disabled, automatic cell-balancing uses BAL_REPETITION to control how many iterations the device repeats. When enabled, balancing continues until the BALANCE_GO register is set to 0 or the balancing list is empty. To stop constant automatic cell-balancing before the balancing list is empty, change this bit to disabled. A cell-balancing done interrupt is triggered if balancing ends because the balancing list is empty.	R/W (can lock to read-only)	0x0 (OTP)	0: Disabled 1: Enabled
BAL_CFG: 0xA7	Bits[7:3]: BAL_REPETITION	Sets the number of repetitions for each execution of the balancing list. When set to 31 repetitions, 32 balancing cycles are executed.	R/W	0x1F (OTP)	LSB: 1 repetition RNG: 0 to 31 repetitions, 0x00: 0 repetitions
BAL_THR: 0xA8	Bits[5:0]: CELL_BAL_MIN	Sets the qualifying minimum cell voltage to run automatic cell-balancing. When a cell is below this level, it is excluded from the balancing list, while other qualifying cells can be balanced if they meet applicable criteria.	R/W	0x21 (OTP)	LSB: 39mV RNG: 2500mV to 4961mV 0x00: 2500mV

BAL_THR: 0xA8	Bit[6]: ABAL_ON_CHARGE	This register is used only when automatic cell-balancing is enabled. 0: Automatic balancing does not run when there is a charge current that exceeds the charge standby threshold 1: Automatic balancing can run when there is a charge current that exceeds the charge standby threshold	R/W	0x0	0: Disabled 1: Enabled
BAL_THR: 0xA8	Bit[7]: ABAL_ON_STBY	This register is used only when automatic cell-balancing is enabled. 0: Automatic balancing does not run when the charge current is between 0 and the standby threshold 1: Automatic balancing can run when the charge current is between 0 and the standby threshold	R/W	0x0	N/A
BAL_THR: 0xA8	Bit[9]: STOP_ON_HOT	0: Do not suspend automatic cell-balancing if the silicon digital die temperature is too hot 1: Suspend automatic cell-balancing if the silicon digital die temperature is too hot	R/W	0x0	N/A
BAL_THR: 0xA8	Bits[12:10]: BAL_MSM_TH	The cell-balancing threshold used by the automatic cell-balancing algorithm.	R/W	0x2 (OTP)	LSB: 9.765mV RNG: 19.5mV to 87.85mV 0: 19.5mV

Memory Control

Address Name and Location	Field Bit Position and Name	Description	Type	Value Reset (OTP/MTP)	Encoding, LSB and Range
MEM_STATUS: 0xB4	Bits[5:3]: MTP_STATUS	Returns the thermometer-coded confirmation status of the MTP: 3'bx1: MTP Page 1 has been fully configured, and the NVM-backed registers can be loaded and use CRC if enabled 3'bx1x: MTP Page 2 has been fully configured, and the NVM-backed registers can be loaded and use CRC if enabled 3'b1xx: MTP Page 3 has been fully configured, and the NVM-backed registers can be loaded and use CRC if enabled	Read-only	0x1	N/A
OTP_CRC_STATUS: 0xB6	Bit[15]: OTP_CRC_ERROR	Reports whether an OTP CRC error has been detected. It is set by a CRC error during OTP restoration, and is reset by the restore command.	Read-only	0x0	0: Not detected 1: Detected
NVM_CRC_STATUS: 0xB7	Bit[15]: MTP_CRC_ERROR	Reports whether an MTP CRC error has been detected. It is set by a CRC error during MTP restoration, and it is reset by the restore command.	Read-only	0x0	0: Not detected 1: Detected

OTP_STORE_CMD: 0xB8	Bit[0]: RESERVED	Reserved. Do not change this register value.	R/W	0x0	N/A
OTP_STORE_CMD: 0xB8	Bit[1]: RESERVED	Reserved. Do not change this register value.	R/W	0x0	N/A
OTP_STORE_CMD: 0xB8	Bit[2]: RESERVED	Reserved. Do not change this register value.	R/W	0x0	N/A
OTP_STORE_CMD: 0xB8	Bit[3]: STORE_MTP_CMD	Store all MTP registers and the MTP CRC code in the NVM if STORE_CMD_ACCESS_CODE has been configured with the correct code. This is a self-clearing register.	R/W	0x0	N/A
OTP_STORE_CMD: 0xB8	Bit[4]: RESERVED	Reserved. Do not change this register value.	R/W	0x0	N/A
OTP_STORE_CMD: 0xB8	Bit[15]: STORE_IN_PROGRESS	Reports whether the store function is in progress. 0: Storage of register data to the MTP memory is not in progress 1: Storage of register data to the MTP memory is in progress	Read-only	0x0	N/A
STORE_CMD_ACCESS_CODE: 0xB9	Bits[15:0]: STORE_CMD_ACCESS_CODE	Before enabling the STORE_MTP_CMD command, 0xA5B6 must be written to this register to allow for the store command. This bit should be cleared once storage is complete.	R/W	0x0000	N/A

LOCK REGISTER MAP

The MP2787 supports the configuration register lock function. This prevents critical safety settings from changing due to unforeseen circumstances

All of the bits in an address that are not populated by a register should be treated as reserved. The value of these reserved bits may be 0 or 1, and should be set to 0 when there are write operations changing other bits in the same address. Setting these reserved bits to 0 does not change the value of the reserved bits.

Once the lock bit is set to 1, it cannot be set to 0 unless the MP2787 is reset.

All of the locking bits support the MTP, and the default value is 0. If the lock bit in the MTP is set to 1, the corresponding setting register cannot be modified anymore. Before setting the MTP command, carefully consider whether to set the locking bit to 1.

Table 9 shows the lock register map.

Table 9: Lock Register Map

Register Location and Bit Position for Locking	Locked Address Name and Location	Locked Field Bit Position and Name
0xAA: Bit[13]	WDT_CFG: 0x10	Bits[15:9]: WDT_BITE_CFG Bits[8:2]: WDT_BARK_CFG Bit[0]: WDT_COM_CTRL
0xAA: Bit[12]	PINS_CFG: 0x0D	Bit[8]: GPIO_LV_CFG Bit[6]: WDT_RST_EN Bit[5]: WDT_RPT Bit[0]: ALERT_POL
0xAA: Bit[11]	GPIO_CFG: 0x0C	Bit[11]: GPIO3_FSEL Bit[10]: GPIO3_PUP Bit[9]: GPIO3_TYPE Bit[8]: GPIO3_IO Bit[6]: GPIO2_PUP Bit[5]: GPIO2_TYPE Bit[4]: GPIO2_IO Bit[2]: GPIO1_PUP Bit[1]: GPIO1_TYPE Bit[0]: GPIO1_IO
0xAA: Bit[8]	RGL_CFG: 0x08	Bit[2]: V3P3_SHDN_EN
0xAA: Bit[6]	STB_CFG: 0x06	Bits[5:4]: STBY_MONITOR_CFG
0xAB: Bit[14]	HR_SCAN2: 0x9E	Bit[8]: NTC4_READ_EN Bit[7]: NTC3_READ_EN Bit[6]: NTC2_READ_EN Bit[5]: NTC1_READ_EN
0xAB: Bit[13]	HR_SCAN2: 0x9E	Bit[2]: GPIO3_READ_EN Bit[1]: GPIO2_READ_EN Bit[0]: GPIO1_READ_EN
0xAB: Bit[12]	HR_SCAN1: 0x9D	Bit[15]: CELL_16_VI_READ_EN Bit[14]: CELL_15_VI_READ_EN Bit[13]: CELL_14_VI_READ_EN Bit[12]: CELL_13_VI_READ_EN

0xAB: Bit[11]	TRIMG_IPCB: 0x9B	Bits[9:0]: I_PCB_GAIN_VALUE
0xAB: Bit[6]	DIE_CFG: 0x46	Bit[1]: DIE_TEMP_DIG_EN
0xAB: Bit[2]	OCFT_CTRL: 0x2A	Bit[5]: OC_CHG_ALARM_EN Bit[4]: OC_DCHG_ALARM_EN Bit[1]: OC_CHG_EN_CTRL Bit[0]: OC_DCHG_EN_CTRL
0xAB: Bit[0]	INT0_EN: 0x19	Bit[15]: INT_ALERT_CTRL
0xAC: Bit[15]	COMM_CFG: 0xA3	Bits[14:8]: DEVICE_ADD
0xAC: Bit[14]	COMM_CFG: 0xA3	Bit[2]: USE_COMM_CRC
0xAC: Bit[13]	HR_SCAN1: 0x9D	Bit[11]: CELL_12_VI_READ_EN Bit[10]: CELL_11_VI_READ_EN
0xAC: Bit[8]	NTC_CFG: 0x47	Bit[10]: NTCB_DYNAMIC_ON
0xAC: Bit[7]	NTC_CFG: 0x47	Bit[7]: NTC4_TYPE_SEL
0xAC: Bit[6]	NTC_CFG: 0x47	Bit[6]: NTC4_EN
0xAC: Bit[5]	NTC_CFG: 0x47	Bit[5]: NTC3_TYPE_SEL
0xAC: Bit[4]	NTC_CFG: 0x47	Bit[4]: NTC3_EN
0xAC: Bit[3]	NTC_CFG: 0x47	Bit[3]: NTC2_TYPE_SEL
0xAC: Bit[2]	NTC_CFG: 0x47	Bit[2]: NTC2_EN
0xAC: Bit[1]	NTC_CFG: 0x47	Bit[1]: NTC1_TYPE_SEL
0xAC: Bit[0]	NTC_CFG: 0x47	Bit[0]: NTC1_EN
0xAD: Bit[15]	CELL_UV: 0x38	Bits[11:8]: CELL_UV_DG Bits[7:0]: CELL_UV
0xAD: Bit[14]	CELL_OV: 0x39	Bits[11:8]: CELL_OV_DG Bits[7:0]: CELL_OV
0xAD: Bit[13]	NTCC_OTHR_DSG: 0x48	Bits[9:0]: NTC_CELL_HOT_DISCH
	NTCC_UTHR_DSG: 0x49	Bits[9:0]: NTC_CELL_COLD_DISCH
	NTCC_OTHR_CHG: 0x4A	Bits[9:0]: NTC_CELL_HOT_CHG
	NTCC_UTHR_CHG: 0x4B	Bits[15:11]: NTC_CELL_HYST Bits[9:0]: NTC_CELL_COLD_CHG
0xAD: Bit[10]	CELLS_CTRL: 0x00	Bits[3:0]: CELL_S_CTRL
0xAD: Bit[4]	SELF_CFG: 0x56	Bit[6]: OTP_CRC_EN
0xAD: Bit[3]	SELF_CFG: 0x56	Bit[3]: ADC_SELF_TEST_EN
0xAD: Bit[2]	SELF_CFG: 0x56	Bit[2]: VDD_EN
0xAD: Bit[1]	SELF_CFG: 0x56	Bit[1]: 3V3_EN
0xAD: Bit[0]	SELF_CFG: 0x56	Bit[0]: REGIN_EN
0xAE: Bit[15]	DSGOC_CFG: 0x2B	Bits[14:8]: OC_DCHG_DG Bit[5]: OC_DCHG_RNG Bits[4:0]: OC_DCHG_LIM
0xAE: Bit[14]	CHGOC_CFG: 0x2C	Bits[14:8]: OC_CHG_DG Bit[5]: OC_CHG_RNG Bits[4:0]: OC_CHG_LIM

0xB0: Bit[15]	PACKFT_CTRL: 0x34	Bit[8]: CELL_DEAD_EN
0xB0: Bit[13]	CELLFT_CTRL: 0x35	Bit[4]: CELL_OV_EN_CTRL Bit[1]: CELL_UV_EN_CTRL
0xB0: Bit[11]	BAL_CFG: 0xA7	Bit[2]: AUTO_BAL_ALWAYS
0xB0: Bit[10]	BAL_CFG: 0xA7	Bit[1]: BALANCE_MODE_REG
0xB0: Bit[9]	BAL_CFG: 0xA7	Bit[0]: BALANCE_MODE_CTRL (I ² C versions only)

APPLICATION INFORMATION

PCB Layout Guidelines

Proper PCB layout is critical to reduce noise, and to optimize the device's accuracy and reliability. For the best results, refer to Figure 21, Figure 22, and Figure 23, and follow the guidelines below:

1. Route the ground plane such that it reduces ground noise and prevents stray current (see Figure 21).

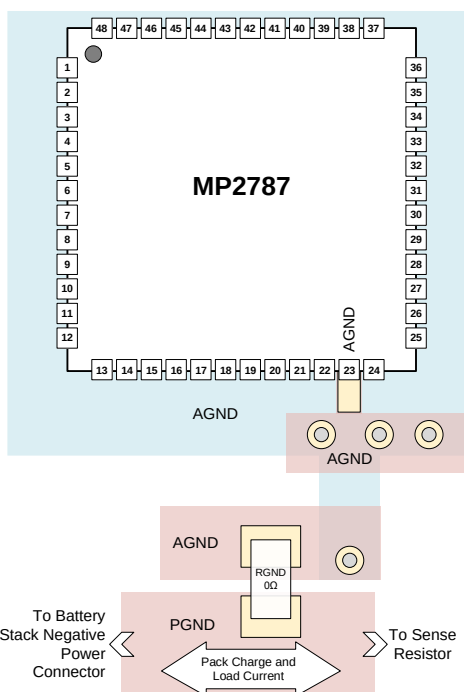


Figure 21: Recommended Layout for AGND

2. Connect the AGND pin to a dedicated ground plane. Then connect the AGND plane to the negative terminal of the battery stack via a 0Ω resistor.
3. Review the schematic and layout to ensure that each component is connected to the correct ground (power vs. signal). For example, the diodes protecting the different power terminals should be connected to PGND.
4. Consider the expected maximum peak load current, then check the PGND length, width, and thickness to ensure that there is an appropriate low-resistance path that prevents voltage drops.
5. The REGIN, VDD, VREF, 3V3, and NTCB pins require external decoupling capacitors, which should be placed as close as possible

to each pin. Minimize the trace inductance from these pins to AGND (see Figure 22).

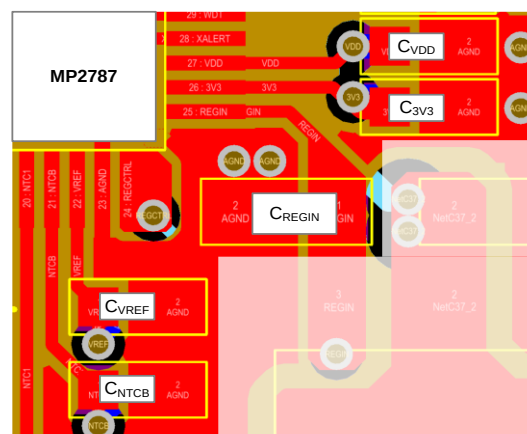


Figure 22: Recommended Layout for Decoupling Capacitors

6. Consider Kelvin connections at the sense resistor. A sense resistor with a dedicated sensing pad is ideal. A simple, two-terminal sense resistor can be used in a 4-wire sensing configuration (see Figure 23).
7. Route the sensing signals to SRP and SRN in parallel to avoid coupling interfering signals.

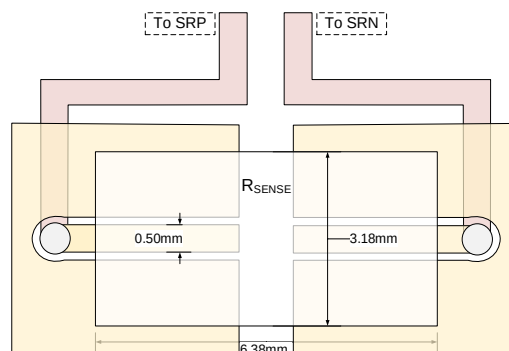


Figure 23: 2512 Sense Resistor Using 4-Wire Footprint

8. Connect the temperature sensors (NTCx) to AGND using the star connection technique to avoid contaminating the voltage reading with the resistive voltage drop caused by the load or charging ground current.
9. Route the two wires connecting the NTCx pins together in twisted pairs to avoid coupling interfering signals. This is recommended if NTCx is located off-board.

TYPICAL APPLICATION CIRCUIT

Figure 24 shows the typical external components and connections required to interface the MP2787 to the battery pack and the system.

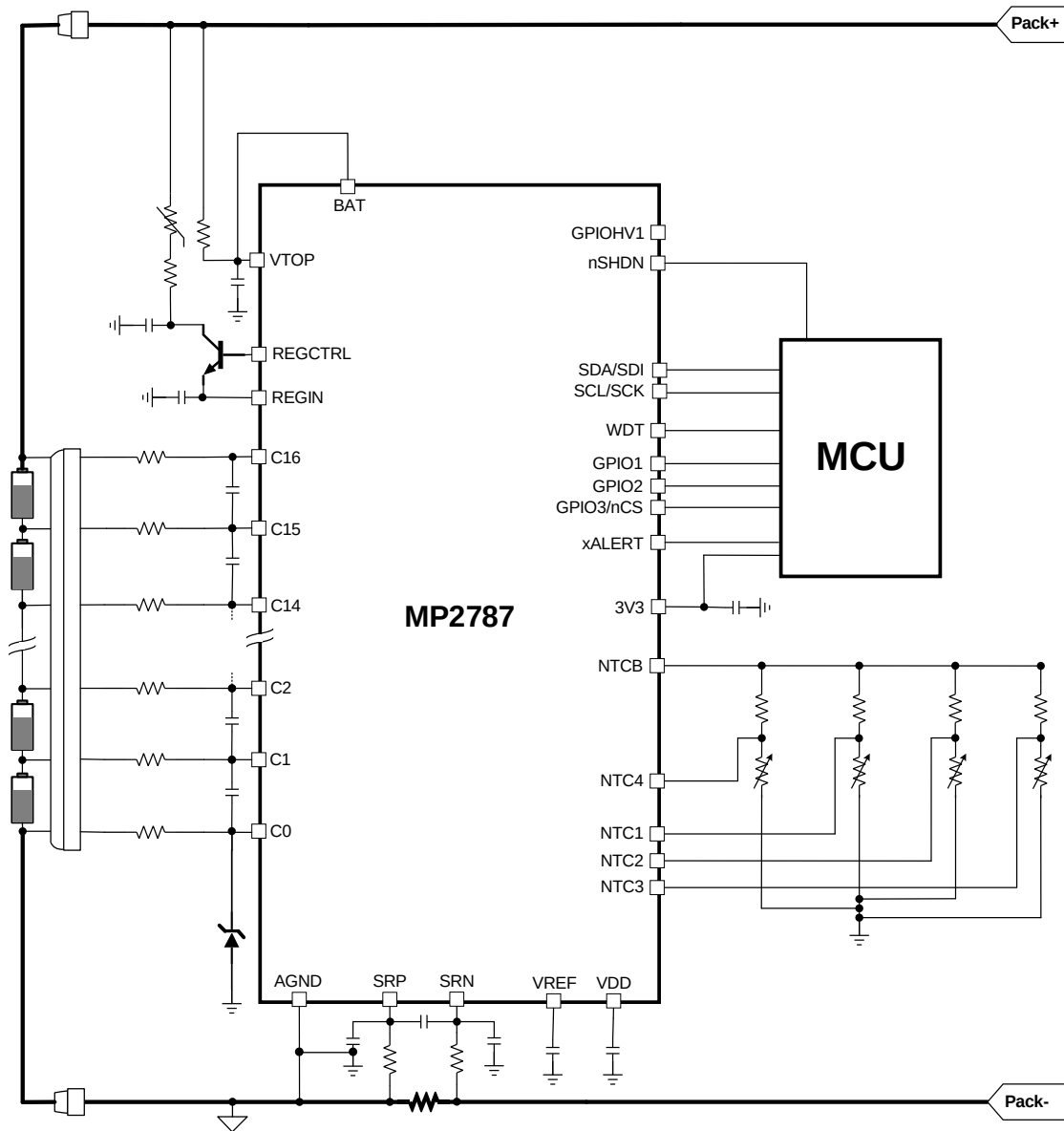


Figure 24: Typical Application Circuit for the MP2787

Figure 25 shows a detailed view of all of the typical recommended external components.

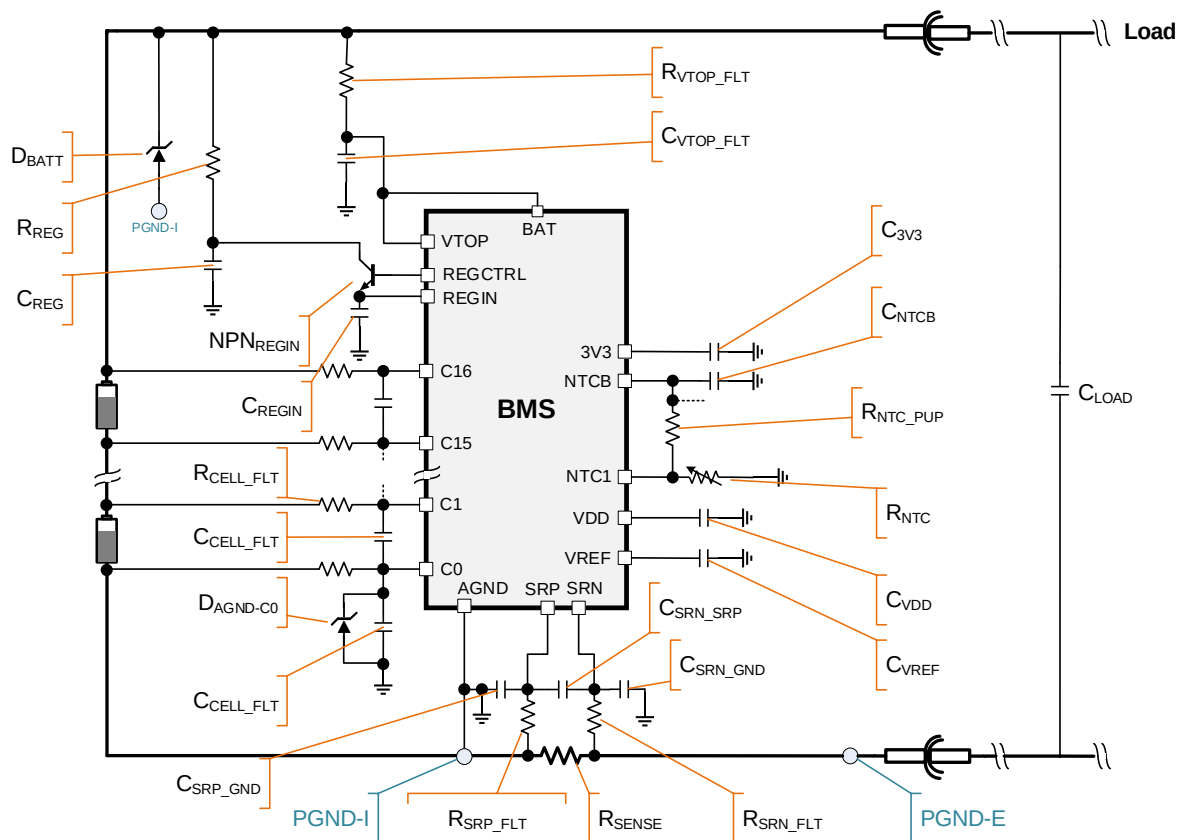


Figure 25: Recommended External Component Diagram

Table 10 on page 97 provides guidance regarding which parameters should be evaluated when selecting each BOM component (see the Configuration for External or Internal Cell-Balancing section on page 98 for more details on cell-balancing). Refer to the MP2787 evaluation board datasheet to see each BOM component.

Table 10: Recommended Components

Parameter Identifier	Parameter Description	Component Description	Parameter Comments	Min	Typ	Max	Unit
General							
D _{AGND-C0}	Recommended Zener diode voltage rating	AGND-C0 Zener diode voltage			3.3		V
Pin Filtering							
C _{VTOP_FLT}	Required capacitance	VTOP-filtering capacitor			470		nF
R _{VTOP_FLT}	Required resistance	VTOP-filtering resistor			20		Ω
Cell Voltage Sensing							
C _{CELL_FLT}	Required capacitance	Cell-filtering capacitor			100		nF
R _{CELL_FLT}	Required resistance	Cell-filtering resistor		20	100		Ω
Current Sensing							
R _{SRN_FLT}	Recommended resistance	SRN filtering resistor			100		Ω
C _{SRN_GND}	Recommended capacitance	SRN-AGND filtering capacitor			100		nF
C _{SRN_SRP}	Recommended capacitance	SRN/P differential filtering capacitor			100		nF
R _{SRP_FLT}	Recommended resistance	SRP filtering resistor			100		Ω
C _{SRP_GND}	Recommended capacitance	SRP-AGND filtering capacitor			100		nF
R _{SENSE}	Recommended resistance	Pack current-sense resistor	The min and max values are recommended, and are based on the current-sense range.	0.2	2	5	mΩ
Regulators							
C _{3V3}	Required capacitance	3.3V capacitor		0.47	1	10	μF
C _{VDD}	Required capacitance	VDD bypass capacitor			1		μF
C _{VREF}	Required capacitance	VREF bypass capacitor			1		μF
C _{REGIN}	Required capacitance	REGIN capacitor			3.3		μF
C _{REG}	Required capacitance	Regulator capacitor			1		μF
R _{REG}	Regulator-limiting resistor value	Regulator-limiting resistor value			500		Ω
Temperature Sensing							
R _{NTC_PUP}	Typical resistance	NTC pull-up	This is a typical value. Generally, the NTC pull-up value should match the NTC thermistor value at 25°C.		10		kΩ
R _{NTC}	Typical resistance	NTC thermistor	If a smaller resistance is used, the NTCB current limit should be evaluated and compared to the total resistance connected to the NTCB in the worst-case scenario (hot temperatures).		10		kΩ
C _{NTCB}	NTCB capacitance	NTCB capacitor				10	nF

Configuration for External or Internal Cell-Balancing

Cell-balancing can be implemented via the internal balancing MOSFETs for up to 58mA. Higher cell-balancing current is possible with external MOSFETs or BJTs (see Figure 26).

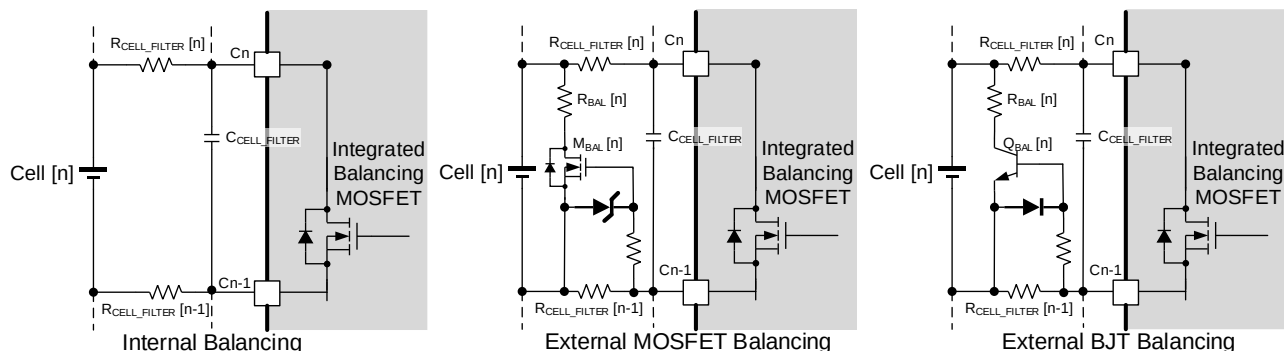


Figure 26: Typical Schematics Internal vs. External Balancing Configurations

The section below describes the recommended components for each of these 3 balancing configurations. Table 11 recommends specific components for internal MOSFET balancing.

Table 11: Recommended Component Selection for Internal MOSFET Balancing

Parameter Identifier	Parameter Description	Component Description	Notes	Min	Typ	Max	Unit
C _{CELL_FILTER}	Required capacitance	Cell-filtering capacitor	Capacitance for internal balancing configuration. If R _{CELL_FILTER} increases, C _{CELL_FILTER} should be proportionally decreased.		100		nF
R _{CELL_FILTER}	Required resistance	Cell-filtering resistor	Resistance for internal balancing configuration. If R _{CELL_FILTER} increases, C _{CELL_FILTER} should be proportionally decreased.	20	100		Ω

The amount of current provided in an internal balancing configuration is easily derived by following the R_{CELL_FILTER} value and considering the integrated MOSFET balancing resistance (R_{DS(ON)_BAL_FET}). For example, for a lithium cell at 4V where R_{CELL_FILTER} = 20Ω, the resistive path allows for 58mA once the integrated balancing resistor MOSFET is enabled.

For higher balancing current, Table 12 and Table 13 on page 99 provide guidance on selecting the MOSFET or BJT circuit components.

Table 12 recommends specific components for external MOSFET balancing.

Table 12: Recommended Component Selection for External MOSFET Balancing

Parameter Identifier	Parameter Description	Component Description	Notes	Min	Typ	Max	Unit
C _{CELL_FILTER}	Required capacitance	Cell-filtering capacitor	Capacitance for external balancing configuration. If R _{CELL_FILTER} increases, C _{CELL_FILTER} should be proportionally decreased.		10		nF
R _{CELL_FILTER}	Required resistance	Cell-filtering resistor	Resistance for external balancing configuration. If R _{CELL_FILTER} increases, C _{CELL_FILTER} should be proportionally decreased.	800	1000		Ω
R _{BAL}	Required resistance	Cell-balancing current-limiting resistor	Adjust this value to set the balancing current, and verify the power dissipation compared to package limit.		43		Ω
V _{TH_MBAL}	MOSFET V _{GS} threshold	External balancing MOSFET M _{BAL}	A higher voltage threshold may prevent the MOSFET from turning on.		1.5	1.8	V

The external balancing N-channel MOSFETs should be selected using Table 12 on page 98. Choose a MOSFET with a sufficiently low V_{GS} threshold. For simplification, assume that the integrated MOSFET-balancing resistance can be ignored. The available V_{GS} that turns on the MOSFET is generated across R_{CELL_FILTER} , meaning its voltage is half of the lithium cell voltage (e.g. $4.2V / 2 = 2.1V$). In this scenario, the V_{GS} threshold for M_{BAL} should be below 2.1V with an appropriate safety margin. The safety margin should allow for voltage drops on the integrated MOSFET-balancing resistor. This drop can be caused by the following:

- Changes in the operating conditions
- Shifts in the M_{BAL} V_{GS} threshold across operating conditions
- Drops that develop across the sensing wires due to the balancing current

Table 13 recommends specific components for external BJT balancing.

Table 13: Recommended Component Selection for External BJT Balancing

Parameter Identifier	Parameter Description	Component Description	Notes	Min	Typ	Max	Unit
C_{CELL_FILTER}	Required capacitance	Cell-filtering capacitor	Capacitance for external balancing configuration. If R_{CELL_FILTER} increases, C_{CELL_FILTER} should be proportionally decreased.		100		nF
R_{CELL_FILTER}	Required resistance	Cell-filtering resistor	Resistance for external balancing configuration. If R_{CELL_FILTER} increases, C_{CELL_FILTER} should be proportionally decreased.	20	100	1000	Ω
R_{BAL}	Required resistance	Cell-balancing current-limiting resistor	Adjust this value to set the balancing current, and verify the power dissipation compared to package limit.		43		Ω
h_{FE_QBAL}	BJT DC current gain	External balancing BJT Q_{BAL}	A lower h_{FE} may limit the external balancing current.	50			

Because R_{BAL} sets the balancing current and dissipates most of the power, ensure that an appropriate resistor and resistor package are selected, especially for resistors below 100 Ω . For example, if $R_{BAL} = 43\Omega$, there is 410mW of power dissipation when the lithium cell voltage = 4.2V. This means that a package of at least 2512 (6432 Metric) should be used.

It is also important to consider the thermal design of the overall board and enclosure. The MP2787 can simultaneously balance only odd or even cells, meaning 8 cells in a 16-cell system. When balancing is run continuously on 8 cells, 3.28W is dissipated as heat with a 43 Ω R_{BAL} . To keep the MP2787 in the allowed temperature range, consider how this additional heat can be dissipated, as it is added to the amount of heat generated by other components.

The MP2787 includes compensation for the voltage drop across R_{CELL_FILTER} via the OTP register $CELL_1K_COMP$. When enabled, this register compensates for an R_{CELL_FILTER} value of 1k Ω . Generally, $CELL_1K_COMP$ should be enabled if R_{CELL_FILTER} is 500 Ω or greater.

Unused Pins

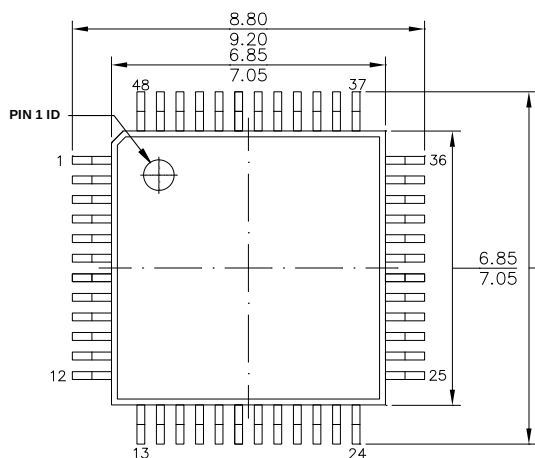
For a particular application, some pins may not be required. Table 14 on page 100 shows how to connect these pins when they are not used.

Table 14: Unused Pins Connection

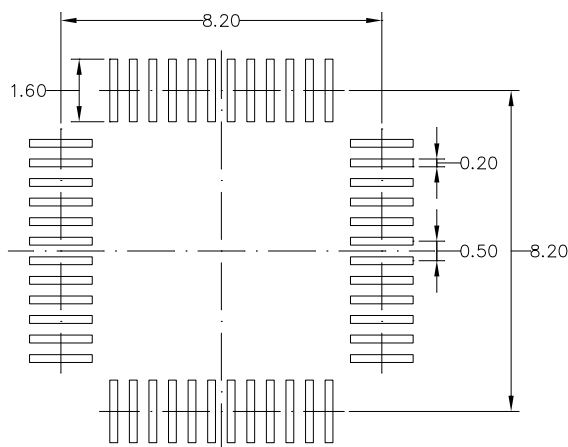
Pin #	Name	Recommendation
1, 2, 3, 4, 5, 6, 46, 47, 48	C8, C9, C10, C11, C12, C13, C14, C15, C16	If fewer than 16 series cells are used, directly connect all unused cell channels to the practical maximum cell channel. For example, if only 10 cells are used, C16-C10 should be connected together.
15, 16	SRP, SRN	If the current-sense function is not required, connect these pins to AGND.
17, 18, 19, 20	NTC1, NTC2, NTC3, NTC4	Float the unused NTC channels.
21	NTCB	If NTC temperature monitoring is not used, float this pin.
28	XALERT	Float this pin if it is not used.
29	WDT	Float this pin if it is not used.
33	SDO	Float this pin if it is not used.
34	GPIO3/nCS	Float this pin if it is not used. Note that the GPIO3 should be set to 20kΩ pull-up mode via the GPIO_CFG register; otherwise there may be additional power consumption.
35, 36	GPIO1, GPIO2	Float this pin if it is not used. Note that the unused GPIO should be set to 20kΩ pull-up mode via the GPIO_CFG register; otherwise there may be additional power consumption.
37	GPIOHV1	Float this pin if it is not used.

PACKAGE INFORMATION

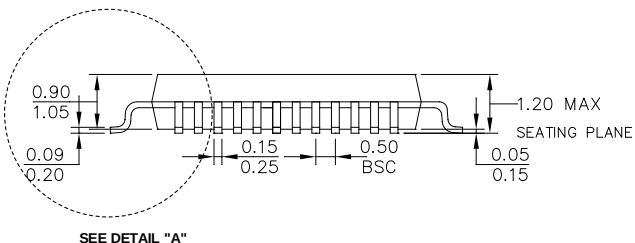
TQFP-48 (7mmx7mm)



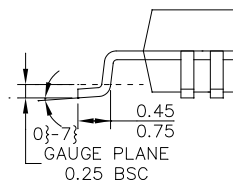
TOP VIEW



RECOMMENDED LAND PATTERN



SIDE VIEW

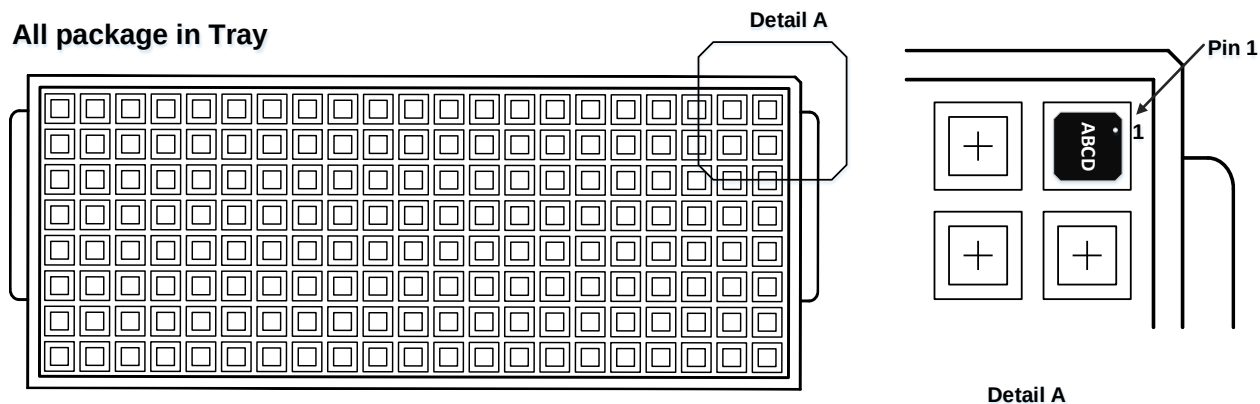


DETAIL "A"

NOTE:

- 1) ALL DIMENSIONS ARE IN MILLIMETERS.
- 2) PACKAGE LENGTH DOES NOT INCLUDE MOLD FLASH, PROTRUSION, OR GATE BURR.
- 3) PACKAGE WIDTH DOES NOT INCLUDE INTERLEAD FLASH OR PROTRUSION.
- 4) LEAD COPLANARITY (BOTTOM OF LEADS AFTER FORMING) SHALL BE 0.10 MILLIMETERS MAX.
- 5) JEDEC REFERENCE IS MO-143.
- 6) DRAWING IS NOT TO SCALE.

CARRIER INFORMATION



Part Number	Package Description	Quantity/ Reel	Quantity/ Tube	Quantity/ Tray	Reel Diameter	Carrier Tape Width	Carrier Tape Pitch
MP2787DFP-xxxx-T	TQFP-48 (7mmx7mm)	N/A	N/A	250pcs	N/A	N/A	N/A

REVISION HISTORY

Revision #	Revision Date	Description	Pages Updated
1.0	8/17/2023	Initial Release	-

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